

View A

Section Plate

CoDa-Standard reading · bars + plan-view scatter · magnitude index

§ FRA France

SECTION PLATE t=0 | Year 2000 | D=9 N=26 pairs=36

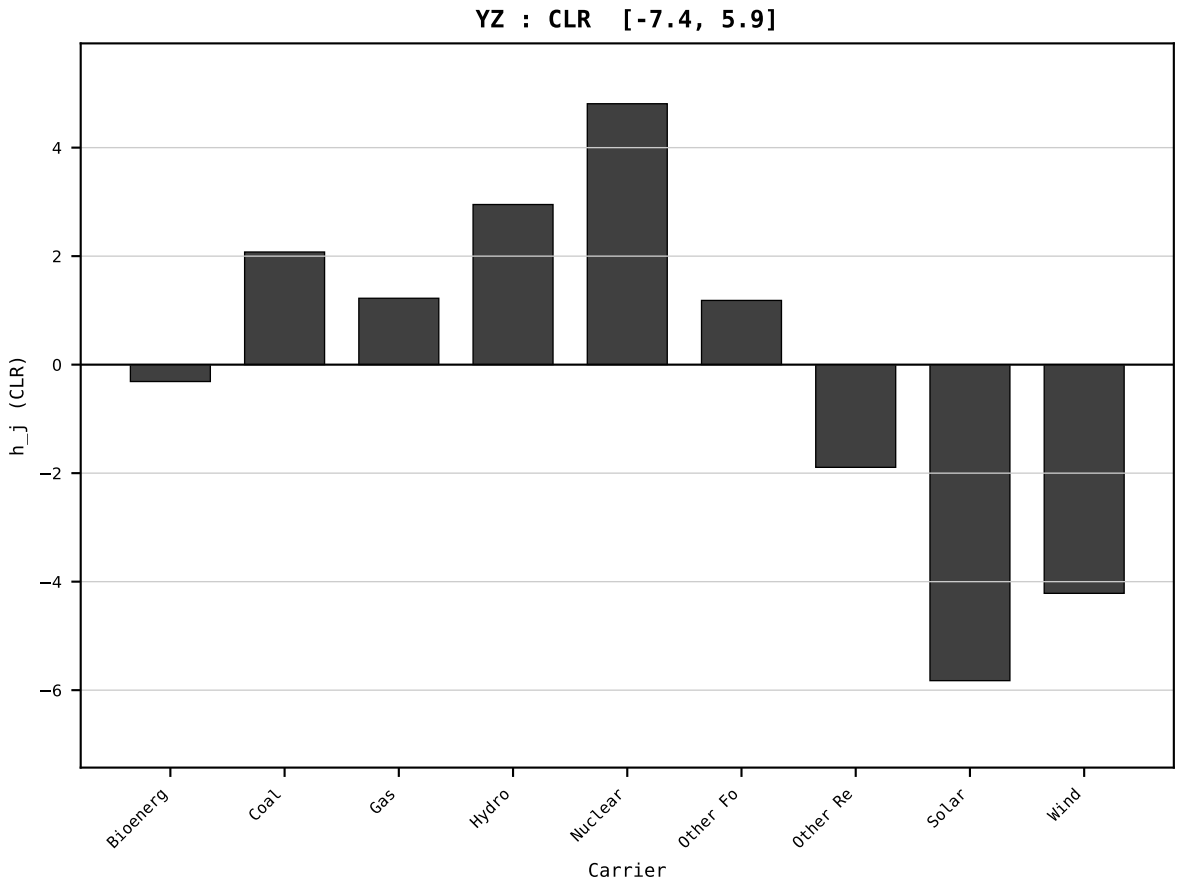
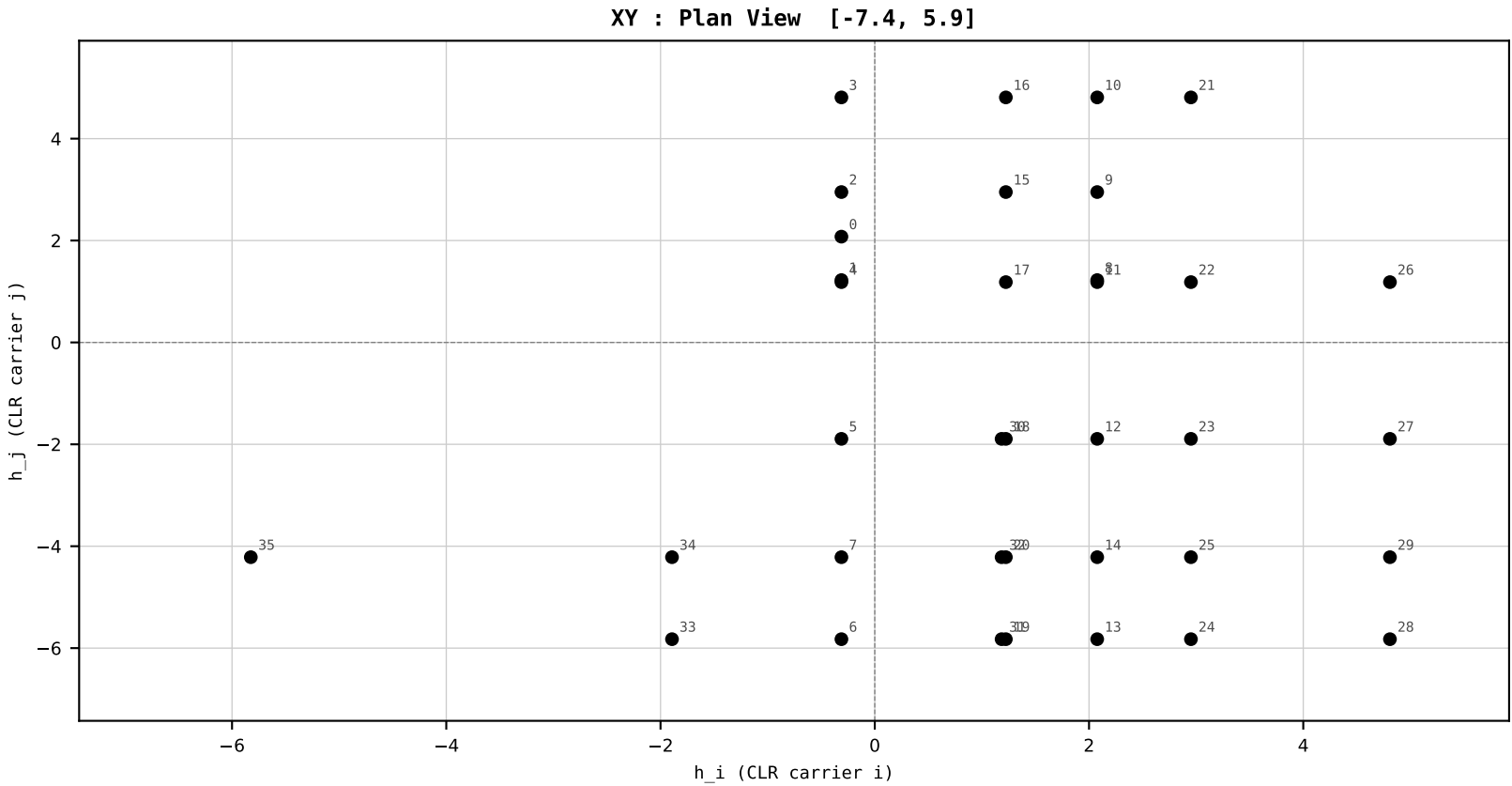
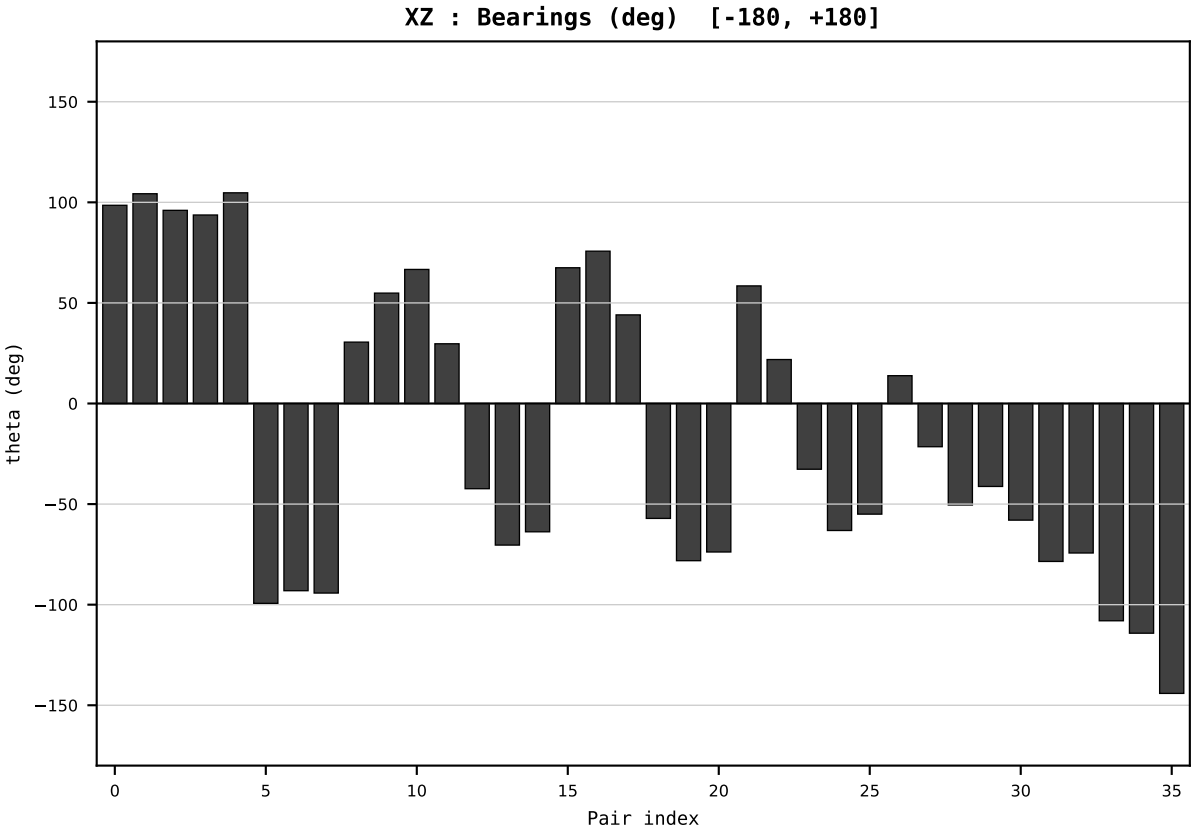
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_FRA_France_generation_TWh.csv
Year : 2000
t : 0 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.636980
Ring = Hs-4
E_metric= 94.4242
kappa_HS= 41516.00
omega = 0.0000 deg
d_A = 0.000000
Helm = ---
Helm d = ---
DR = 10.634 HLR
DR ratio= 41516.0x

XZ [-180,180]
CLR[-7.4,5.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=1 | Year 2001 | D=9 N=26 pairs=36

HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

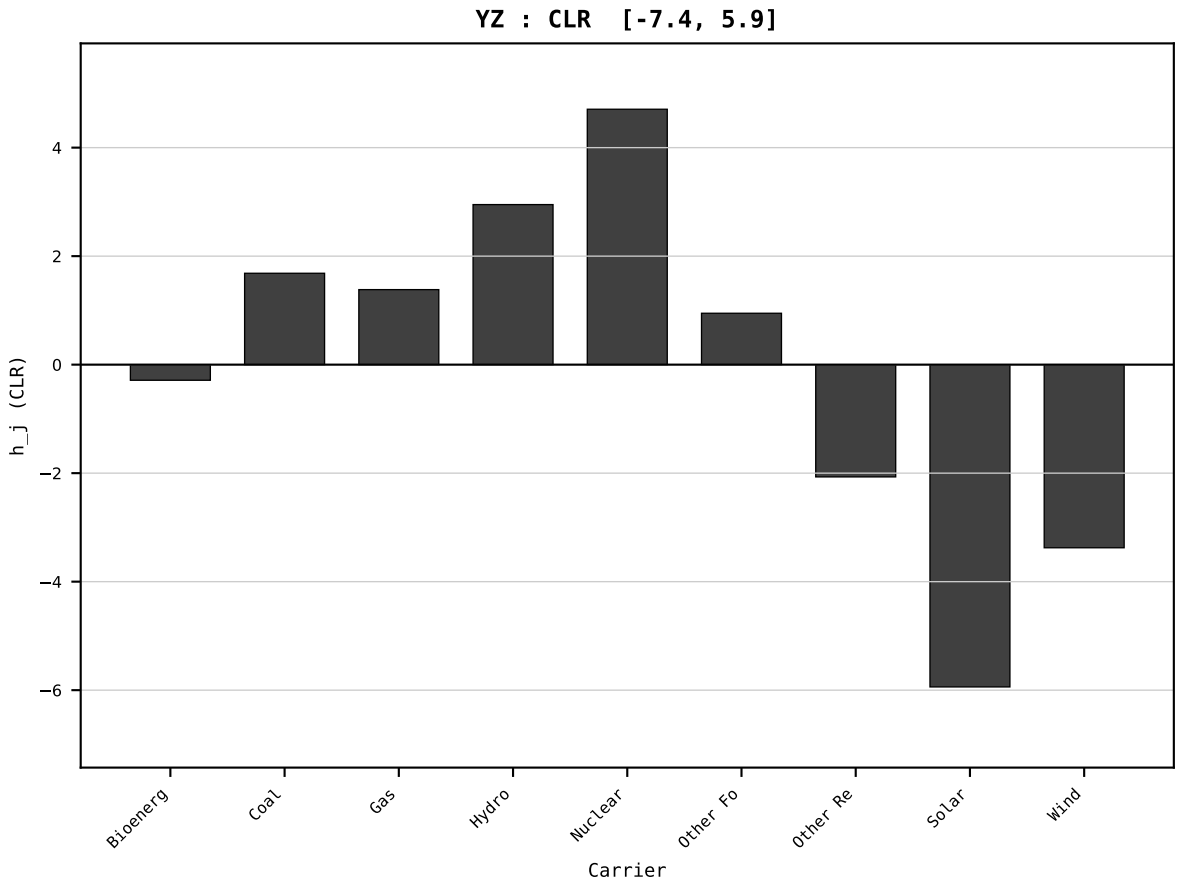
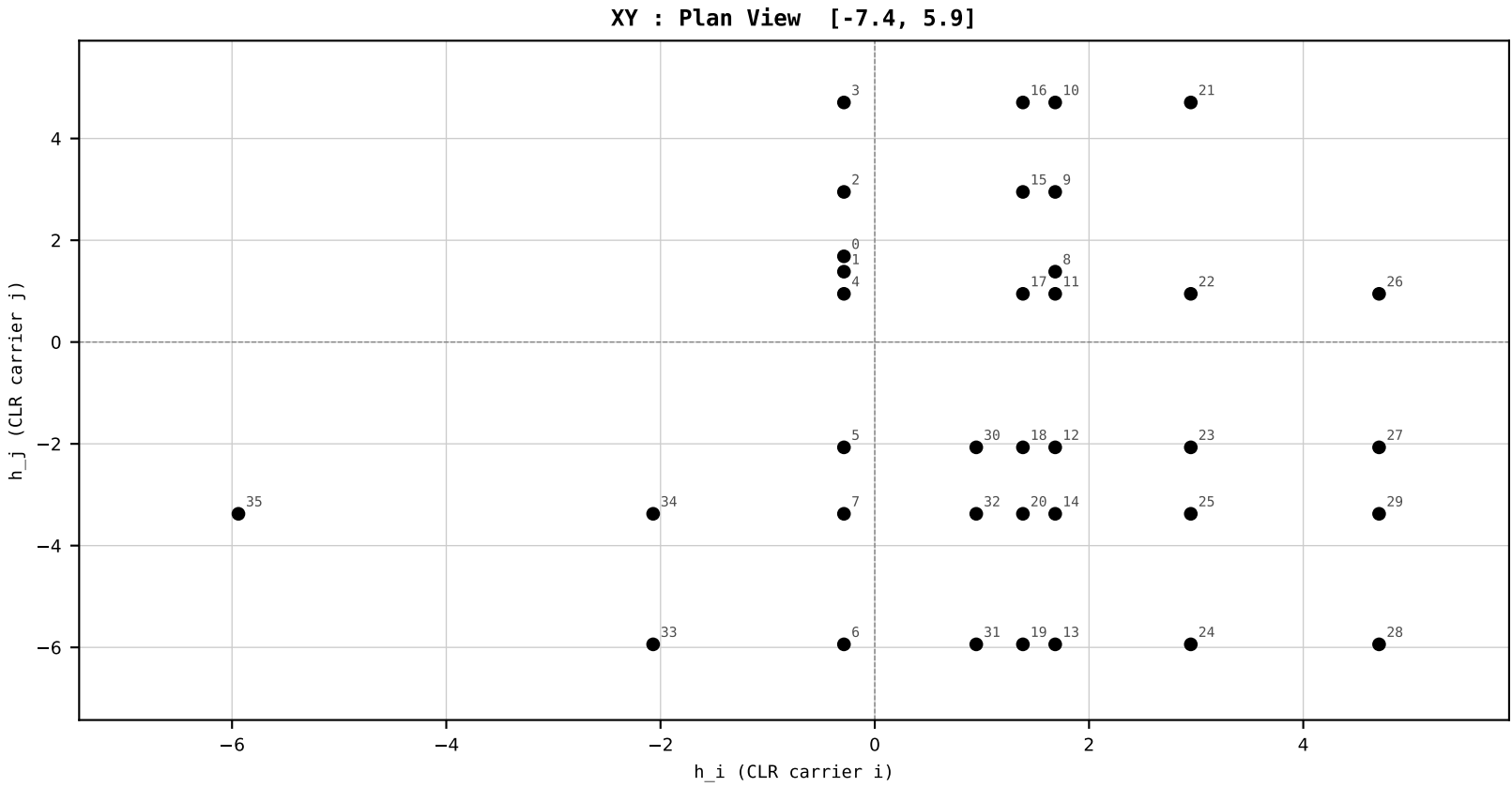
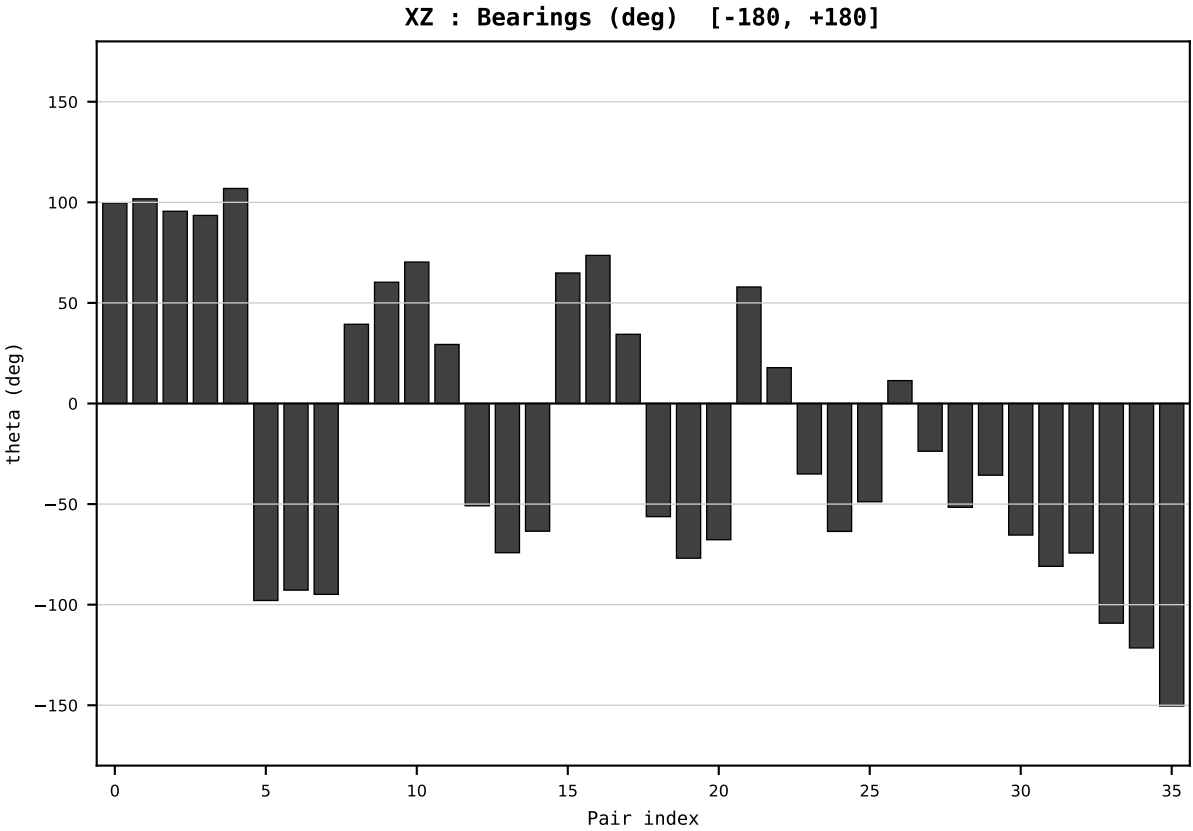
Dataset : ember_FRA_France_generation_TWh.csv
Year : 2001
t : 1 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.636809
Ring = Hs-4
E_metric= 87.5562
kappa_HS= 42108.00
omega = 5.5939 deg
d_A = 0.997829
Helm = Wind
Helm d = +0.839979
DR = 10.648 HLR
DR ratio= 42108.0x

XZ [-180,180]
CLR[-7.4,5.9]
Hs [0, 1]

PAIR INDEX:

0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=2 | Year 2002 | D=9 N=26 pairs=36

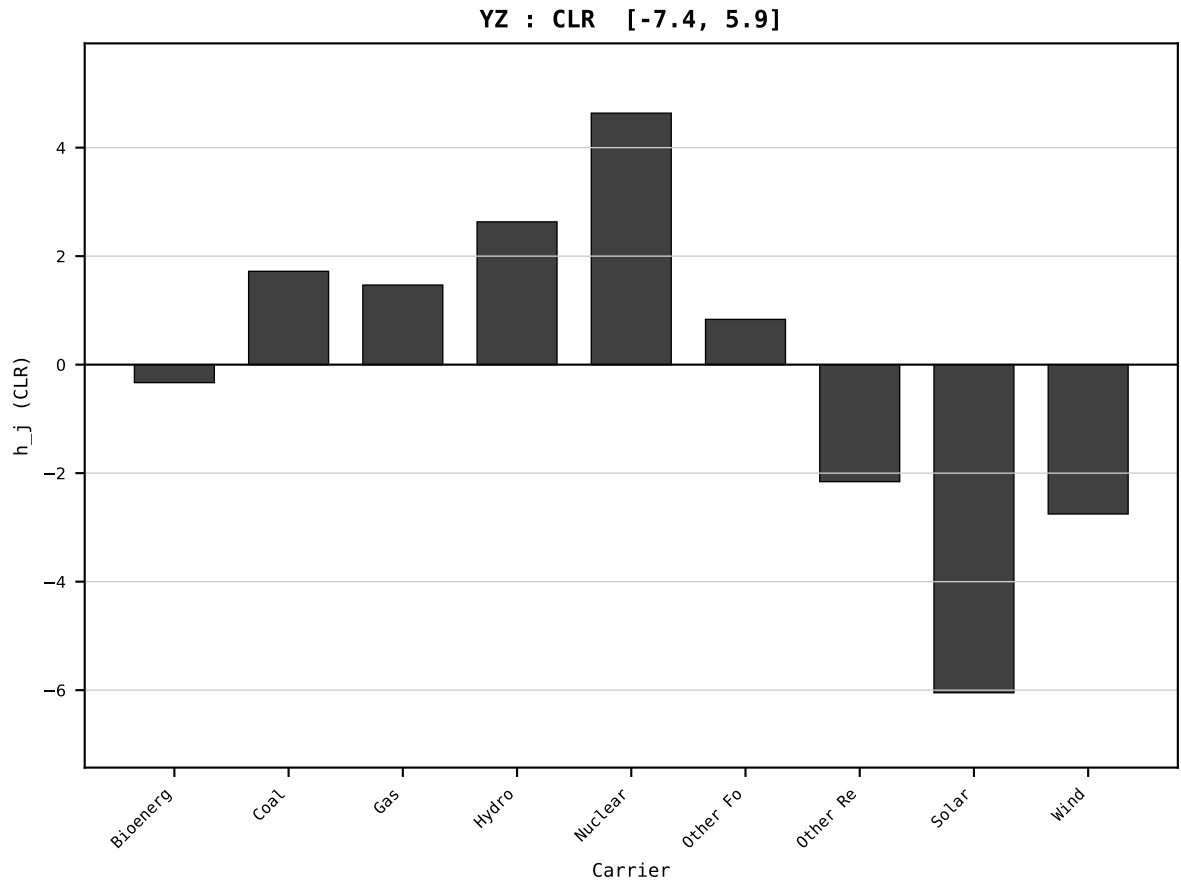
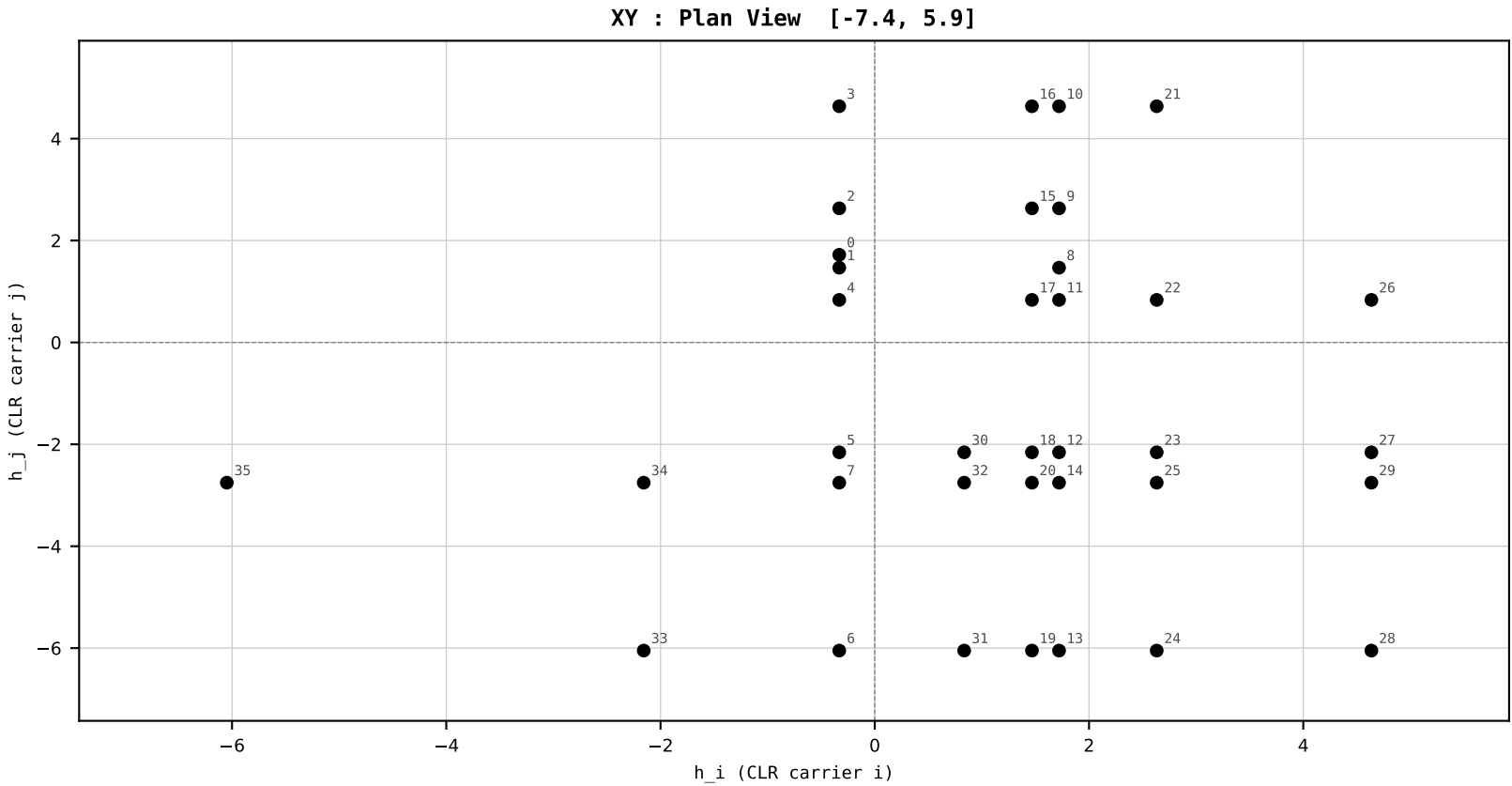
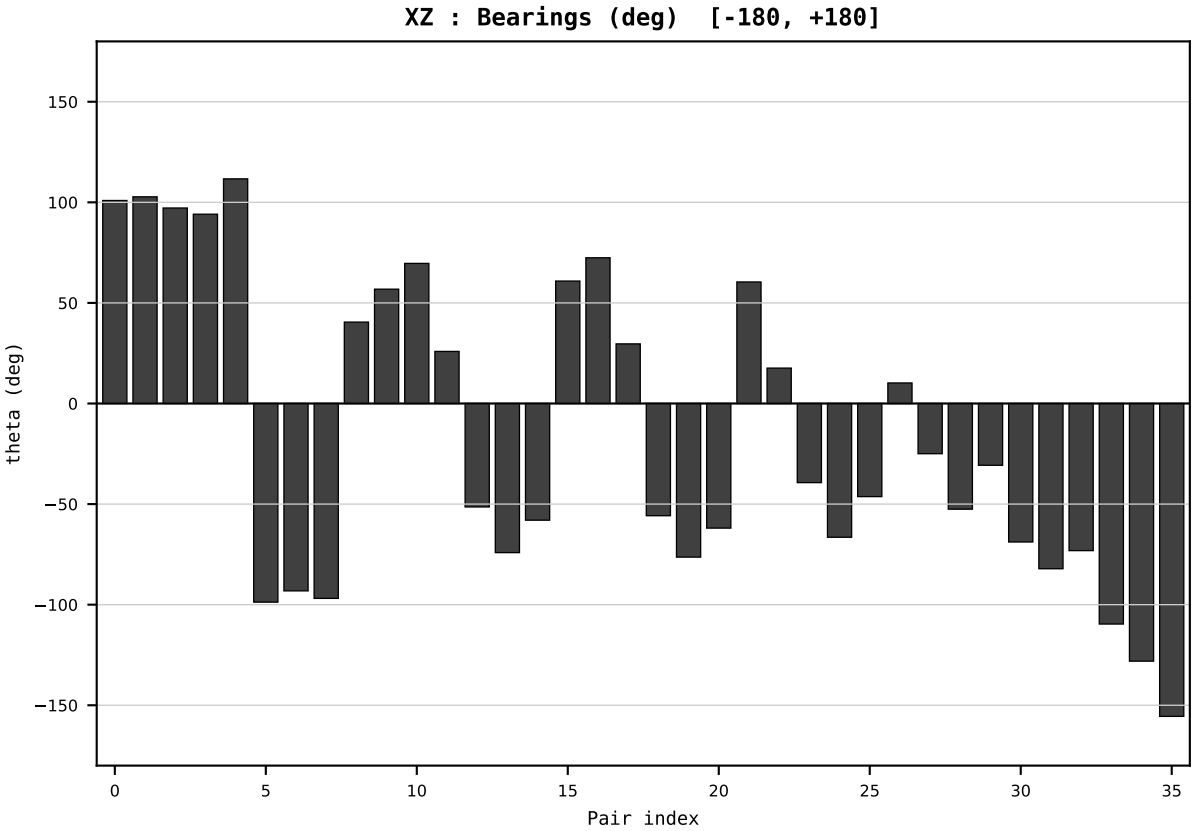
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_FRA_France_generation_TWh.csv
Year : 2002
t : 2 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.644046
Ring = Hs-4
E_metric= 83.1543
kappa_HS= 43676.00
omega = 4.2963 deg
d_A = 0.732324
Helm = Wind
Helm d = +0.622462
DR = 10.685 HLR
DR ratio= 43676.0x

XZ [-180,180]
CLR[-7.4,5.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=3 | Year 2003 | D=9 N=26 pairs=36

HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

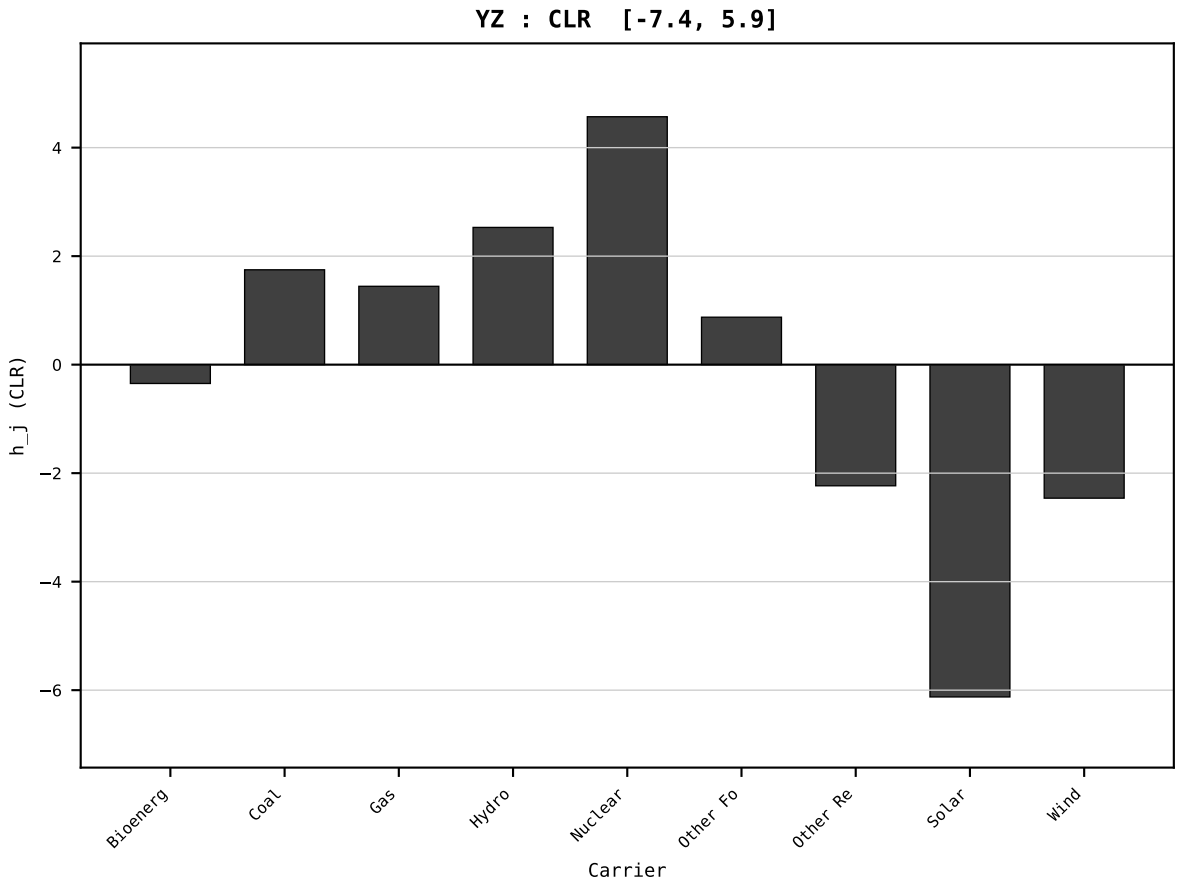
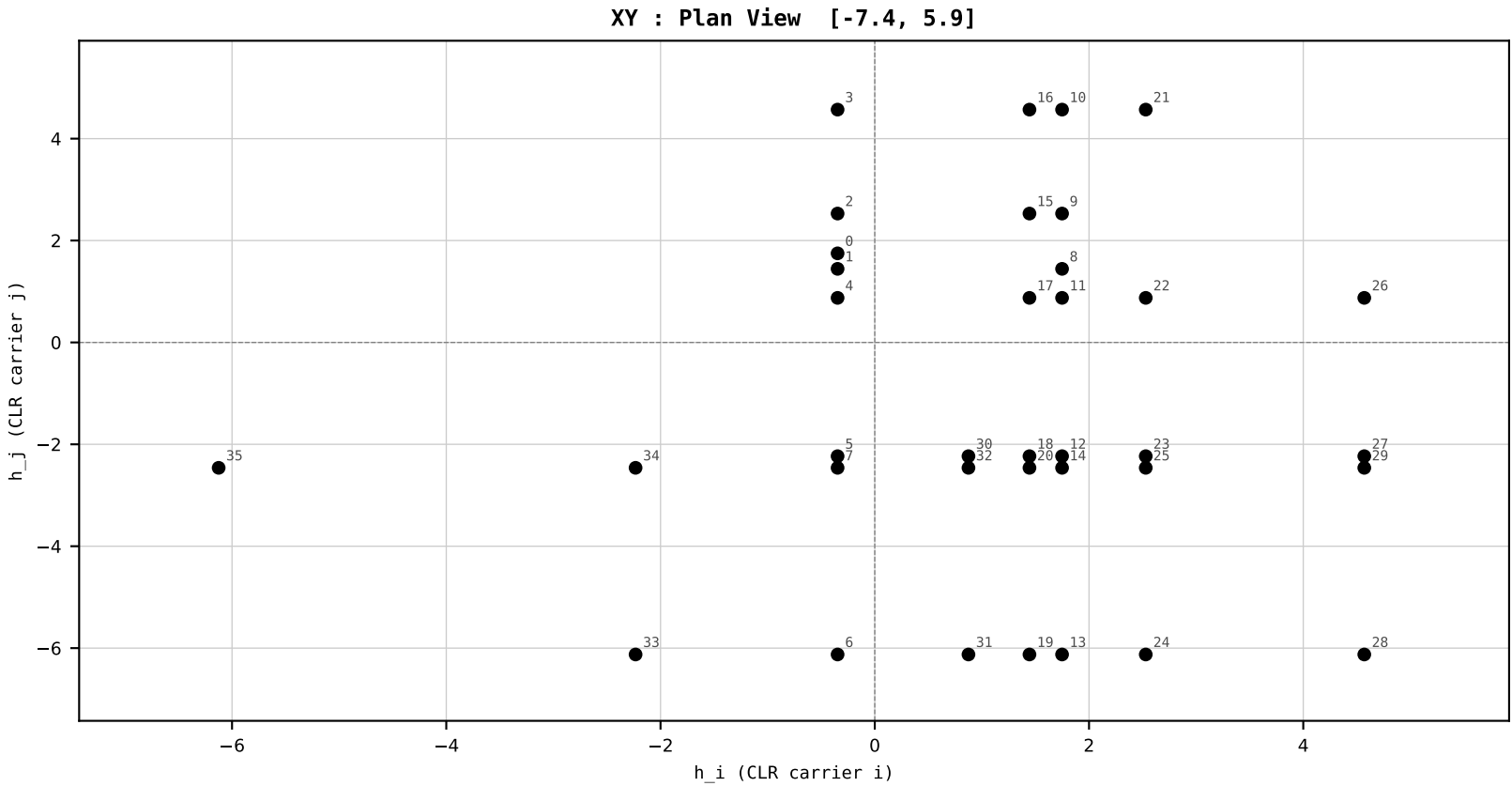
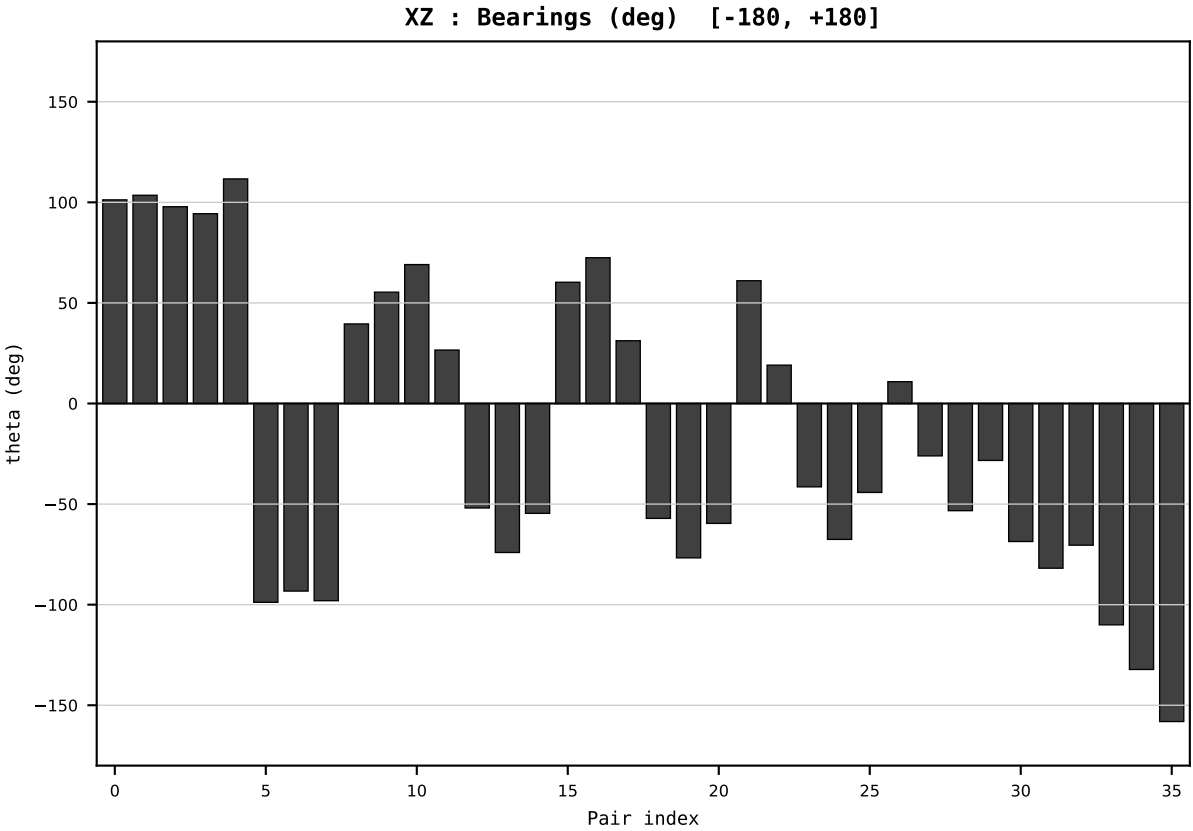
Dataset : ember_FRA_France_generation_TWh.csv
Year : 2003
t : 3 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.636320
Ring = Hs-4
E_metric= 81.8645
kappa_HS= 44107.00
omega = 2.0886 deg
d_A = 0.338625
Helm = Wind
Helm d = +0.291606
DR = 10.694 HLR
DR ratio= 44107.0x

XZ [-180,180]
CLR[-7.4,5.9]
Hs [0, 1]

PAIR INDEX:

0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=4 | Year 2004 | D=9 N=26 pairs=36

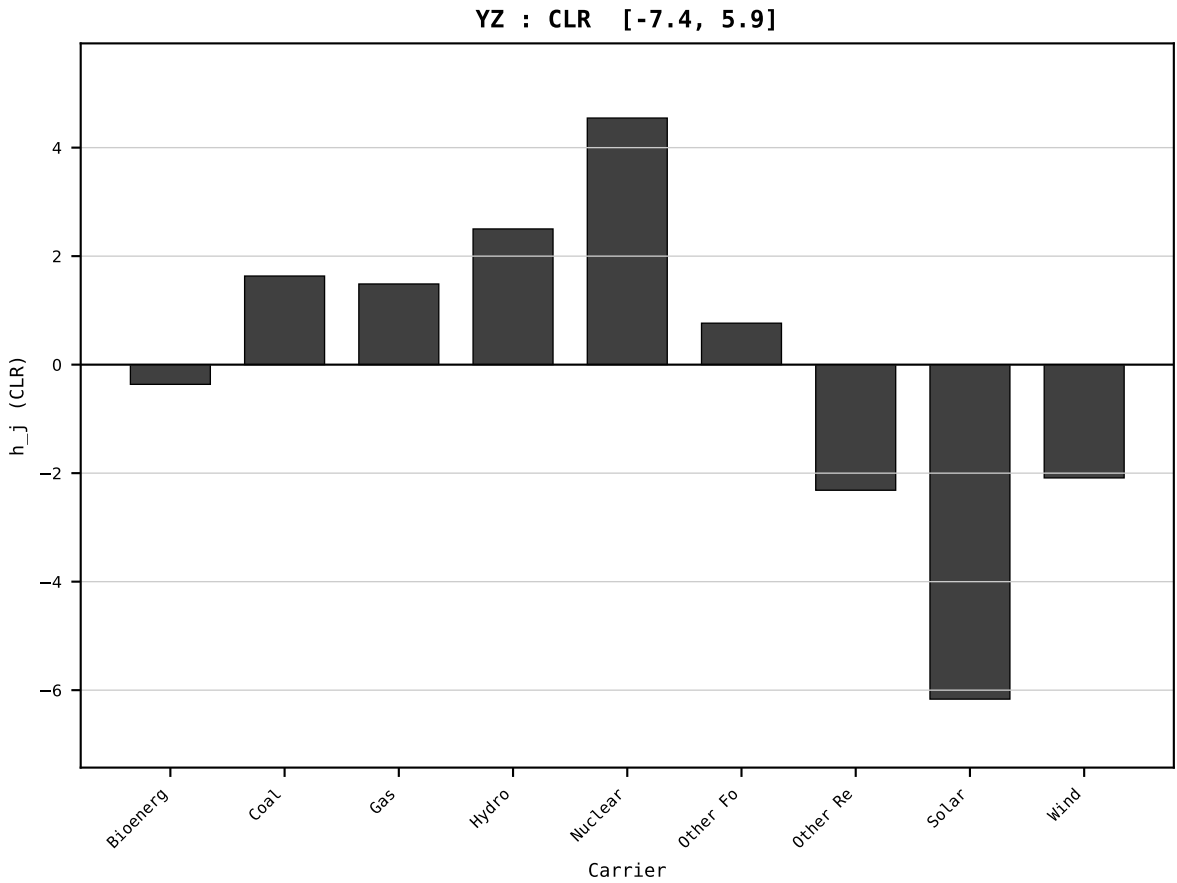
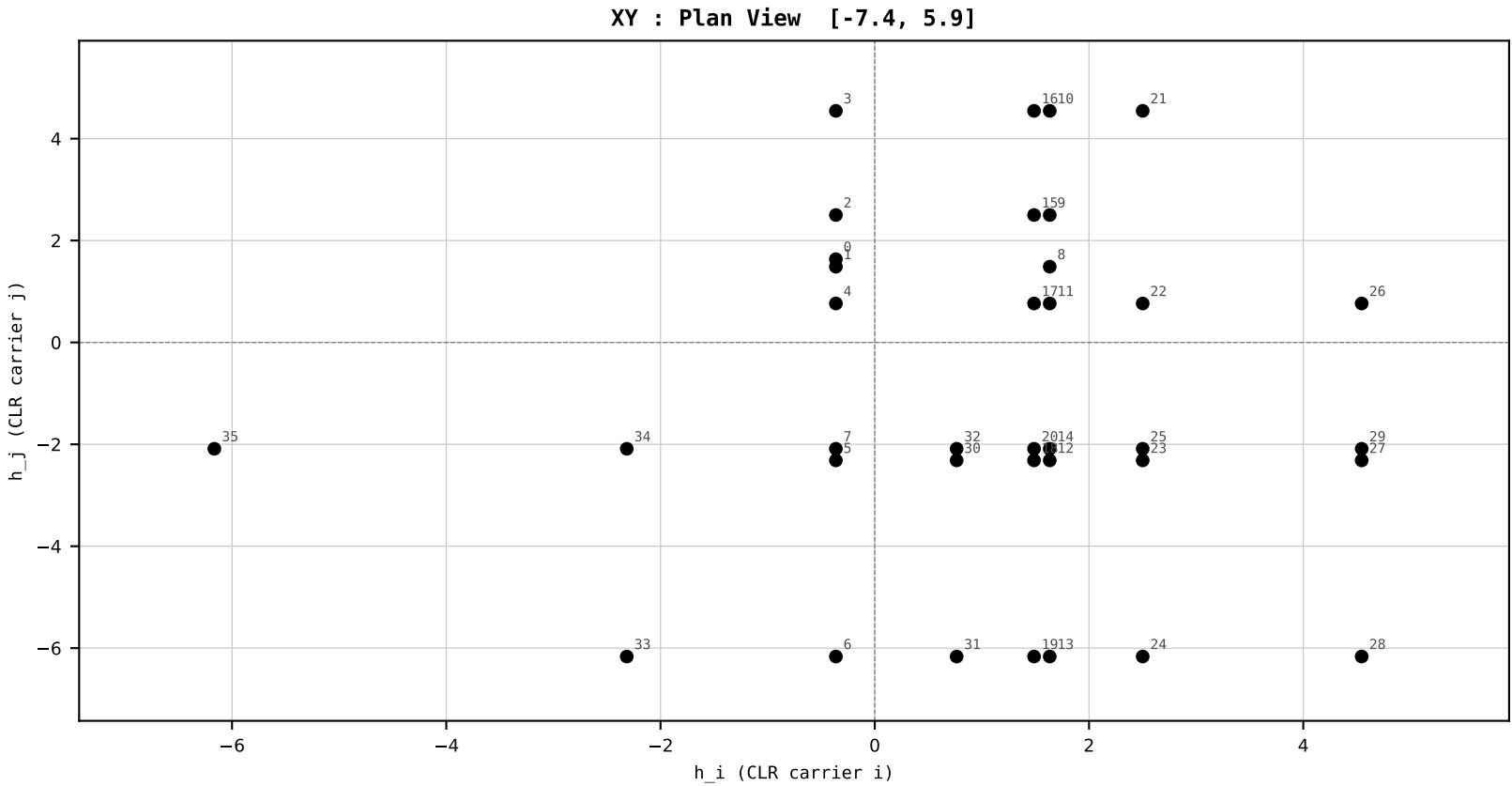
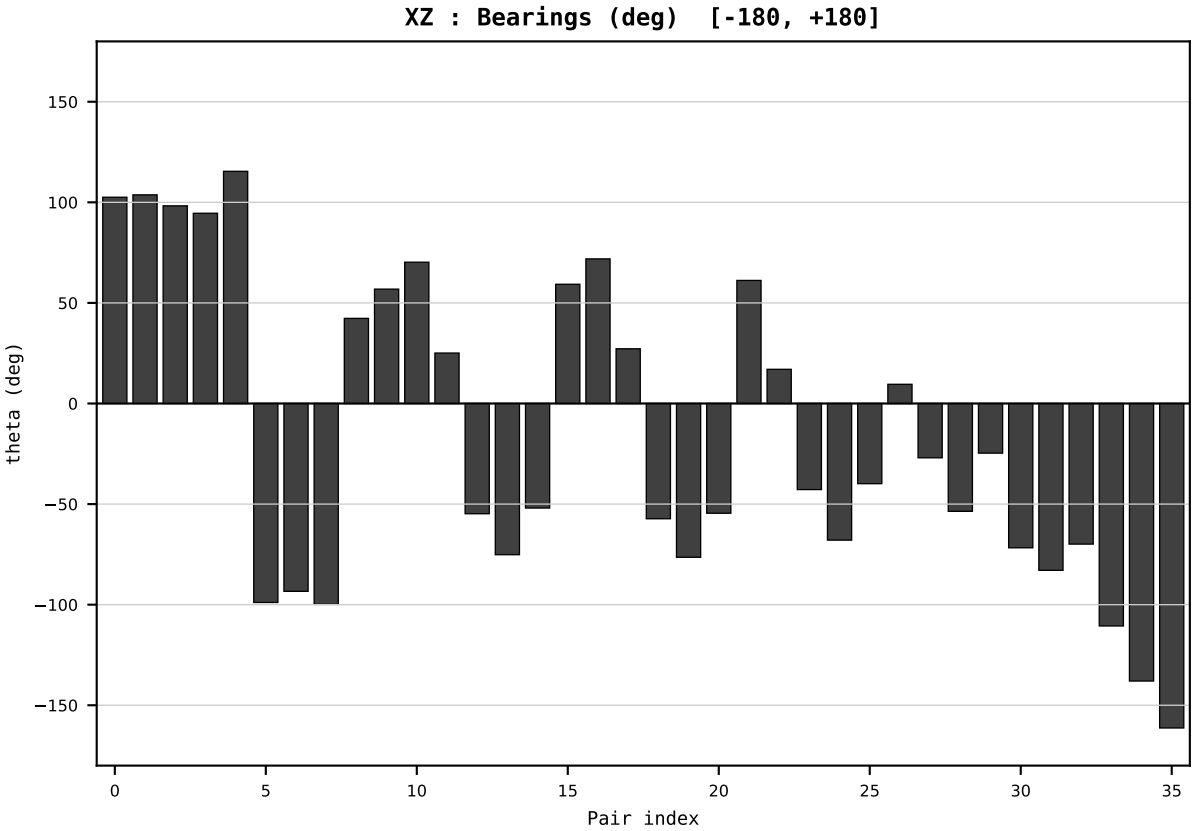
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_FRA_France_generation_TWh.csv
Year : 2004
t : 4 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.639536
Ring = Hs-4
E_metric= 80.2373
kappa_HS= 44824.00
omega = 2.6139 deg
d_A = 0.420503
Helm = Wind
Helm d = +0.373595
DR = 10.710 HLR
DR ratio= 44824.0x

XZ [-180,180]
CLR[-7.4,5.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=5 | Year 2005 | D=9 N=26 pairs=36

HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

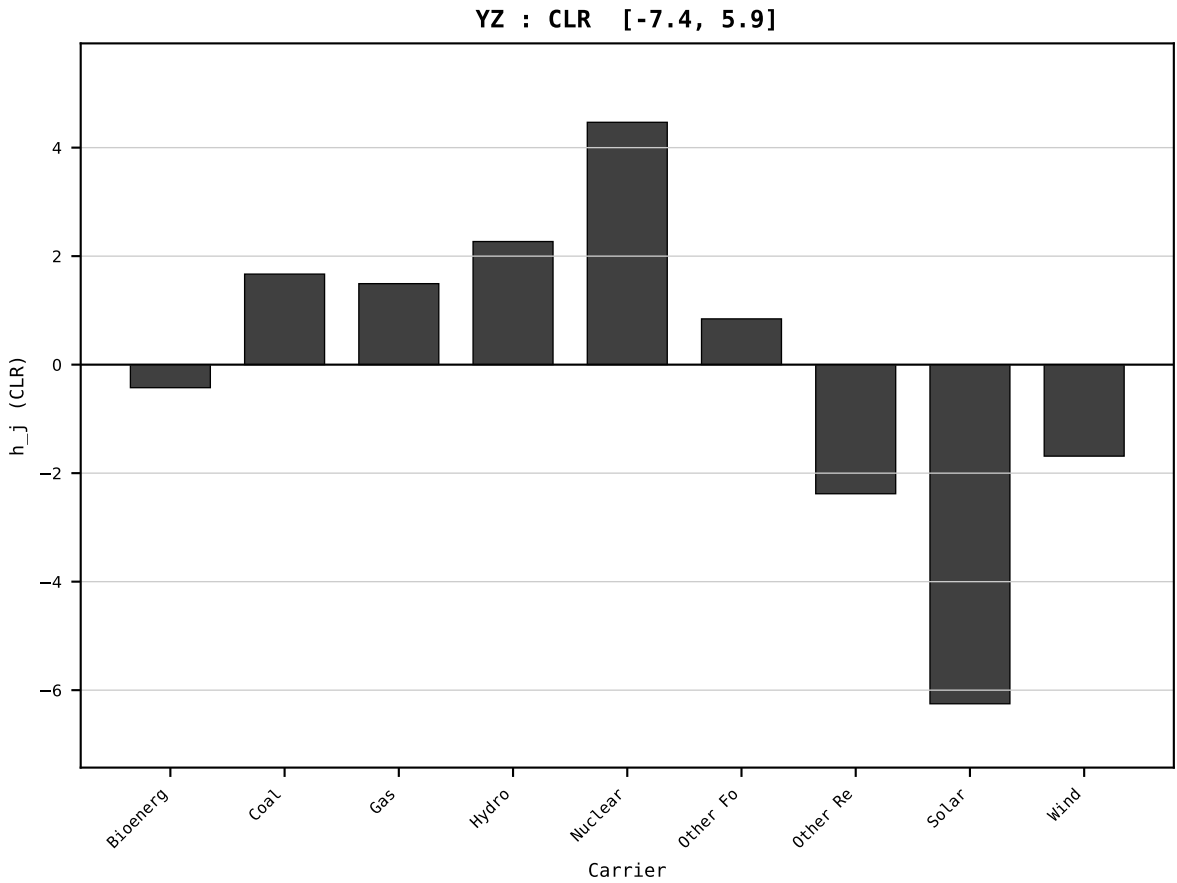
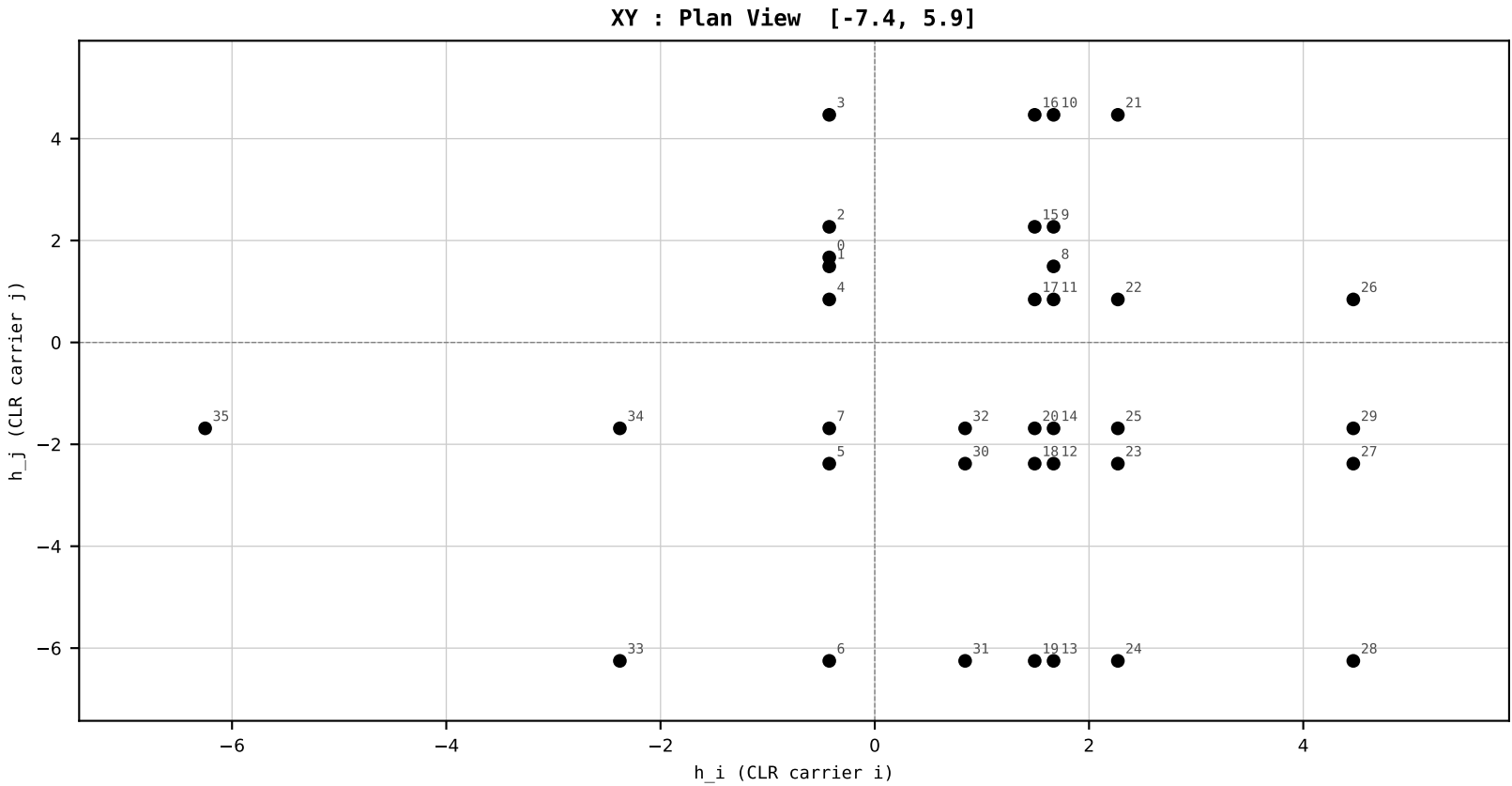
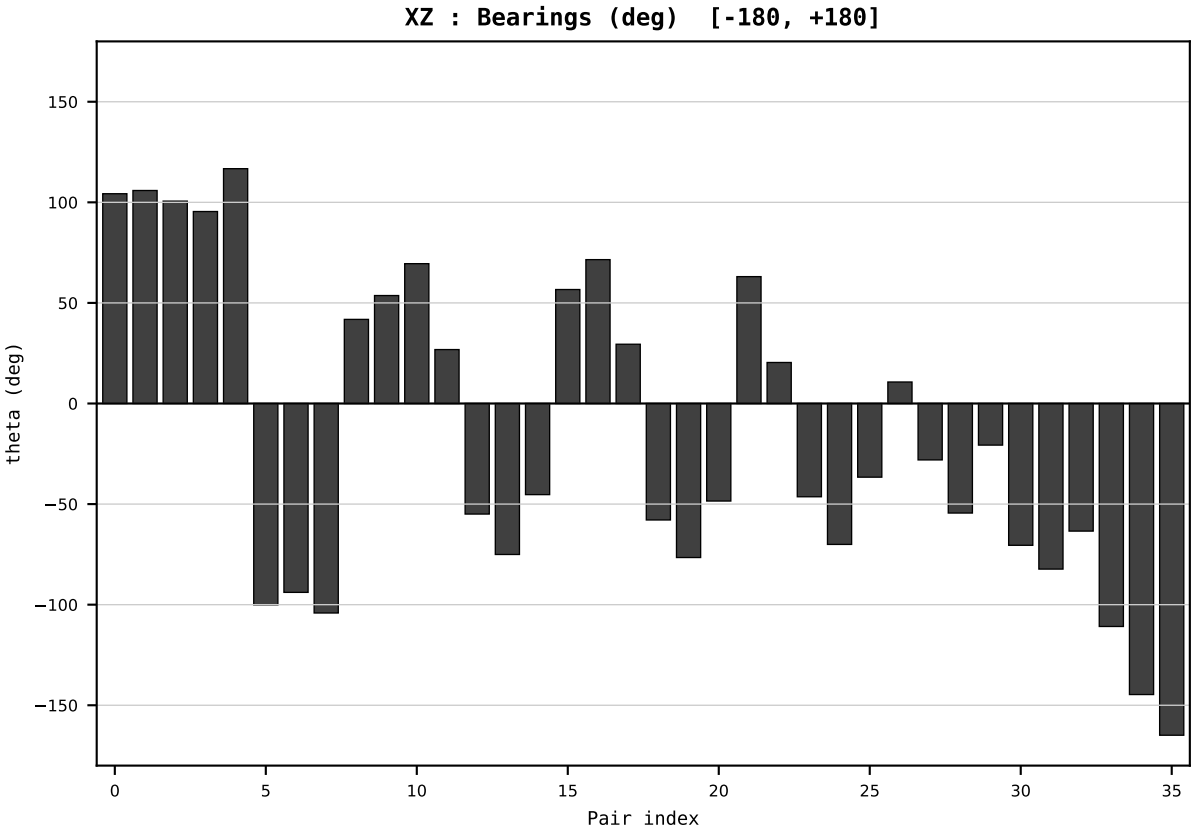
Dataset : ember_FRA_France_generation_TWh.csv
Year : 2005
t : 5 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.634489
Ring = Hs-4
E_metric= 78.5884
kappa_HS= 45153.00
omega = 3.1185 deg
d_A = 0.493709
Helm = Wind
Helm d = +0.401223
DR = 10.718 HLR
DR ratio= 45153.0x

XZ [-180,180]
CLR[-7.4,5.9]
Hs [0, 1]

PAIR INDEX:

0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=6 | Year 2006 | D=9 N=26 pairs=36

HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

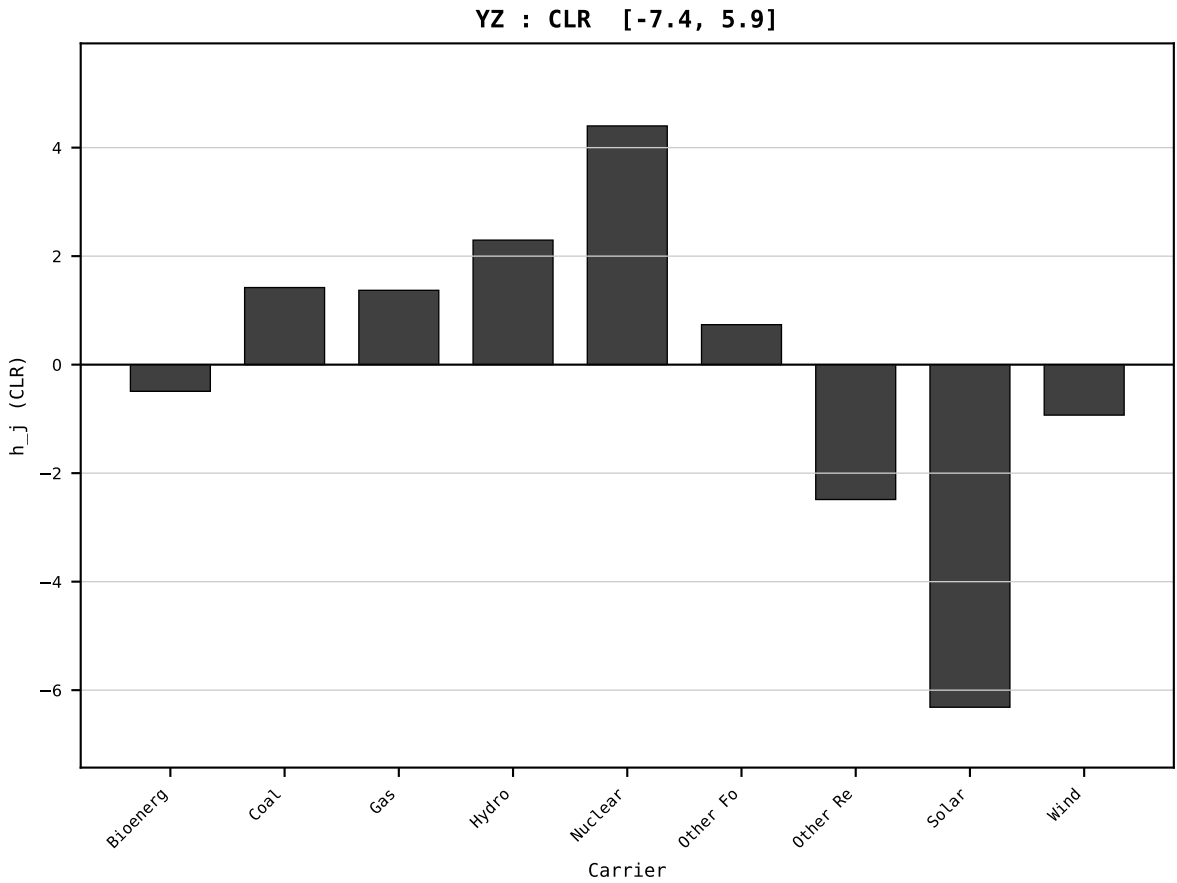
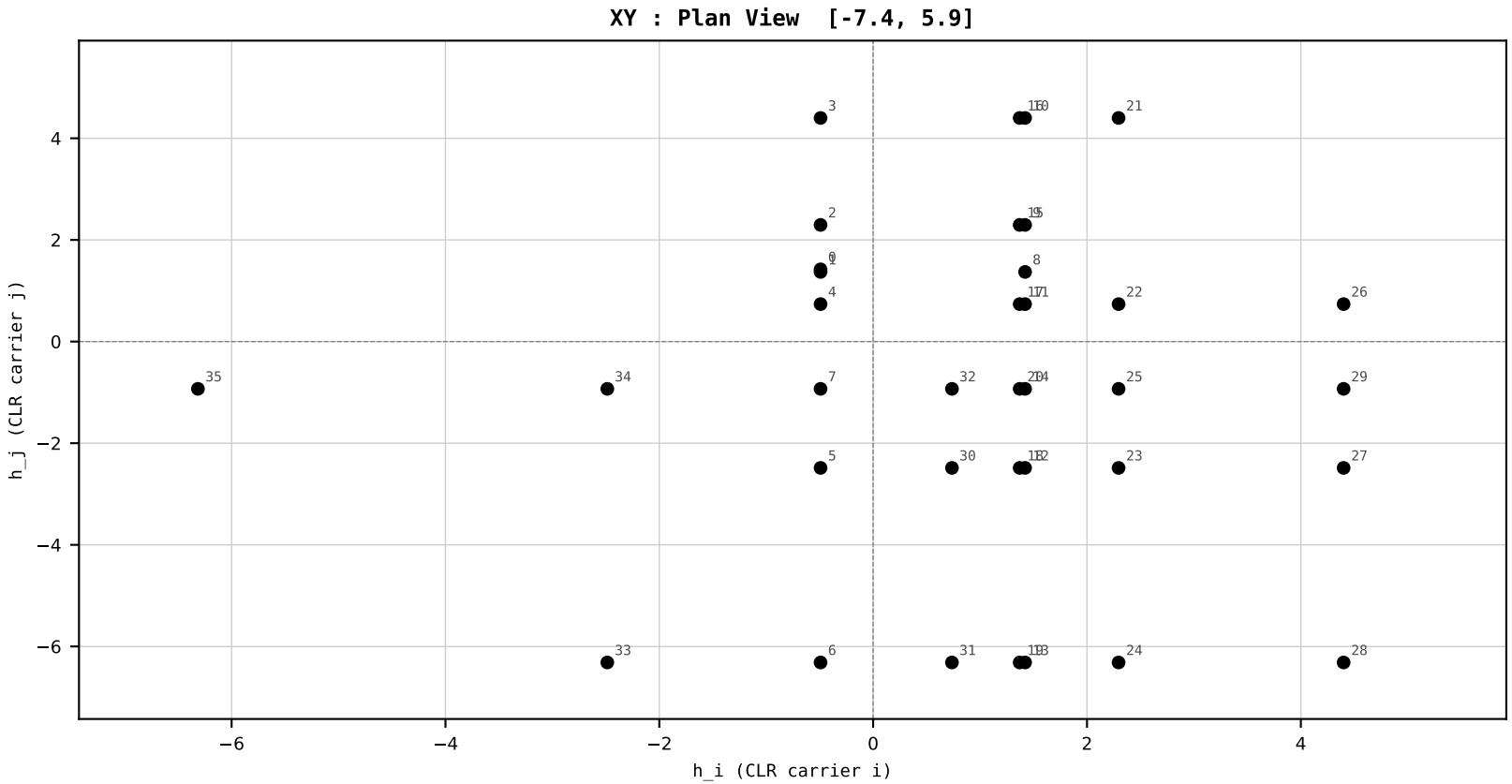
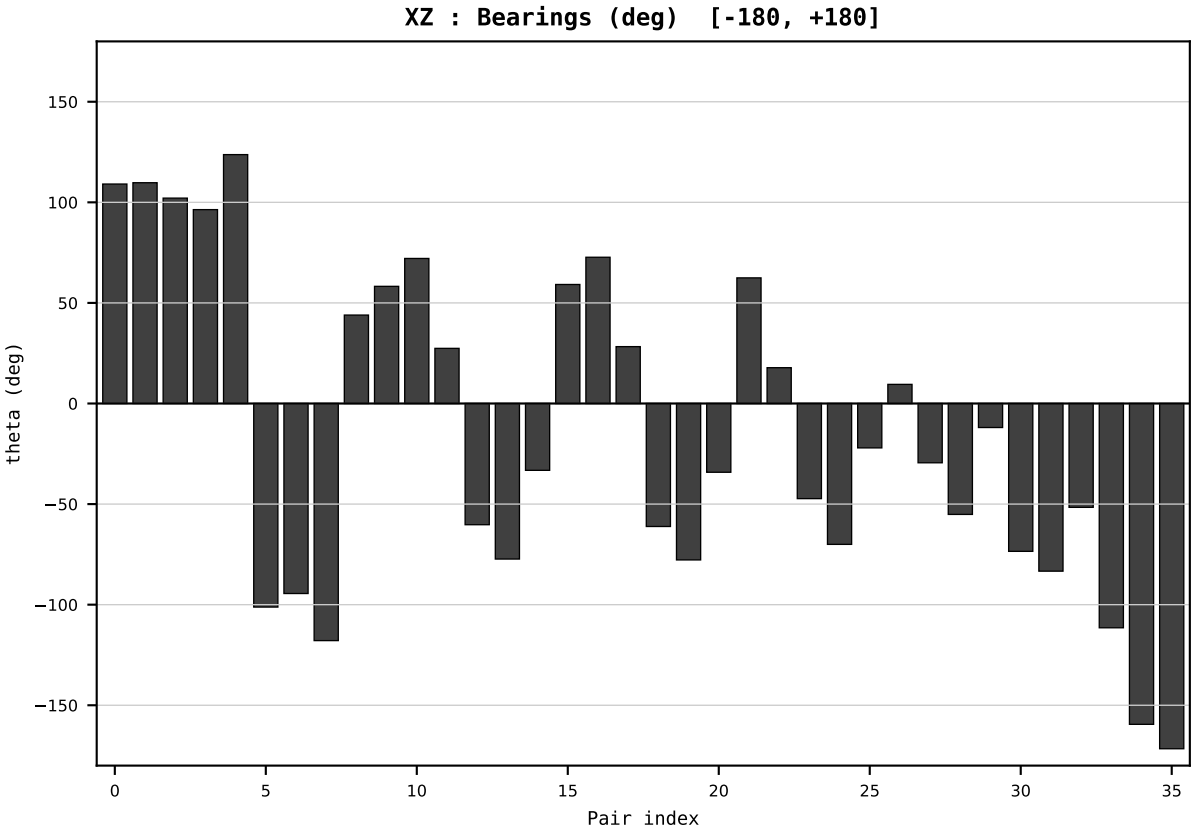
Dataset : ember_FRA_France_generation_TWh.csv
Year : 2006
t : 6 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.635362
Ring = Hs-4
E_metric= 76.2355
kappa_HS= 45019.00
omega = 5.3196 deg
d_A = 0.827413
Helm = Wind
Helm d = +0.756002
DR = 10.715 HLR
DR ratio= 45019.0x

XZ [-180,180]
CLR[-7.4,5.9]
Hs [0, 1]

PAIR INDEX:

0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=7 | Year 2007 | D=9 N=26 pairs=36

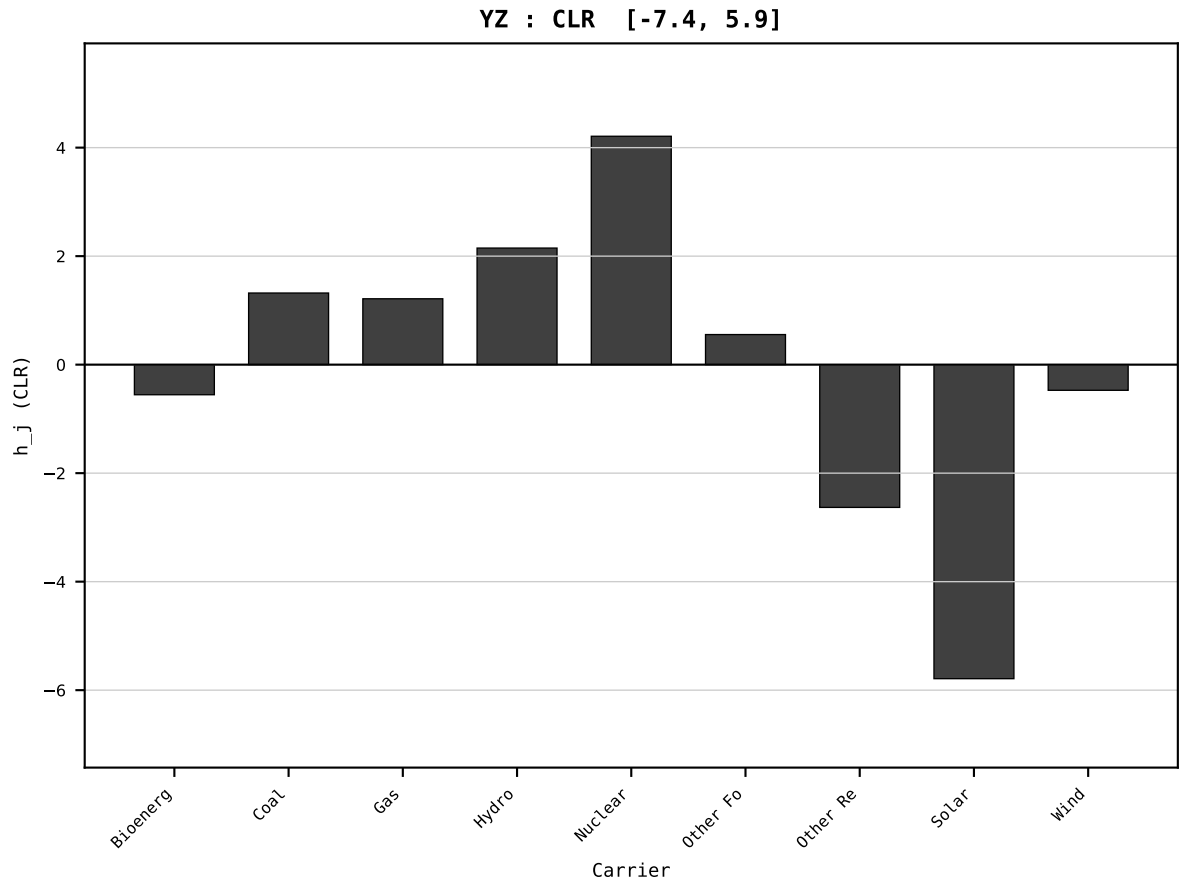
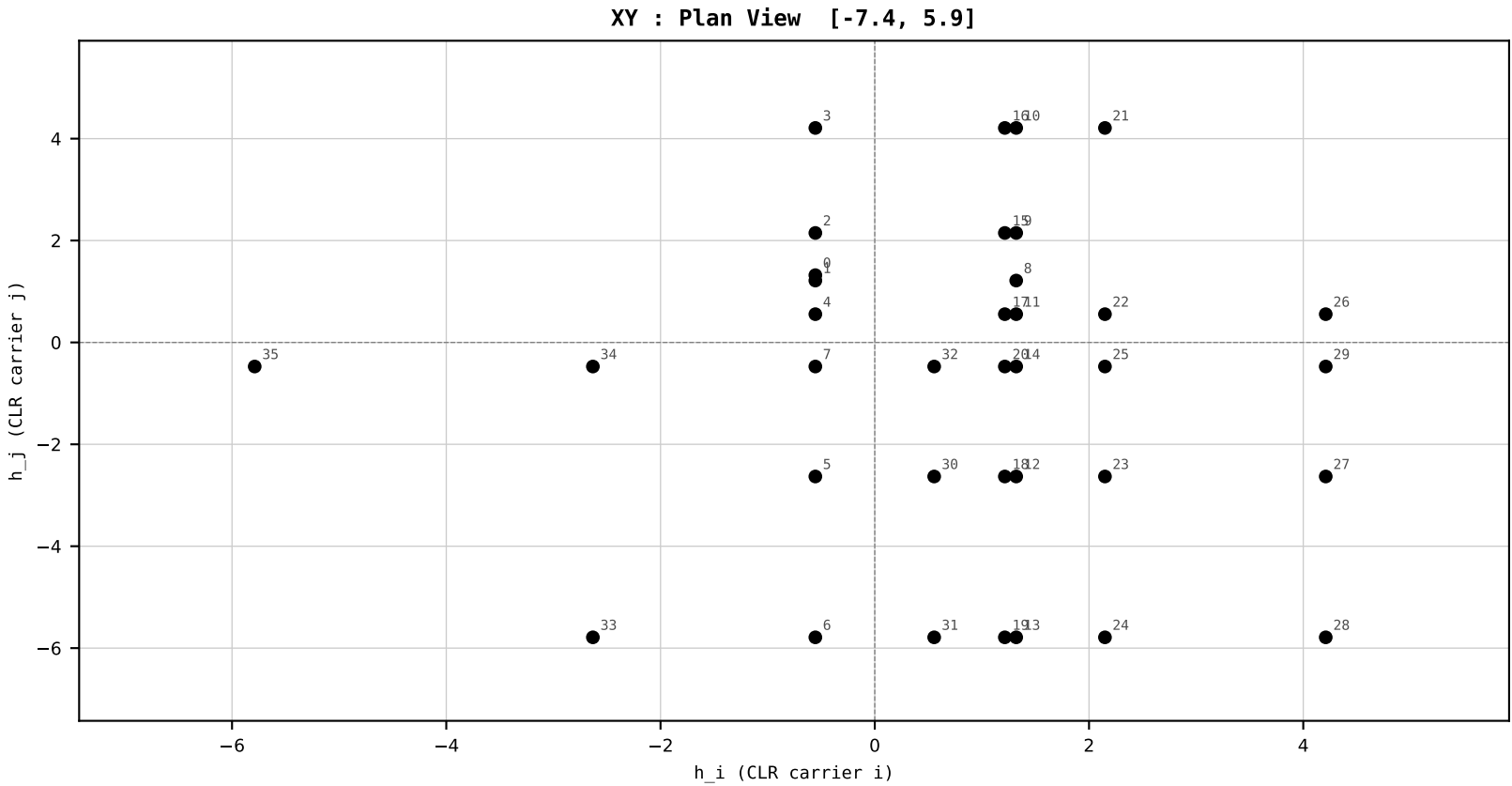
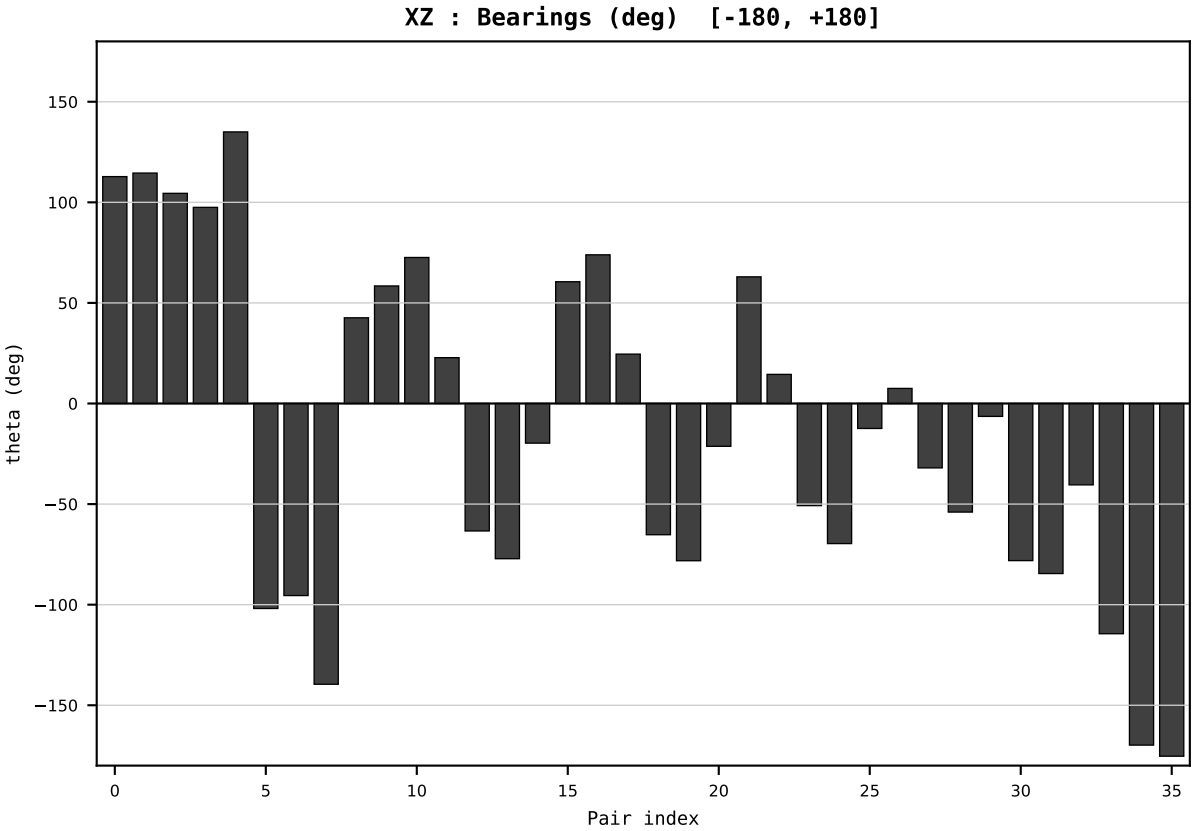
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_FRA_France_generation_TWh.csv
Year : 2007
t : 7 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.618062
Ring = Hs-4
E_metric= 66.8255
kappa_HS= 21986.50
omega = 3.8806 deg
d_A = 0.798187
Helm = Solar
Helm d = +0.526469
DR = 9.998 HLR
DR ratio= 21986.5x

XZ [-180,180]
CLR[-7.4,5.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=8 | Year 2008 | D=9 N=26 pairs=36

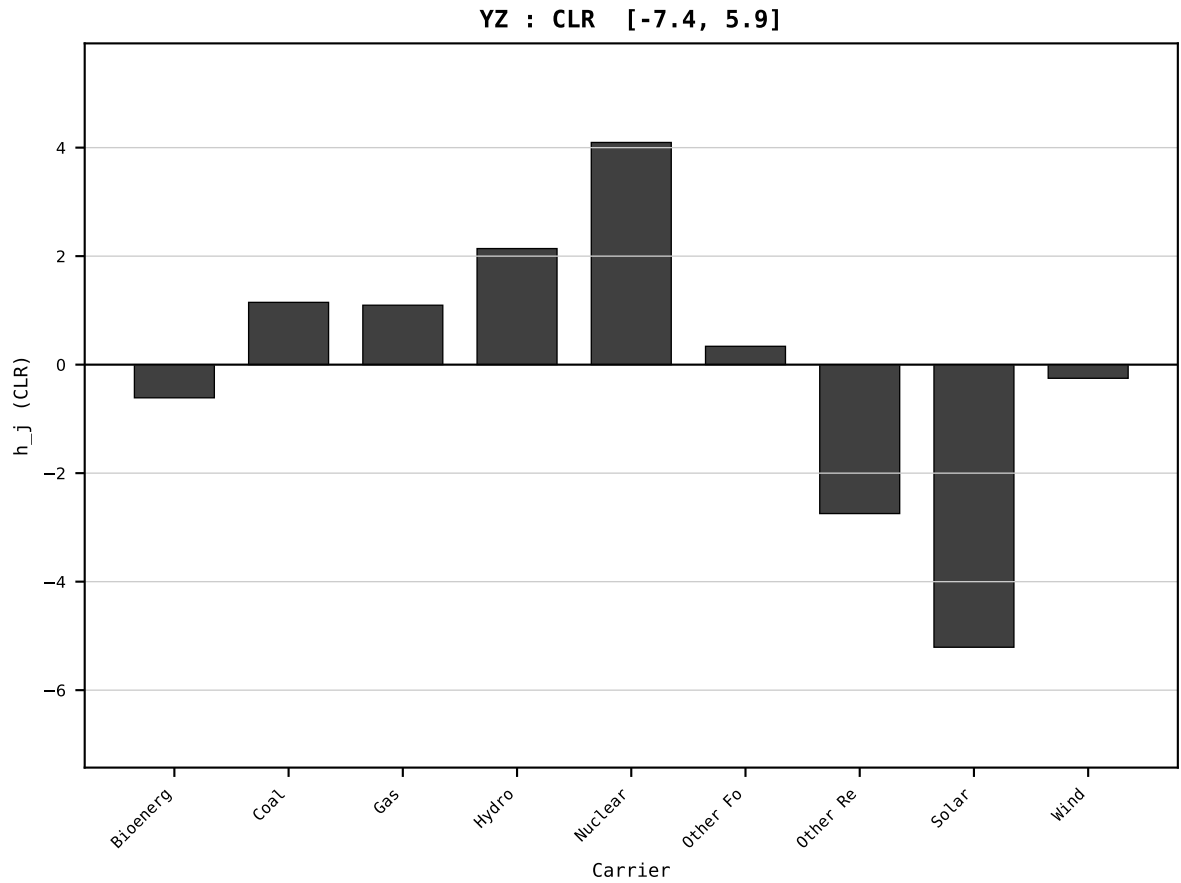
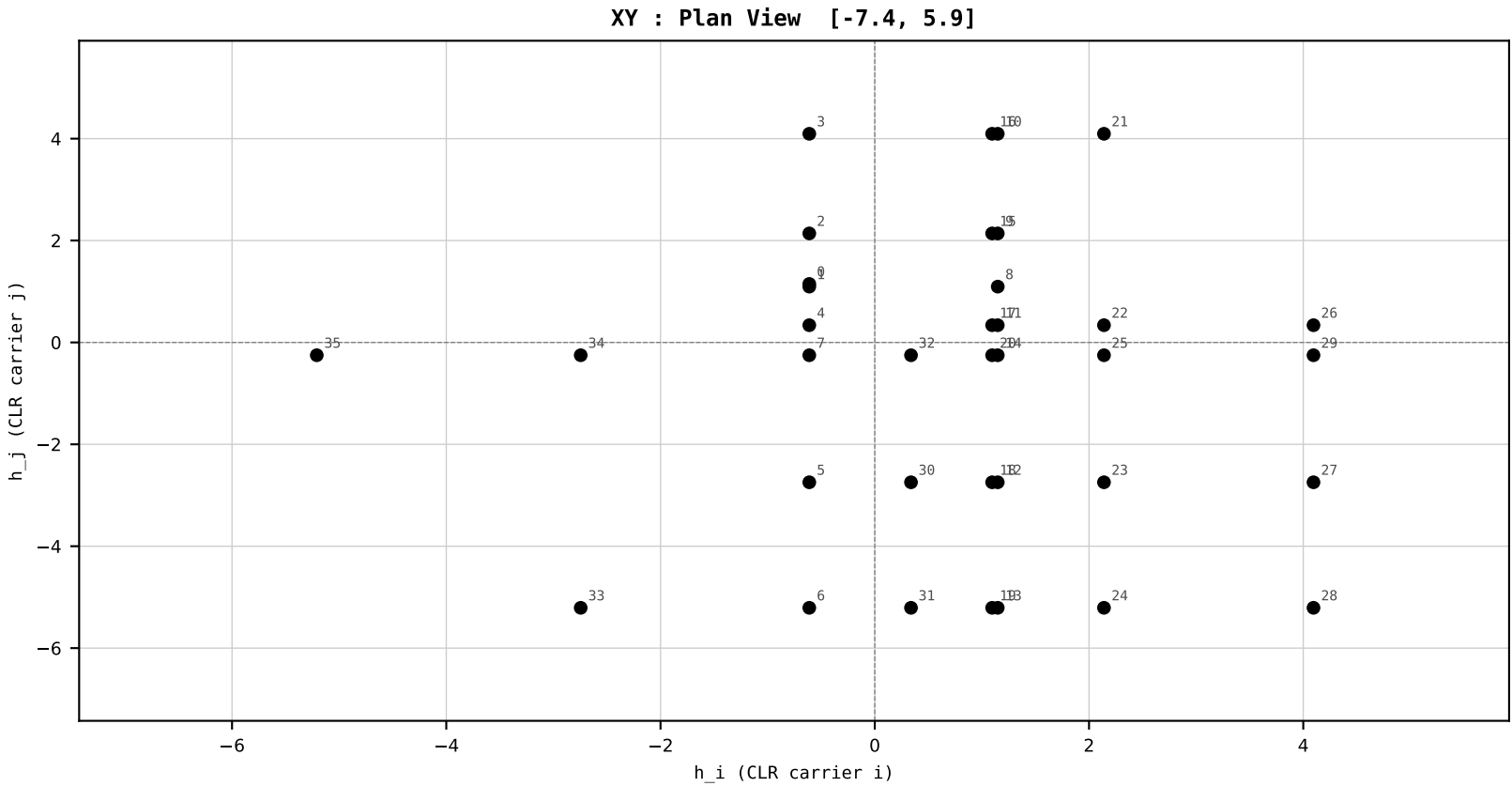
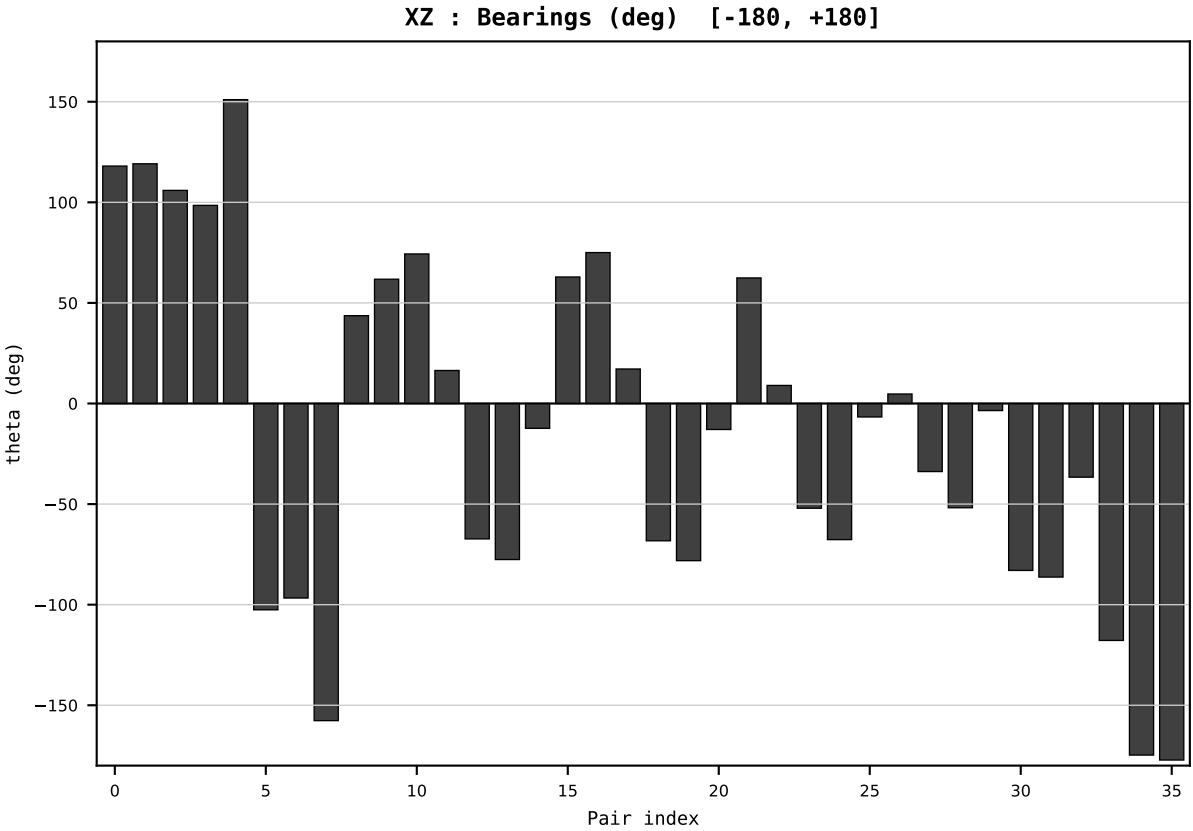
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_FRA_France_generation_TWh.csv
Year : 2008
t : 8 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.610524
Ring = Hs-4
E_metric= 59.0888
kappa_HS= 10986.25
omega = 3.7337 deg
d_A = 0.710392
Helm = Solar
Helm d = +0.579497
DR = 9.304 HLR
DR ratio= 10986.3x

XZ [-180,180]
CLR[-7.4,5.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



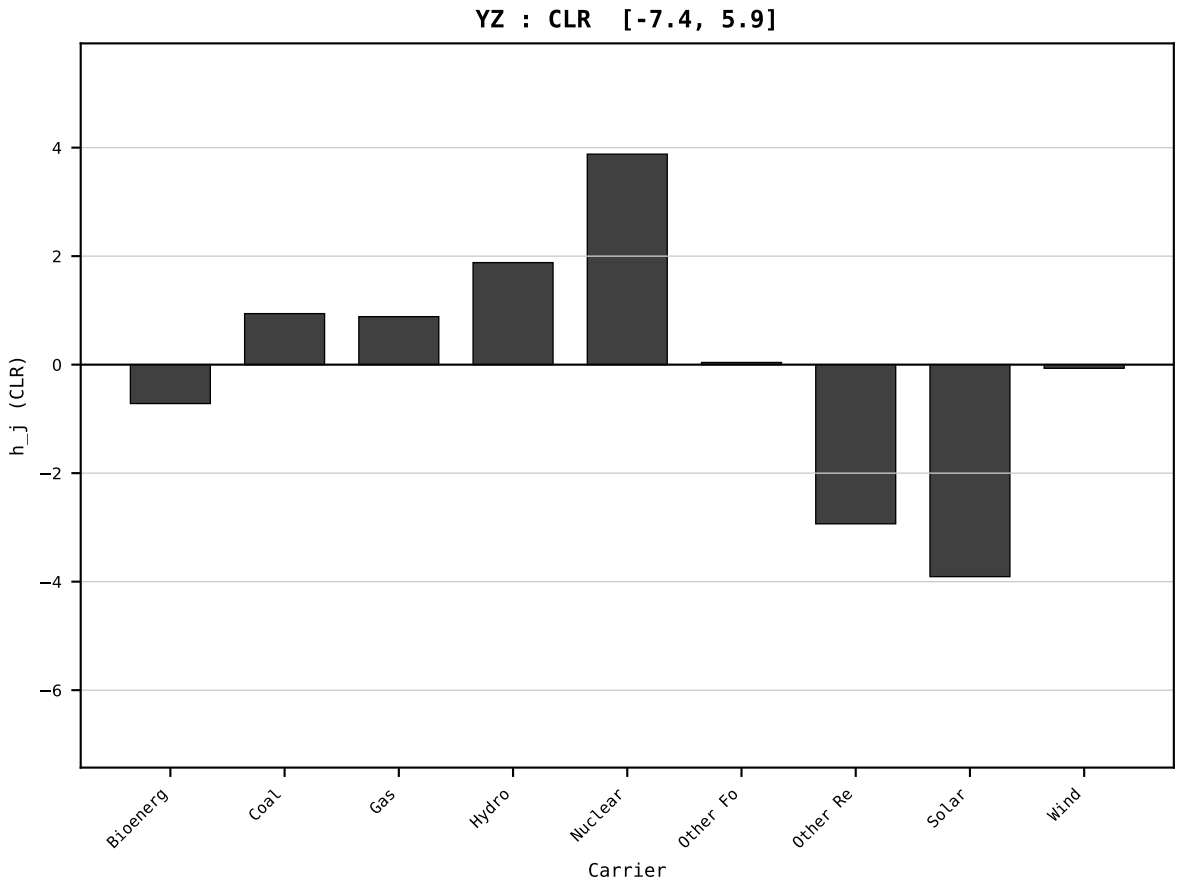
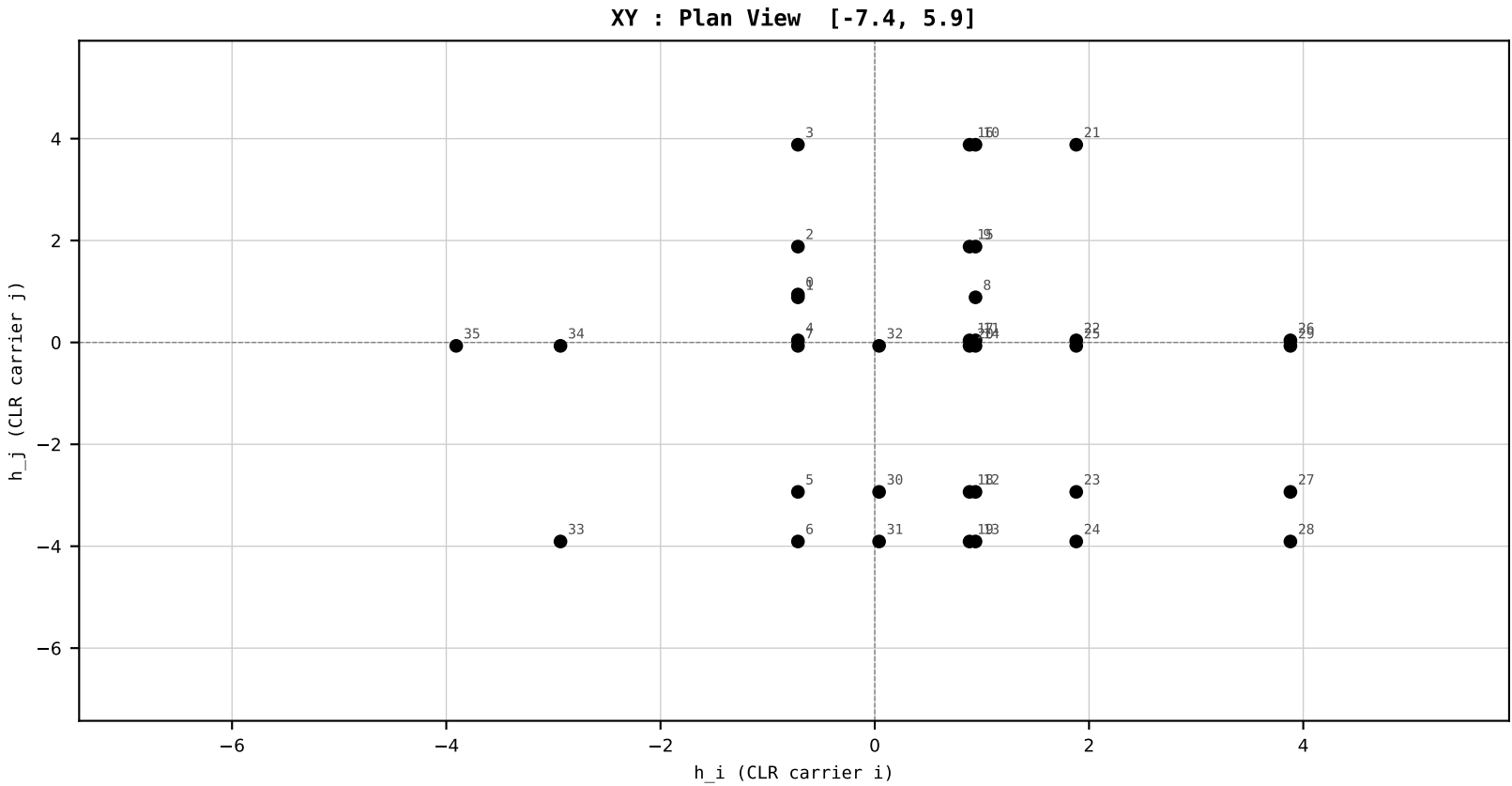
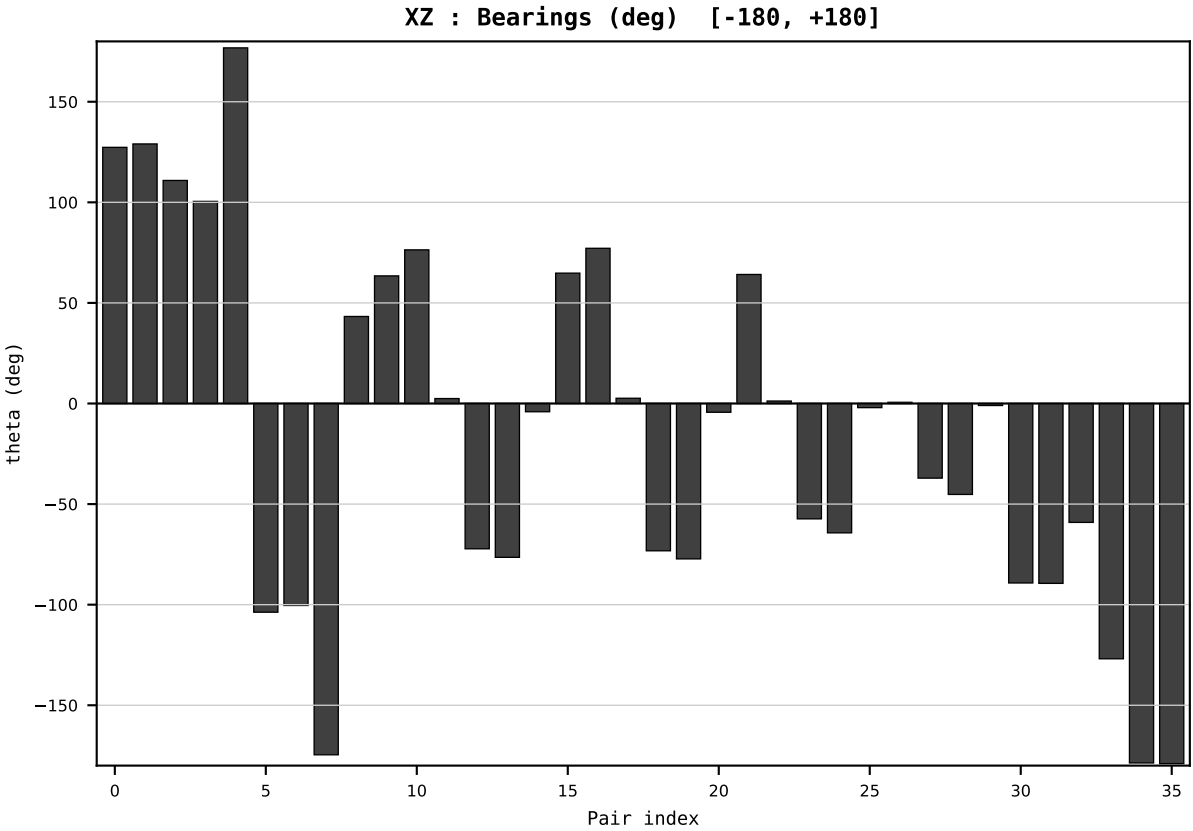
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_FRA_France_generation_TWh.csv
Year : 2009
t : 9 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.604621
Ring = Hs-4
E_metric= 44.6563
kappa_HS= 2410.24
omega = 8.2198 deg
d_A = 1.436738
Helm = Solar
Helm d = +1.301456
DR = 7.787 HLR
DR ratio= 2410.2x

XZ [-180,180]
CLR[-7.4,5.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=10 | Year 2010 | D=9 N=26 pairs=36

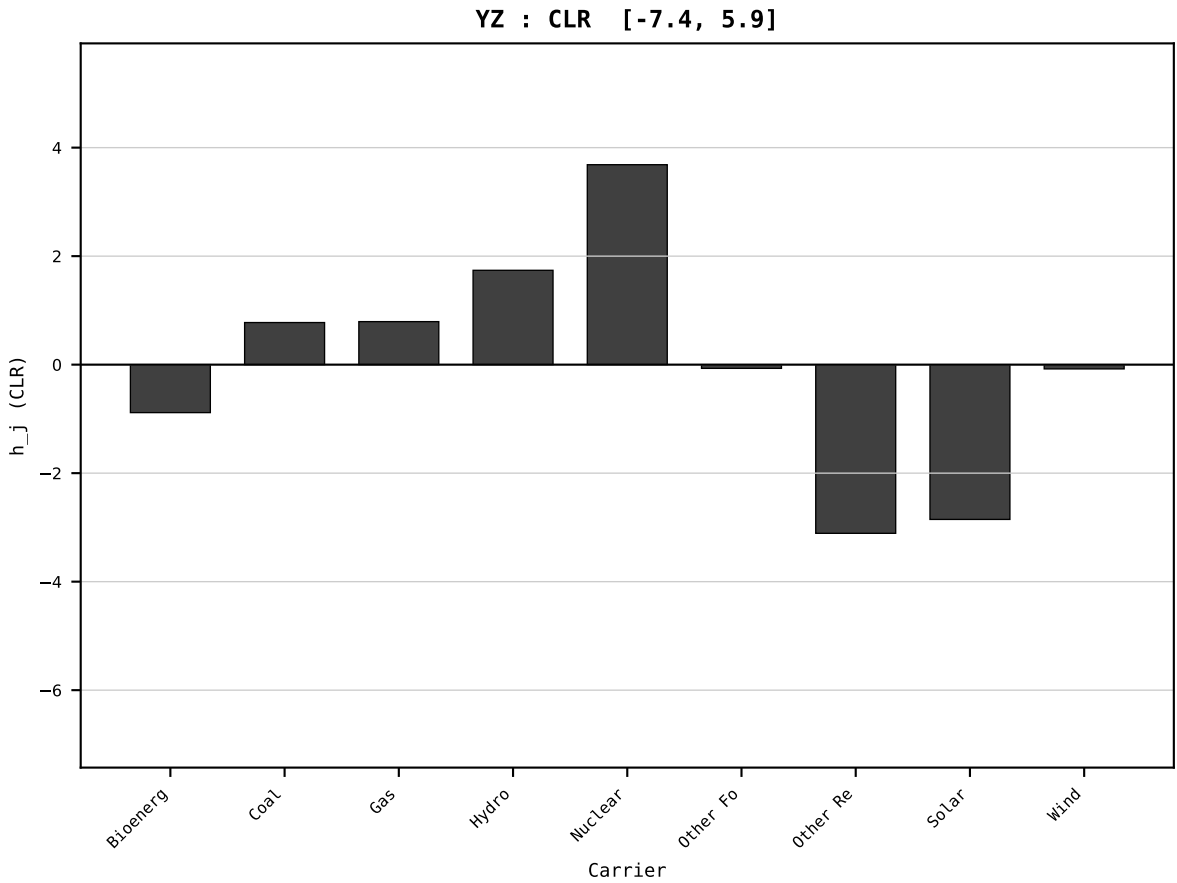
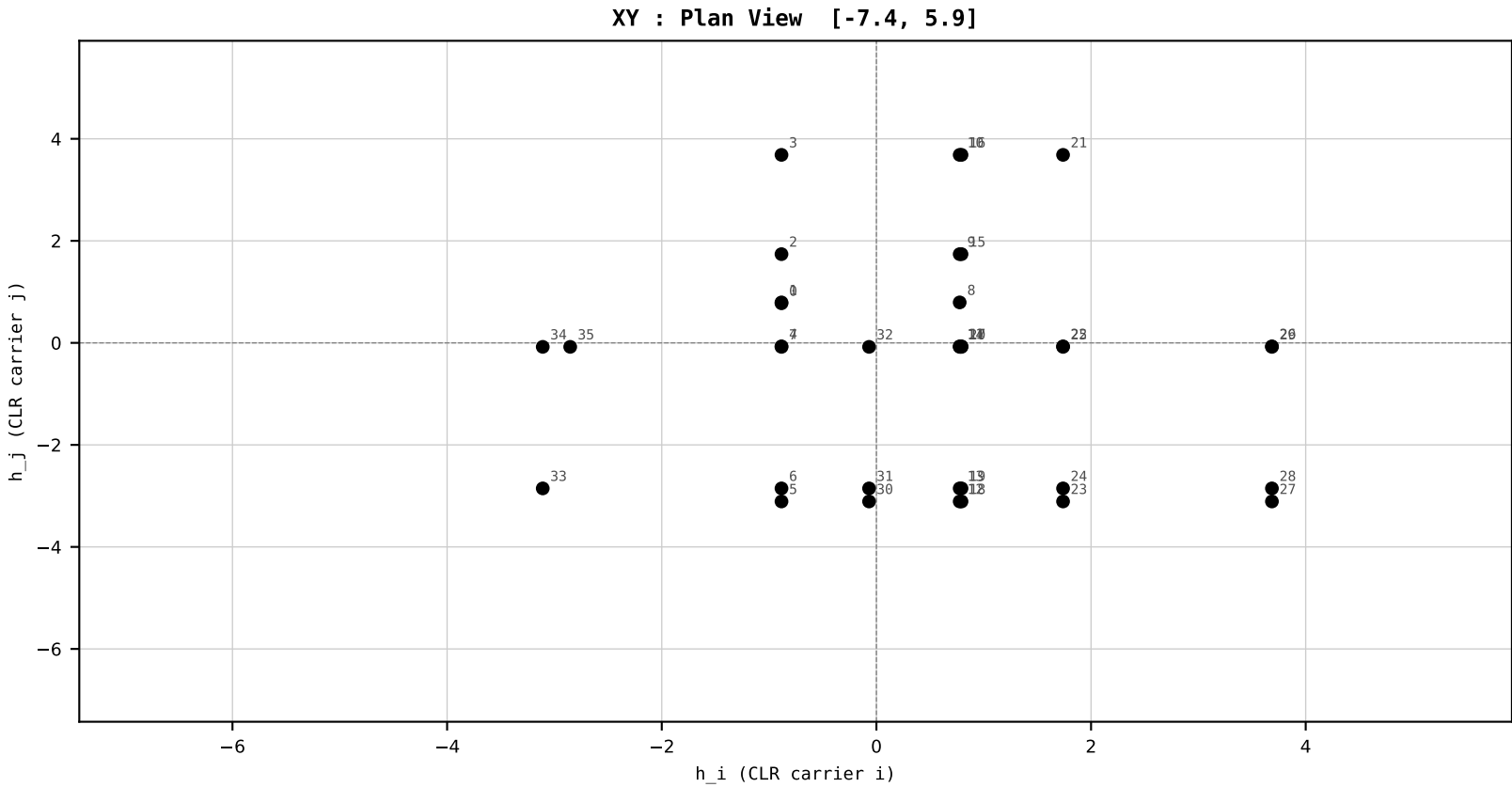
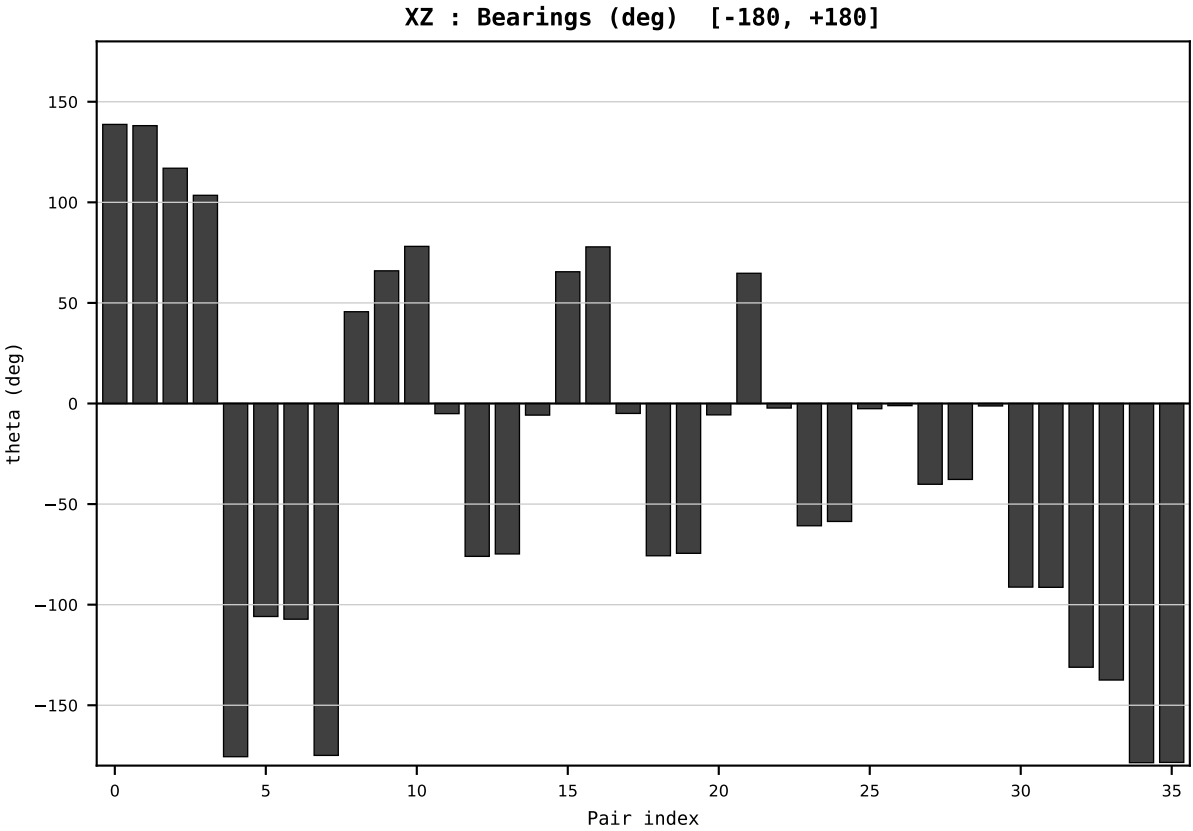
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_FRA_France_generation_TWh.csv
Year : 2010
t : 10 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.586166
Ring = Hs-4
E_metric= 36.4358
kappa_HS= 892.75
omega = 8.3636 deg
d_A = 1.129480
Helm = Solar
Helm d = +1.054432
DR = 6.794 HLR
DR ratio= 892.7x

XZ [-180,180]
CLR[-7.4,5.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=11 | Year 2011 | D=9 N=26 pairs=36

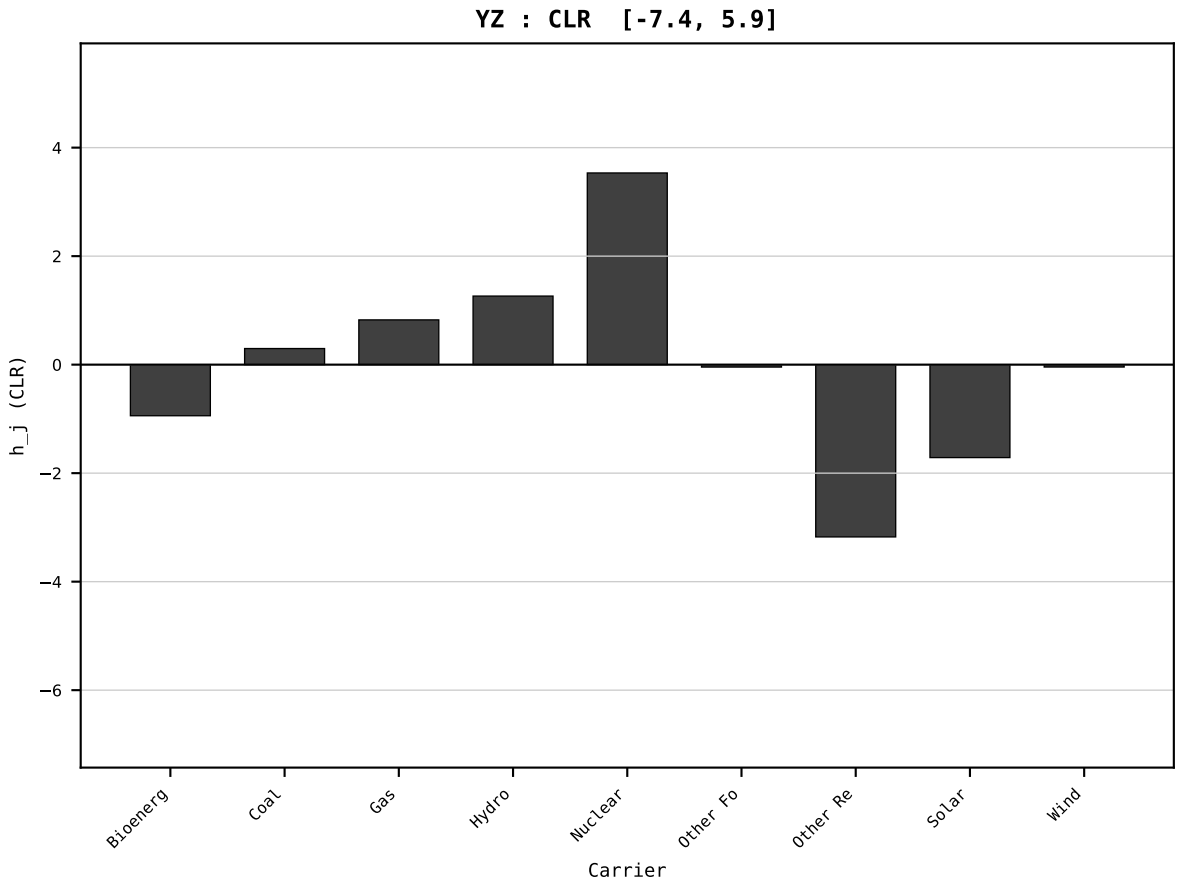
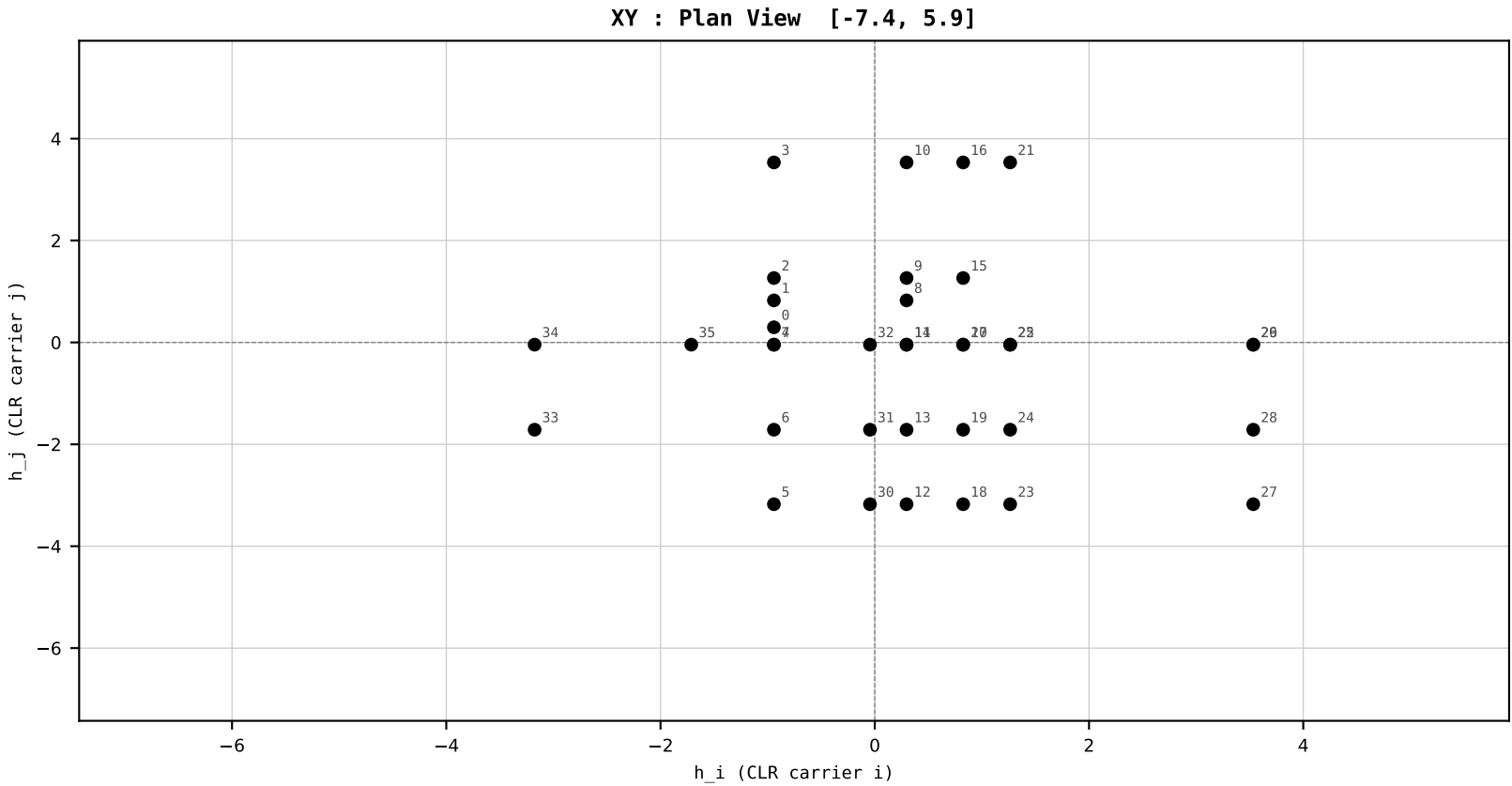
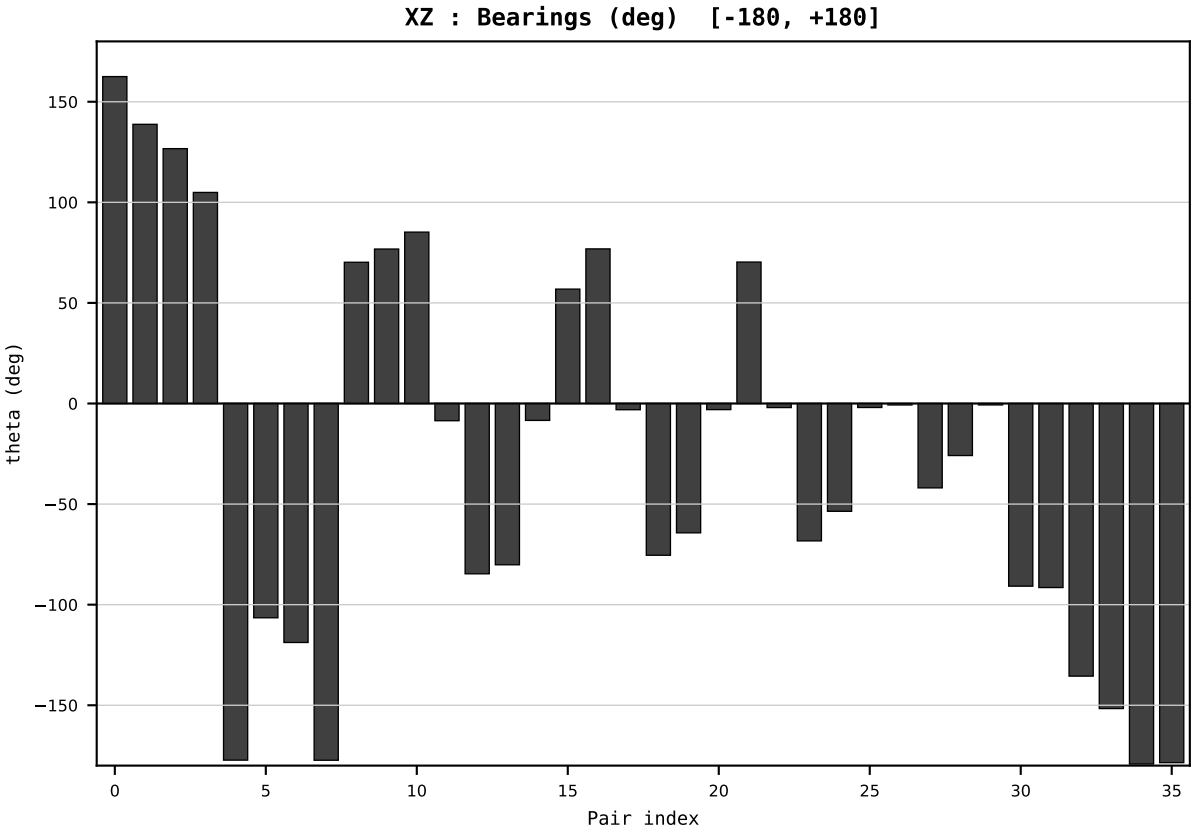
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_FRA_France_generation_TWh.csv
Year : 2011
t : 11 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.592444
Ring = Hs-4
E_metric= 28.7571
kappa_HS= 819.24
omega = 11.6539 deg
d_A = 1.337292
Helm = Solar
Helm d = +1.139834
DR = 6.708 HLR
DR ratio= 819.2x

XZ [-180,180]
CLR[-7.4,5.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



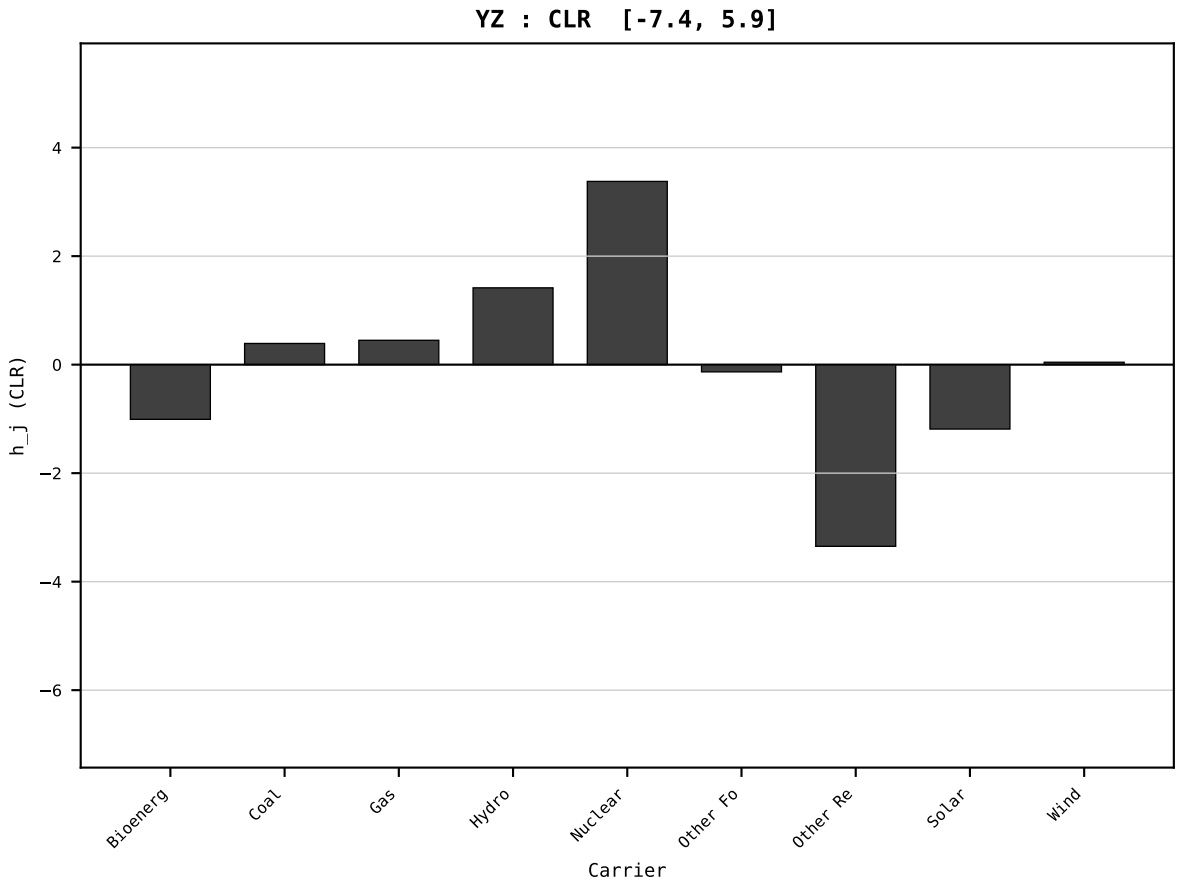
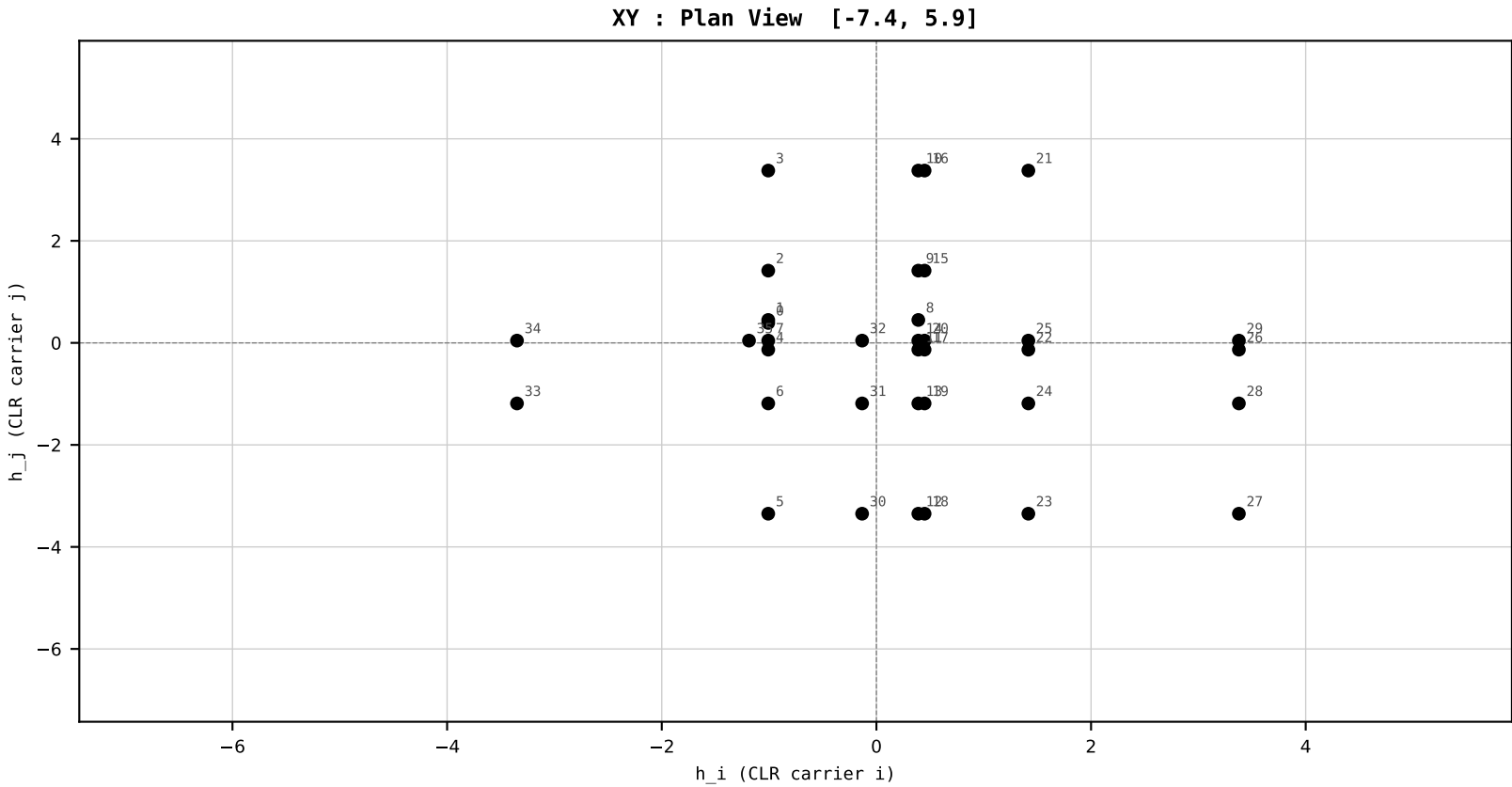
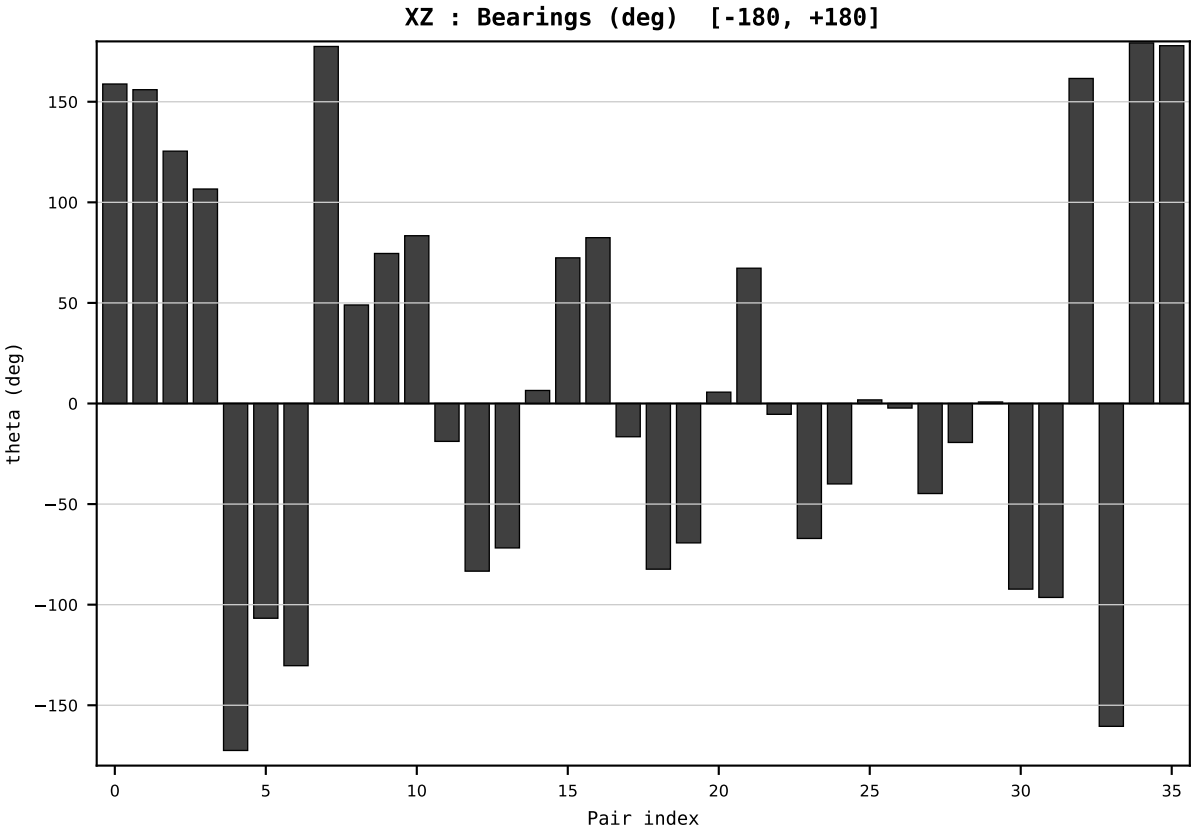
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_FRA_France_generation_TWh.csv
Year : 2012
t : 12 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.555897
Ring = Hs-4
E_metric= 27.4266
kappa_HS= 834.14
omega = 7.7165 deg
d_A = 0.724138
Helm = Solar
Helm d = +0.526031
DR = 6.726 HLR
DR ratio= 834.1x

XZ [-180,180]
CLR[-7.4,5.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



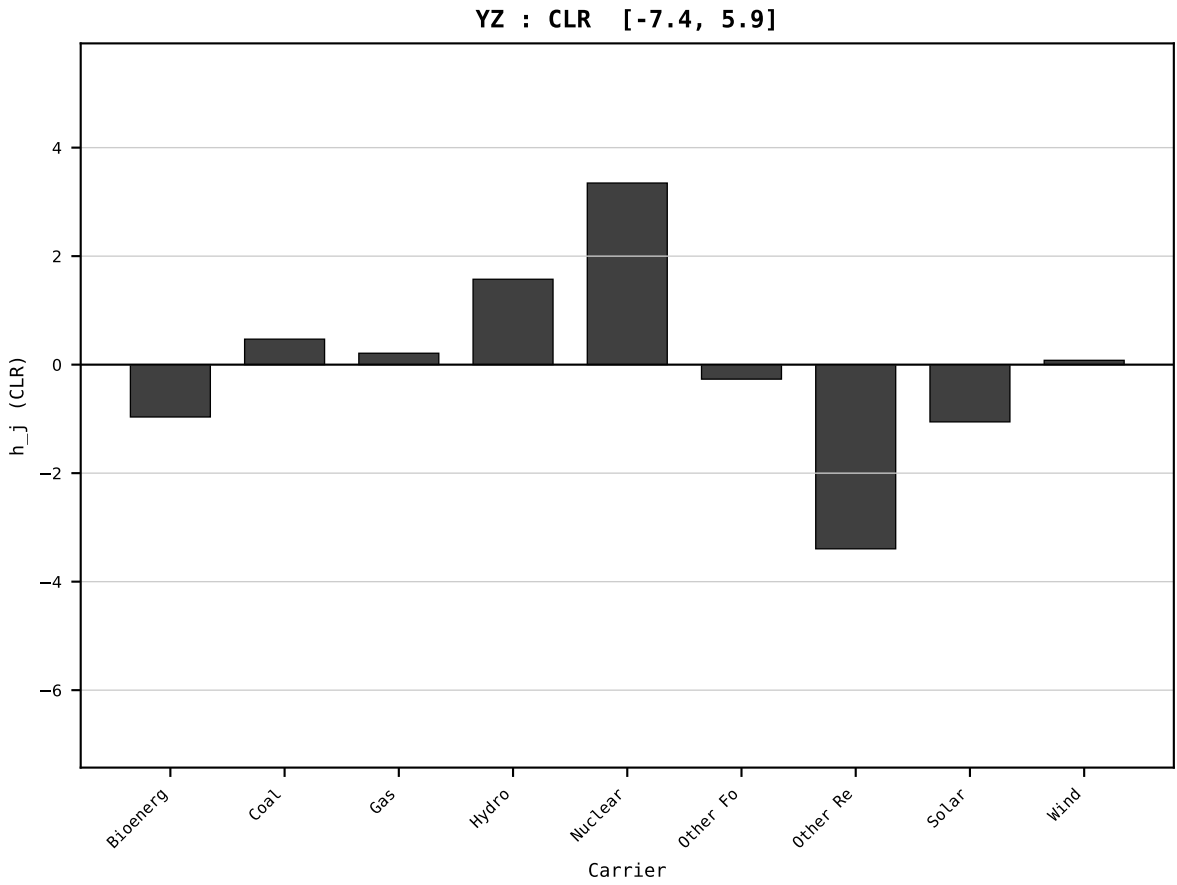
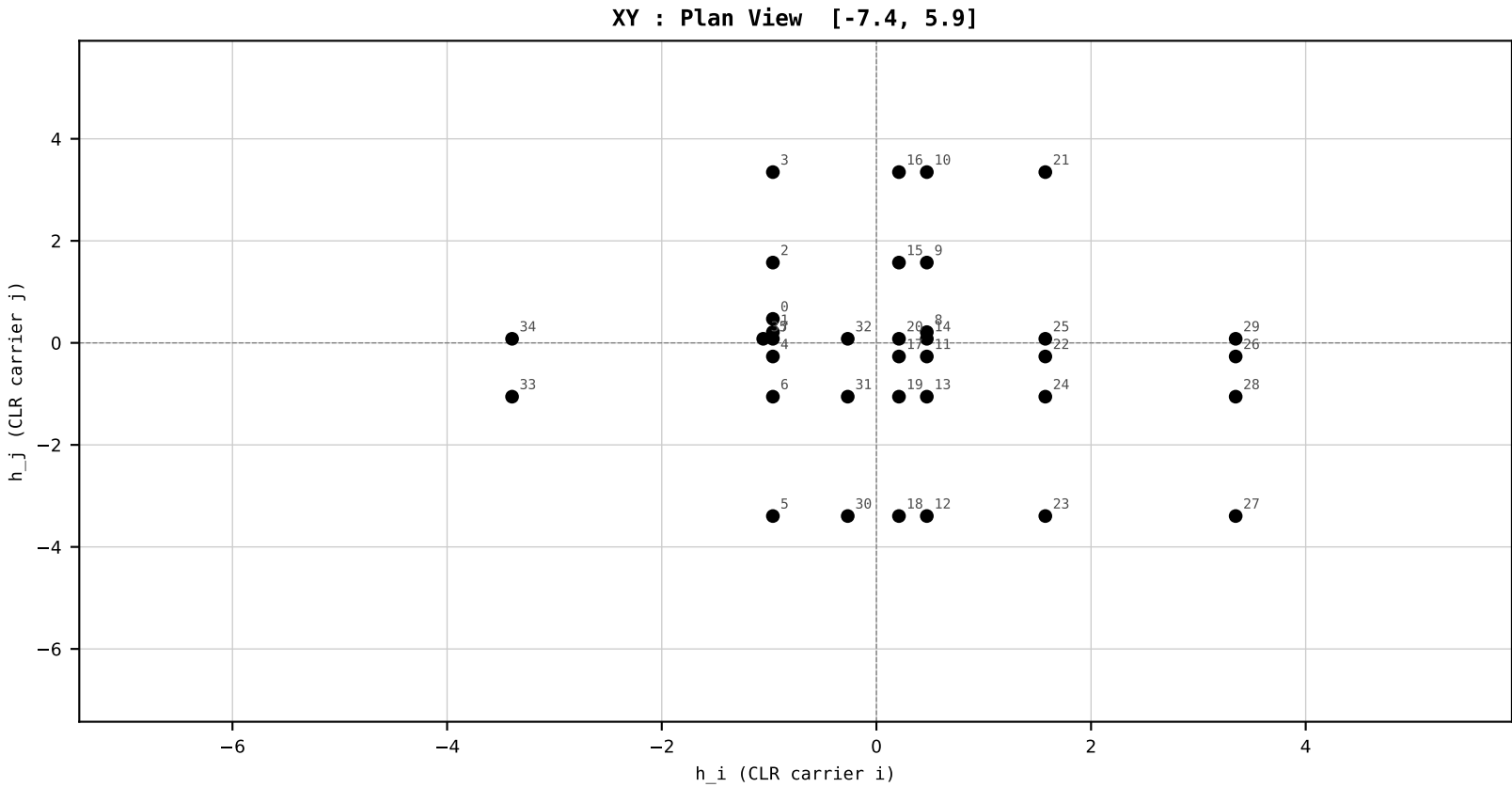
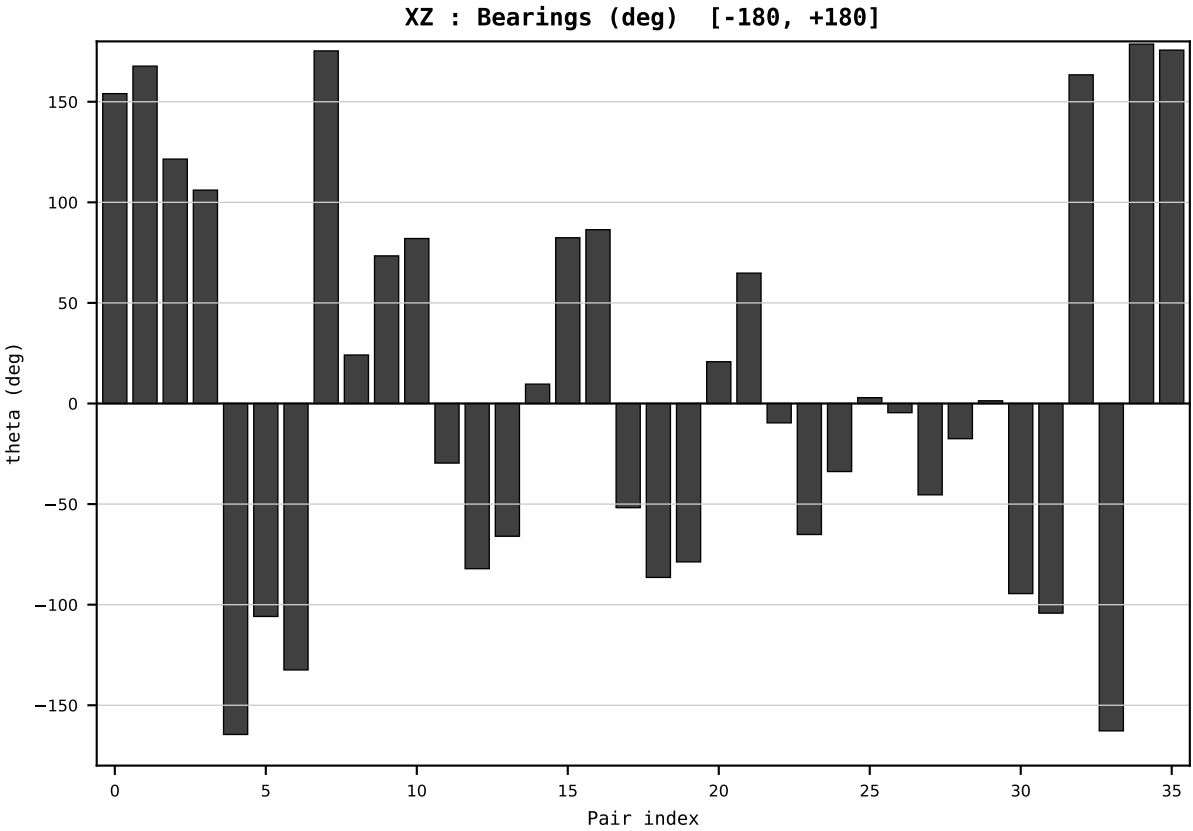
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_FRA_France_generation_TWh.csv
Year : 2013
t : 13 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.545141
Ring = Hs-4
E_metric= 27.5924
kappa_HS= 847.36
omega = 3.9329 deg
d_A = 0.360297
Helm = Gas
Helm d = -0.238712
DR = 6.742 HLR
DR ratio= 847.4x

XZ [-180,180]
CLR[-7.4,5.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=14 | Year 2014 | D=9 N=26 pairs=36

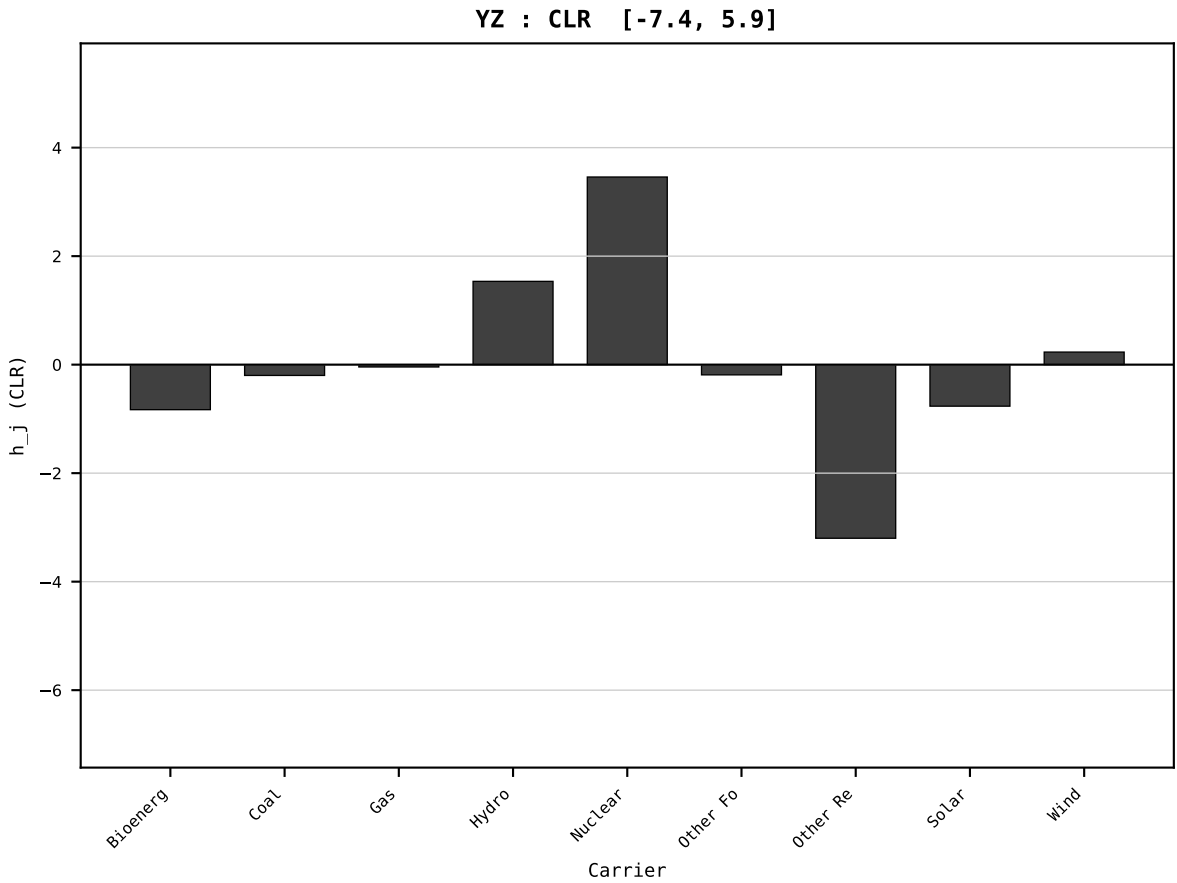
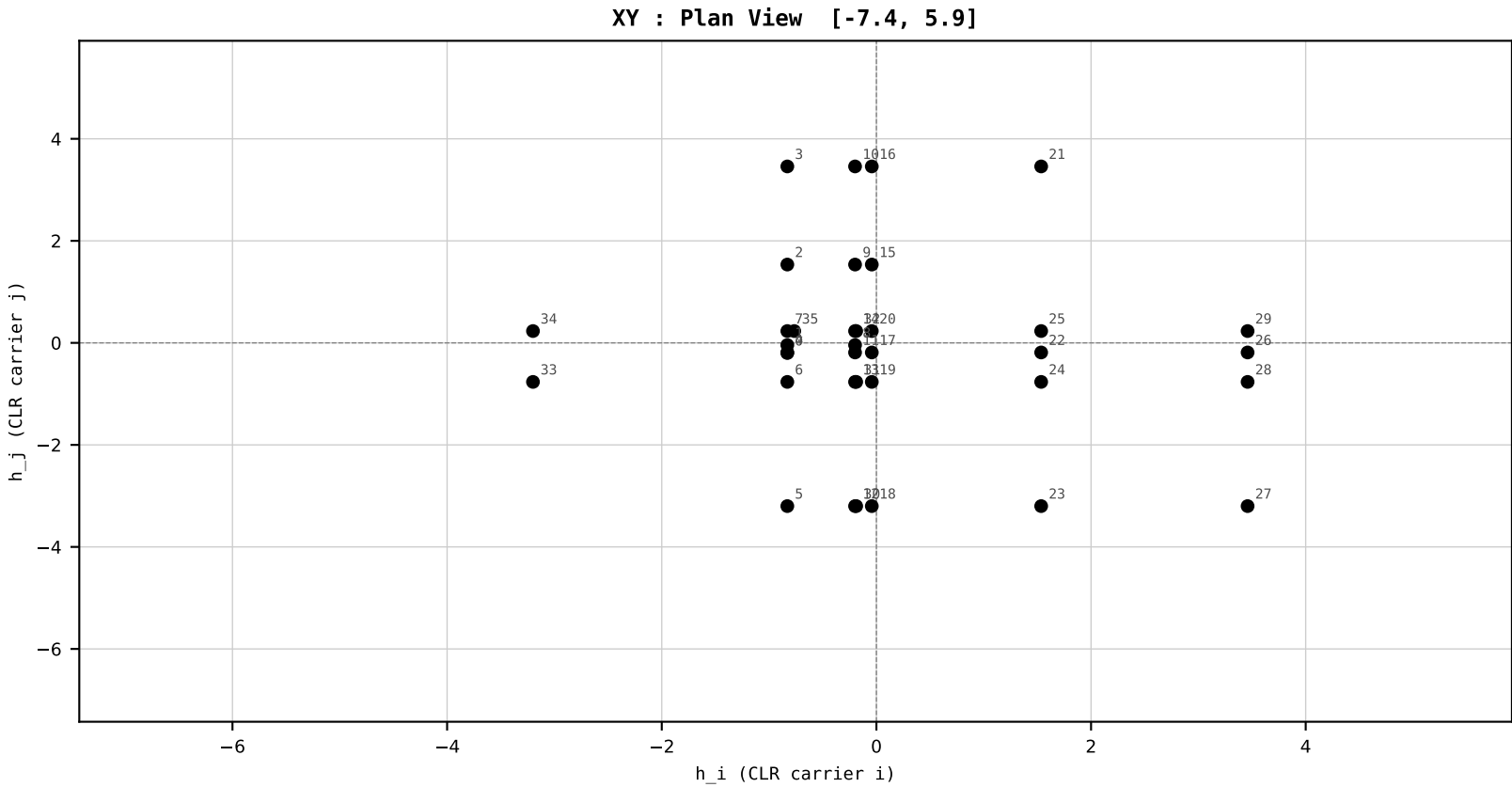
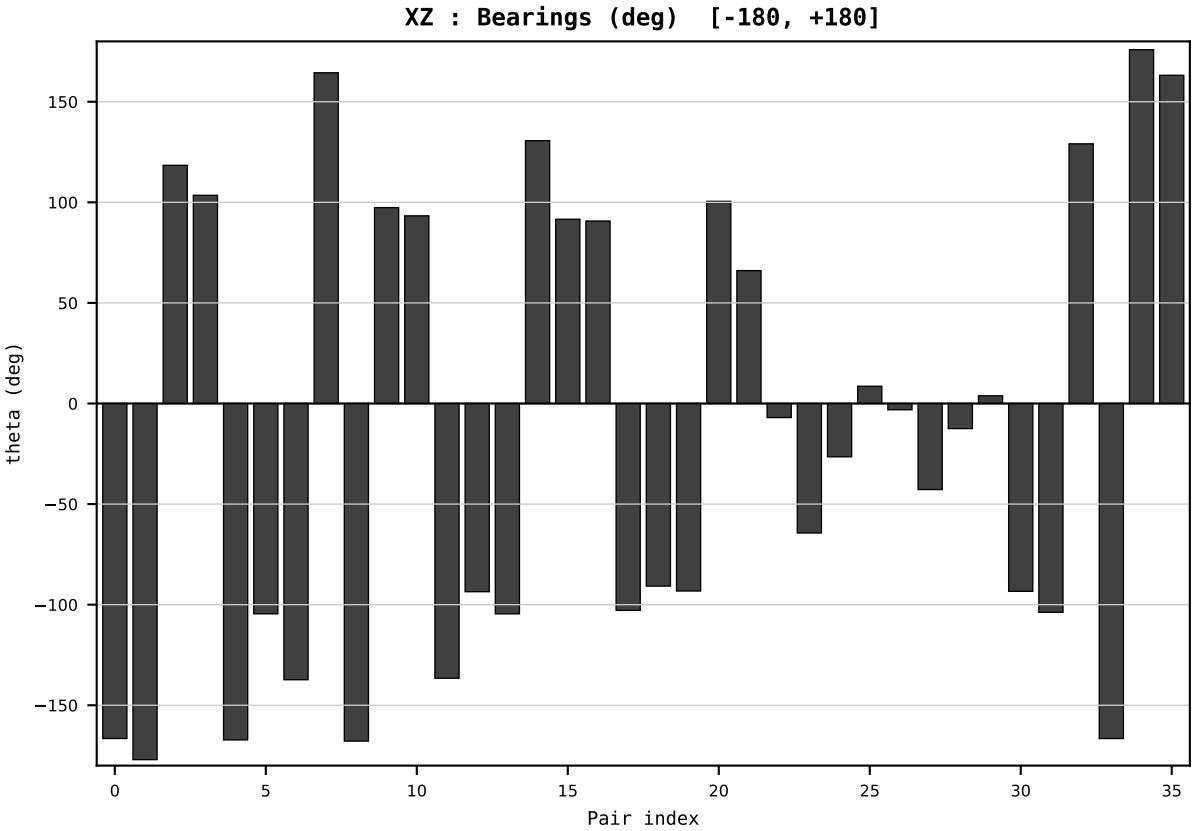
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_FRA_France_generation_TWh.csv
Year : 2014
t : 14 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.589178
Ring = Hs-4
E_metric= 25.9637
kappa_HS= 779.43
omega = 9.0709 deg
d_A = 0.833206
Helm = Coal
Helm d = -0.668681
DR = 6.659 HLR
DR ratio= 779.4x

XZ [-180,180]
CLR[-7.4,5.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



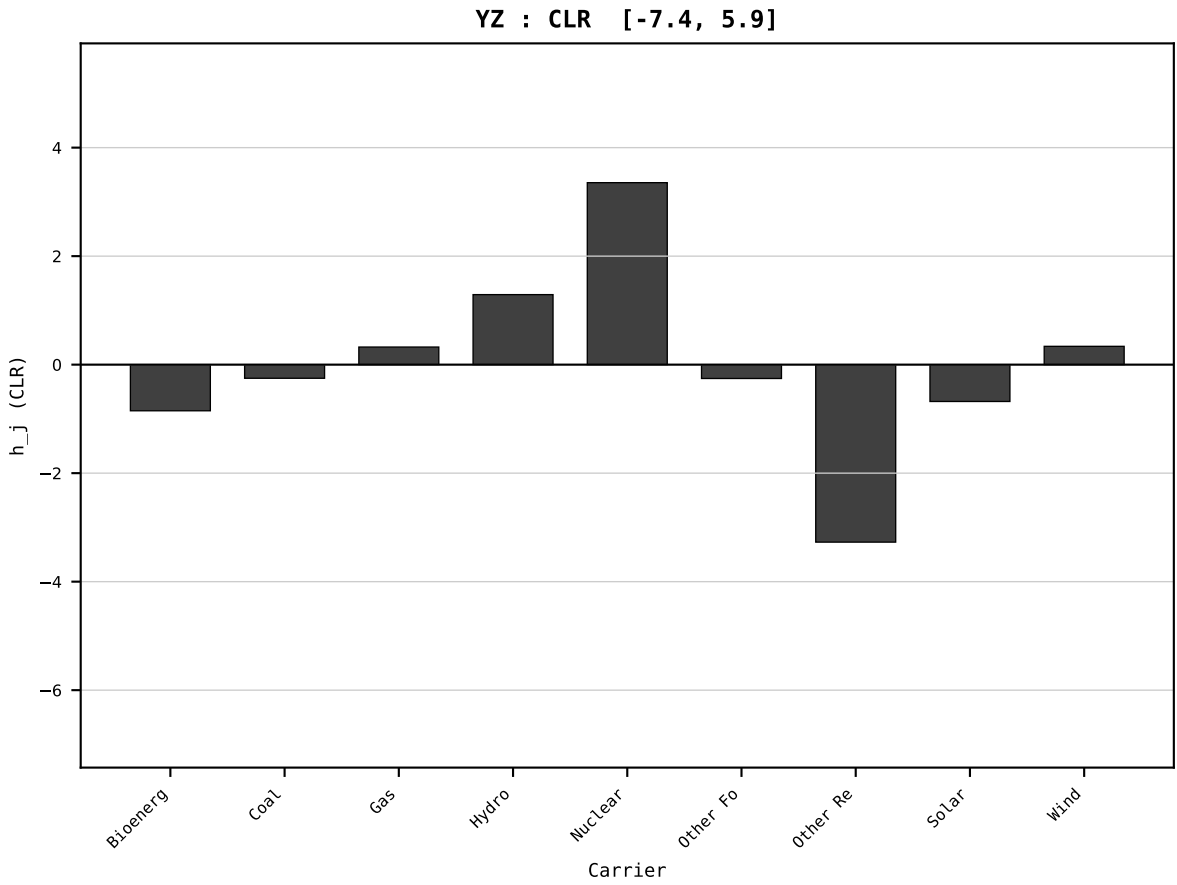
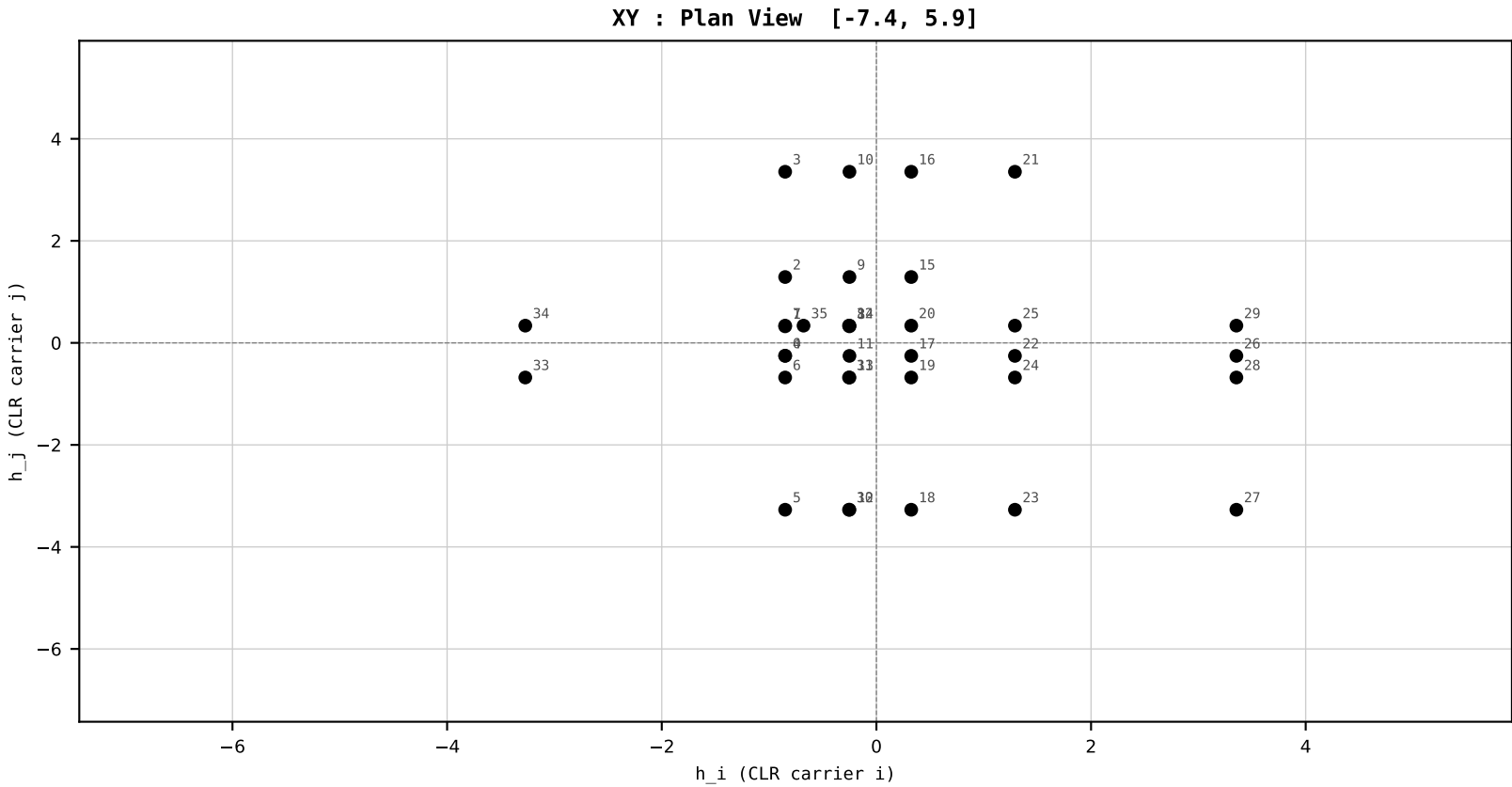
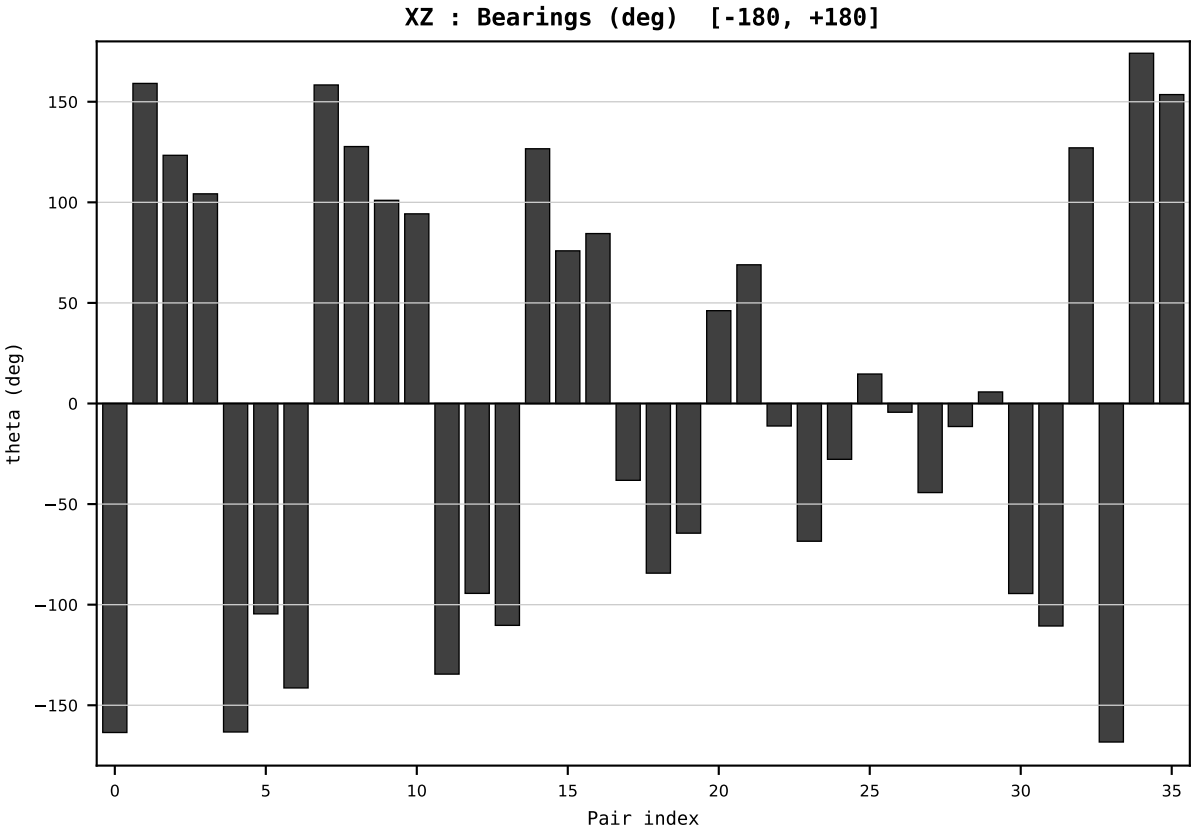
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_FRA_France_generation_TWh.csv
Year : 2015
t : 15 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.565912
Ring = Hs-4
E_metric= 25.1505
kappa_HS= 754.19
omega = 5.4442 deg
d_A = 0.486840
Helm = Gas
Helm d = +0.367403
DR = 6.626 HLR
DR ratio= 754.2x

XZ [-180,180]
CLR[-7.4,5.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



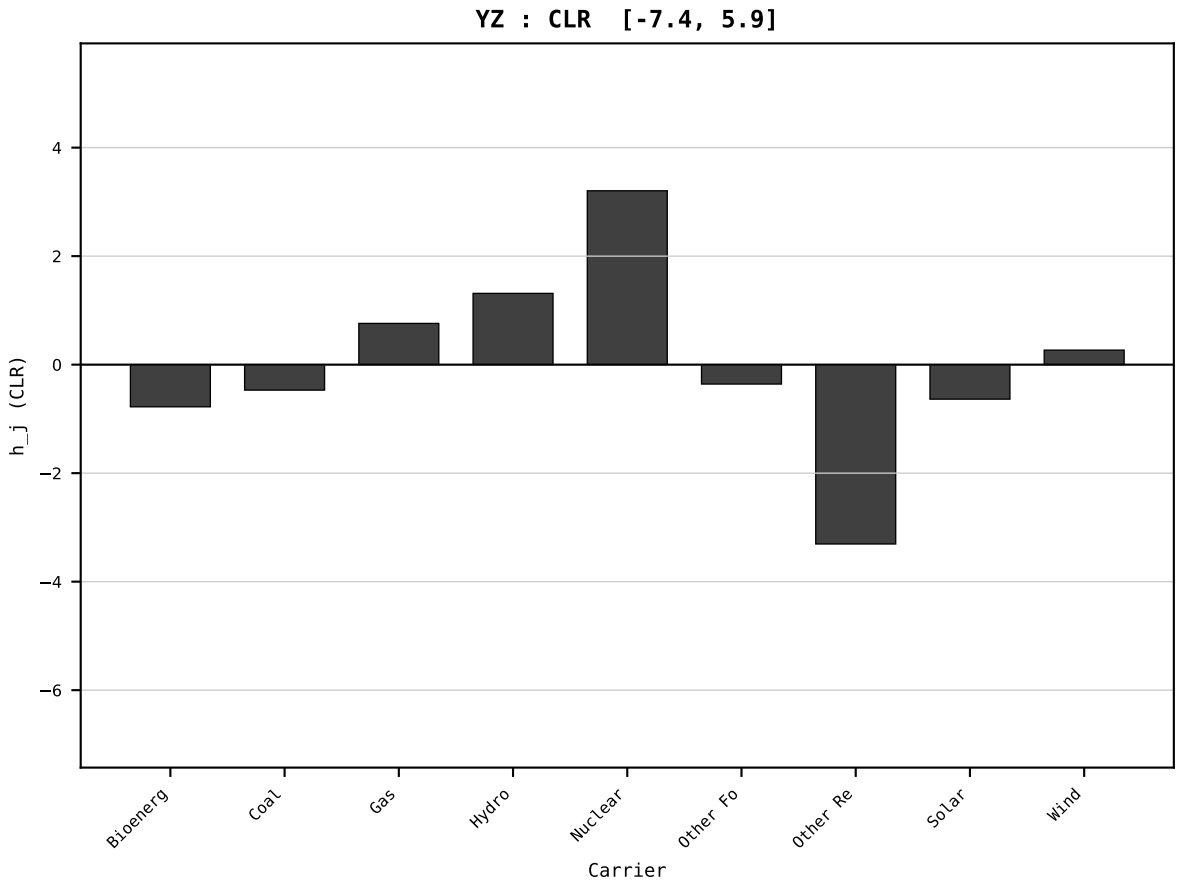
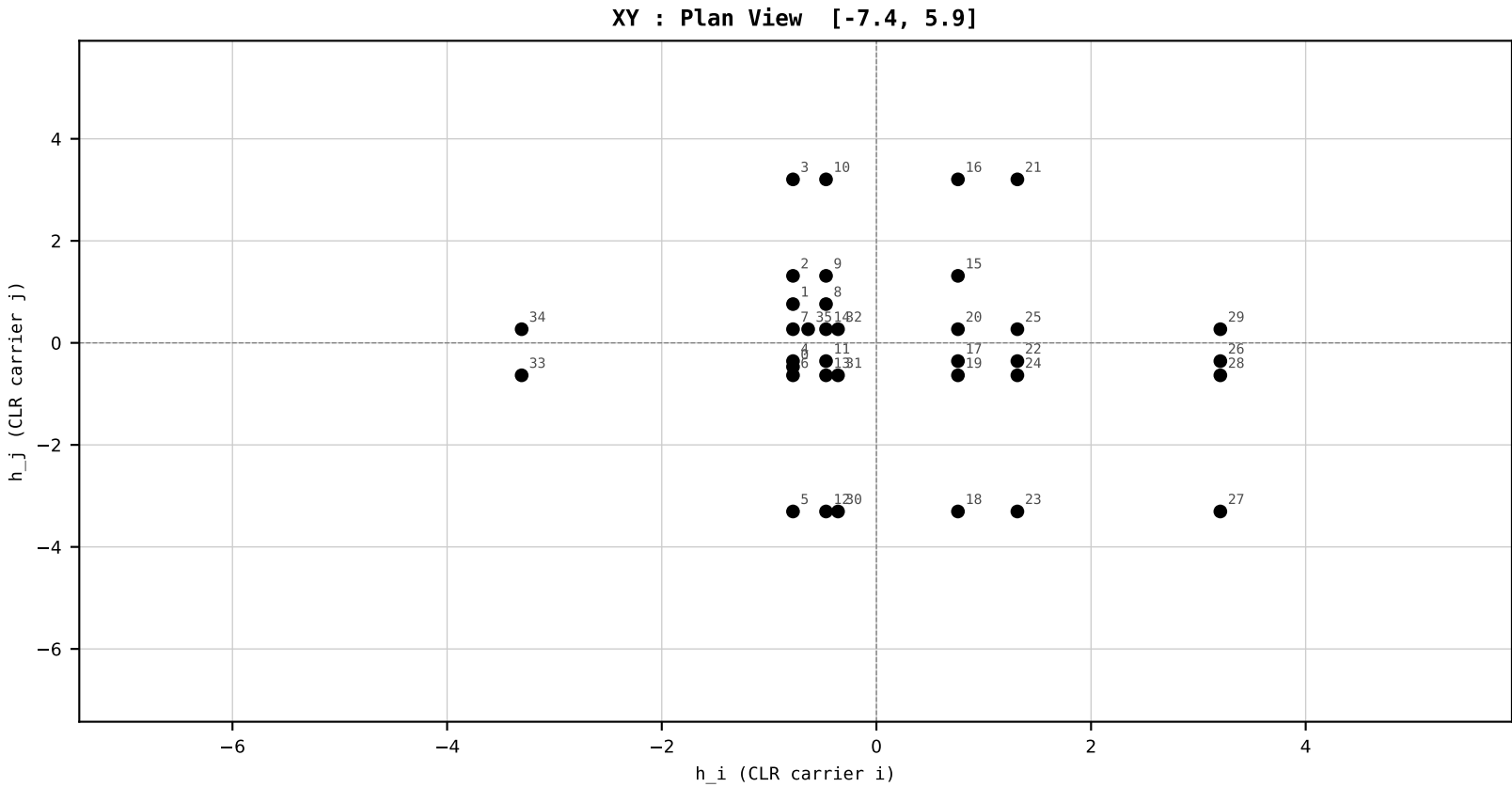
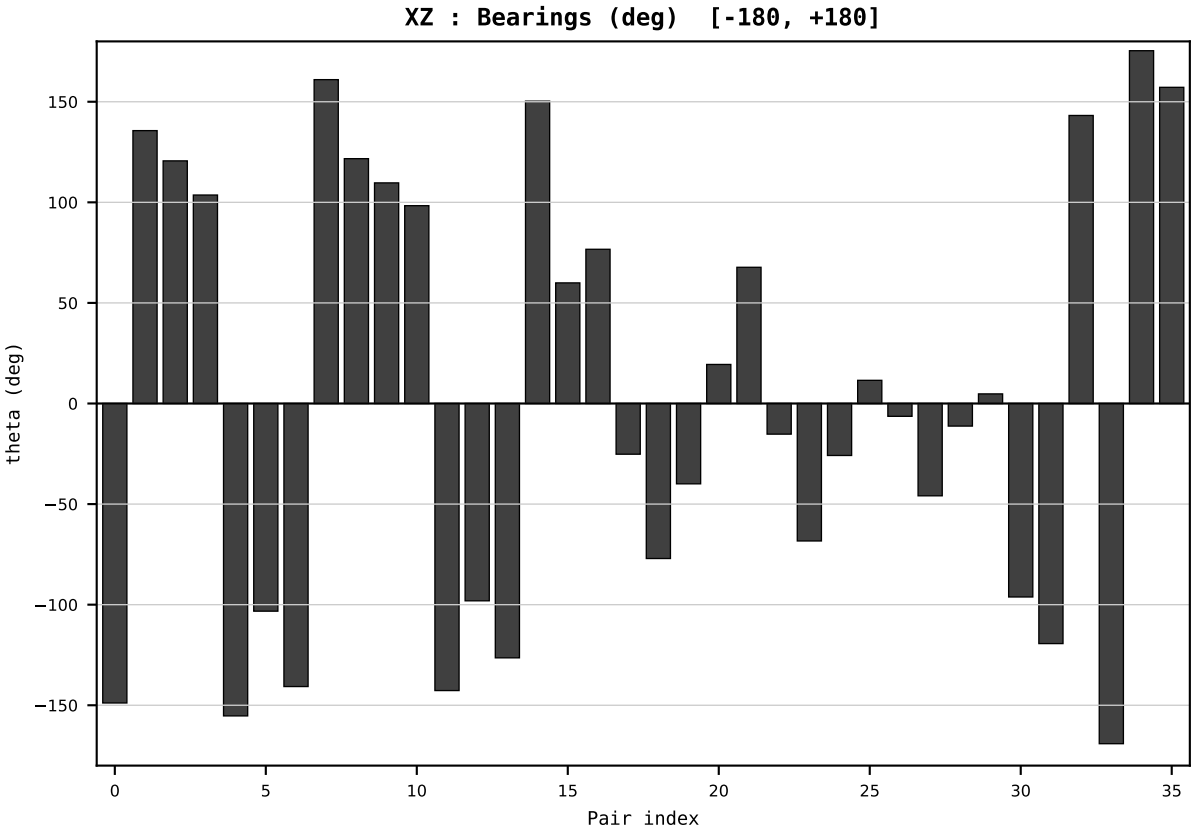
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_FRA_France_generation_TWh.csv
Year : 2016
t : 16 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.518507
Ring = Hs-4
E_metric= 24.9278
kappa_HS= 672.00
omega = 6.0973 deg
d_A = 0.532714
Helm = Gas
Helm d = +0.435313
DR = 6.510 HLR
DR ratio= 672.0x

XZ [-180,180]
CLR[-7.4,5.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=17 | Year 2017 | D=9 N=26 pairs=36

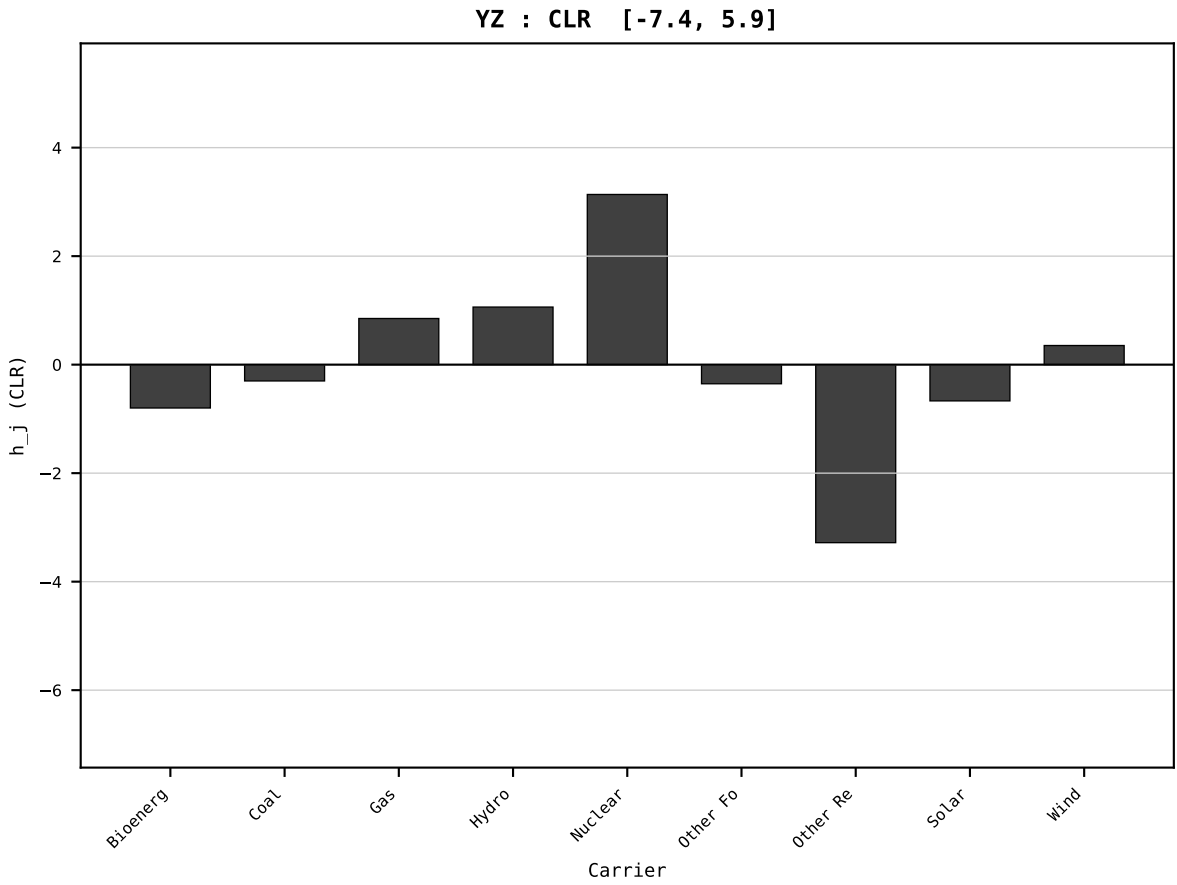
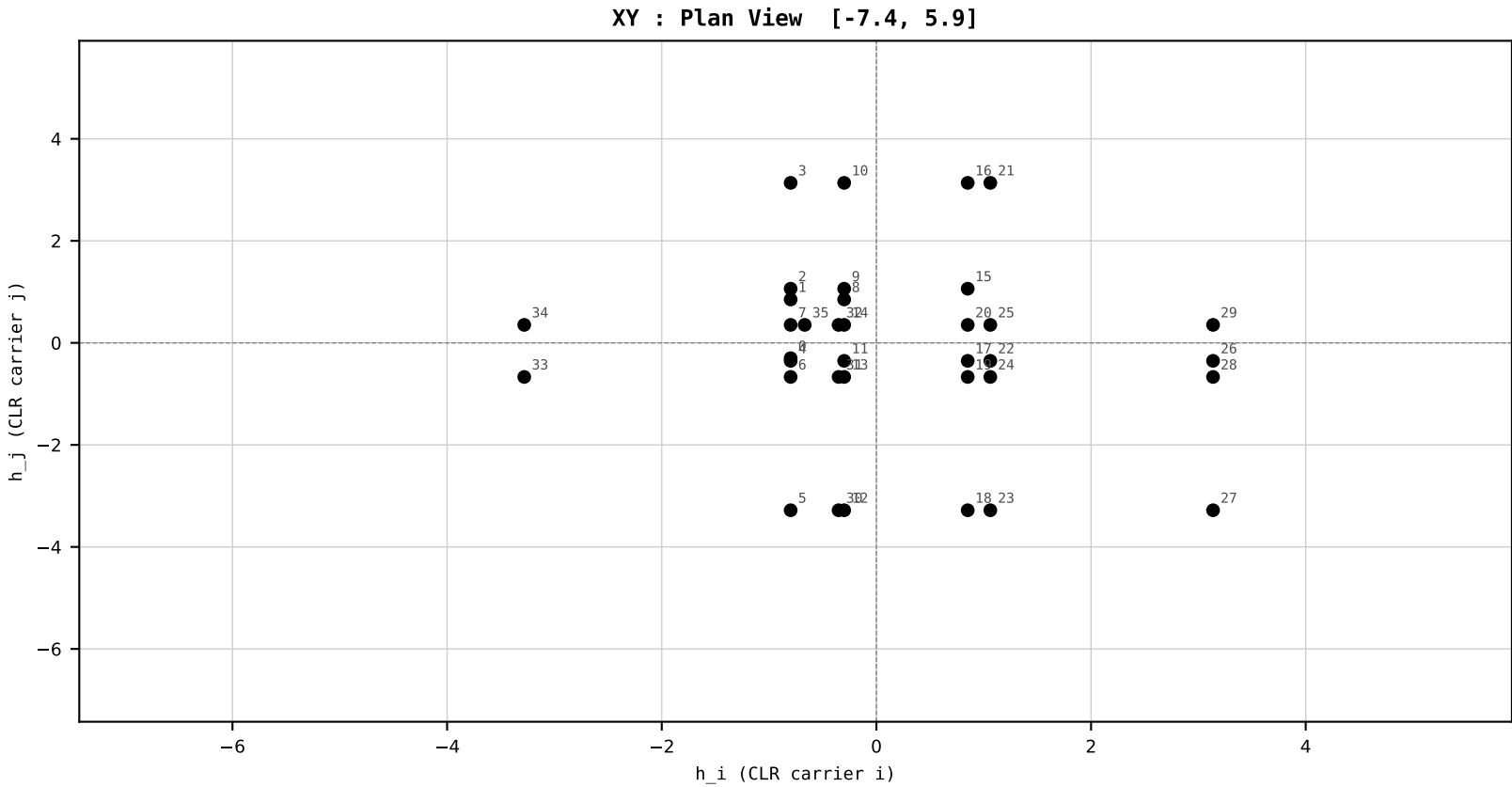
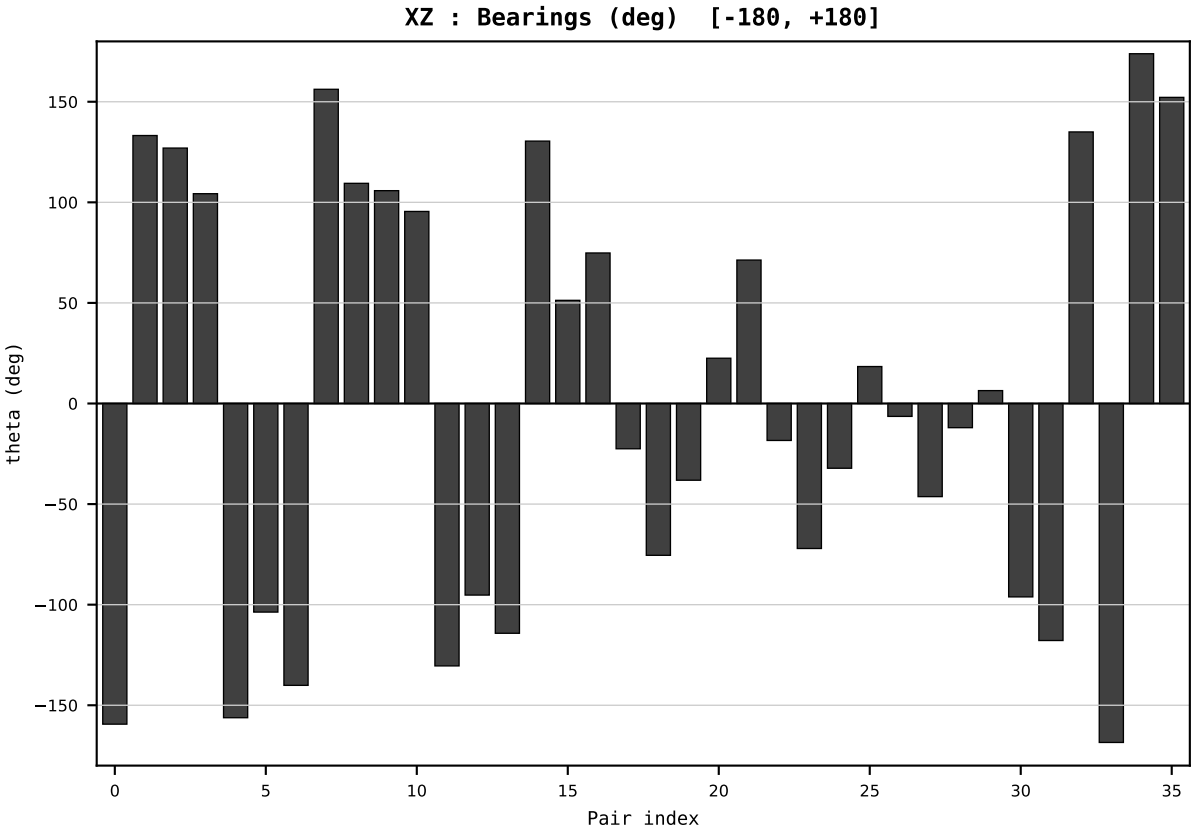
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_FRA_France_generation_TWh.csv
Year : 2017
t : 17 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.504621
Ring = Hs-4
E_metric= 23.8804
kappa_HS= 612.86
omega = 3.7247 deg
d_A = 0.338103
Helm = Hydro
Helm d = -0.252129
DR = 6.418 HLR
DR ratio= 612.9x

XZ [-180,180]
CLR[-7.4,5.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



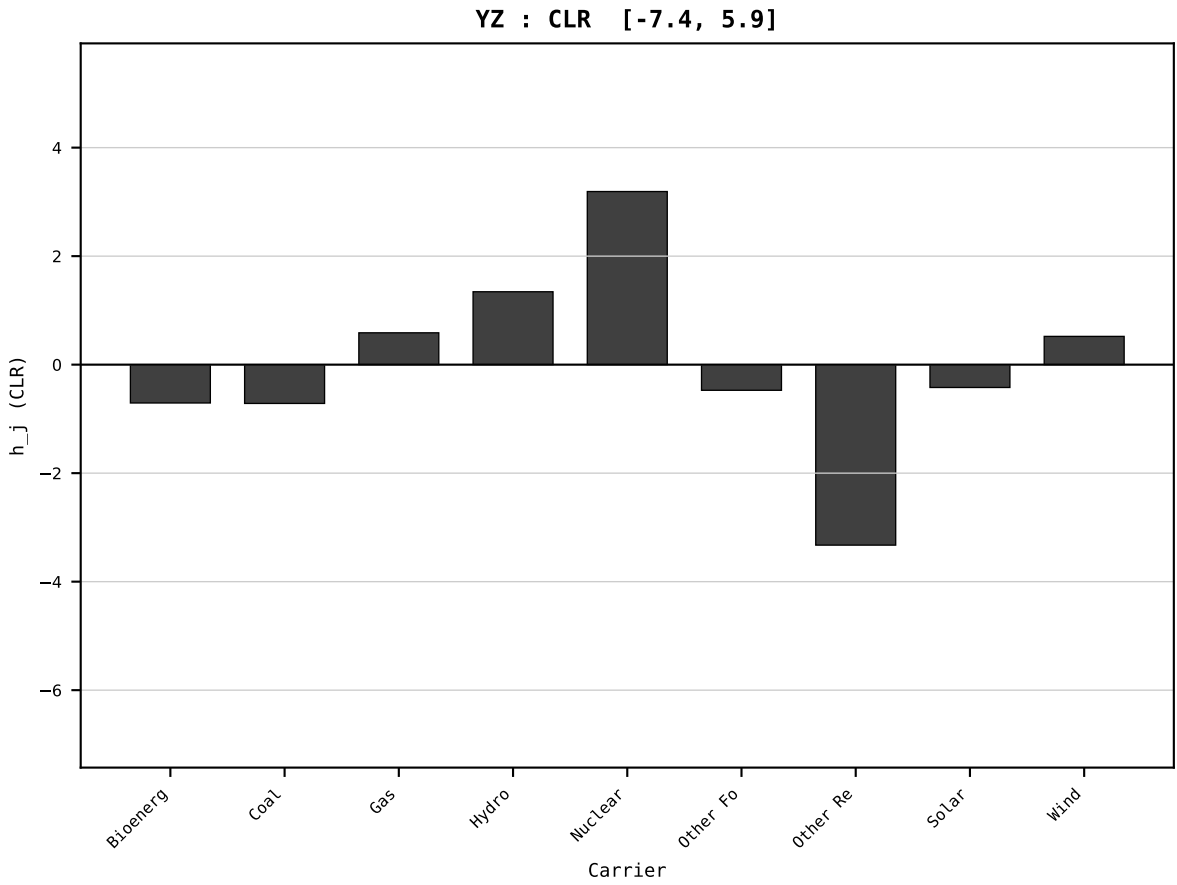
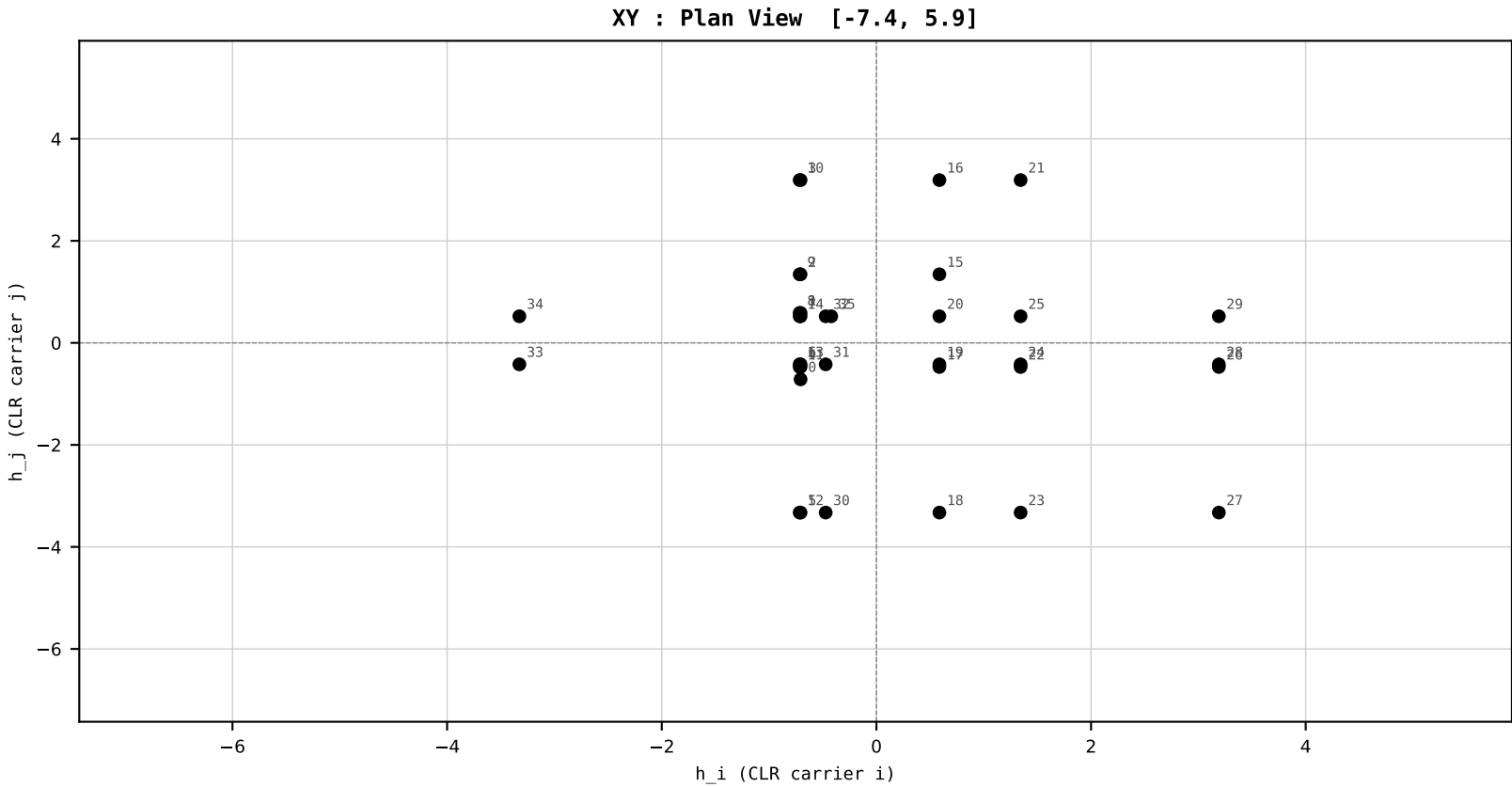
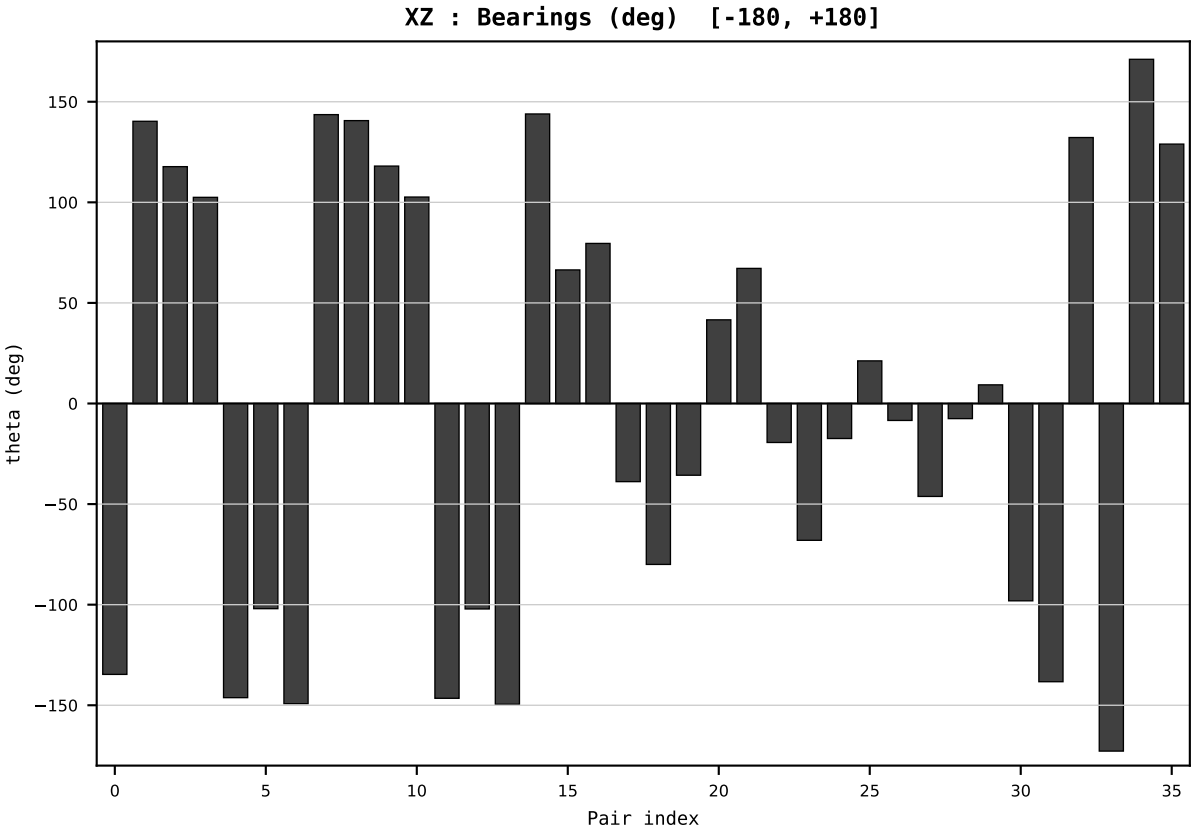
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_FRA_France_generation_TWh.csv
Year : 2018
t : 18 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.513194
Ring = Hs-4
E_metric= 25.0810
kappa_HS= 676.95
omega = 7.5442 deg
d_A = 0.662124
Helm = Coal
Helm d = -0.414639
DR = 6.518 HLR
DR ratio= 677.0x

XZ [-180,180]
CLR[-7.4,5.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



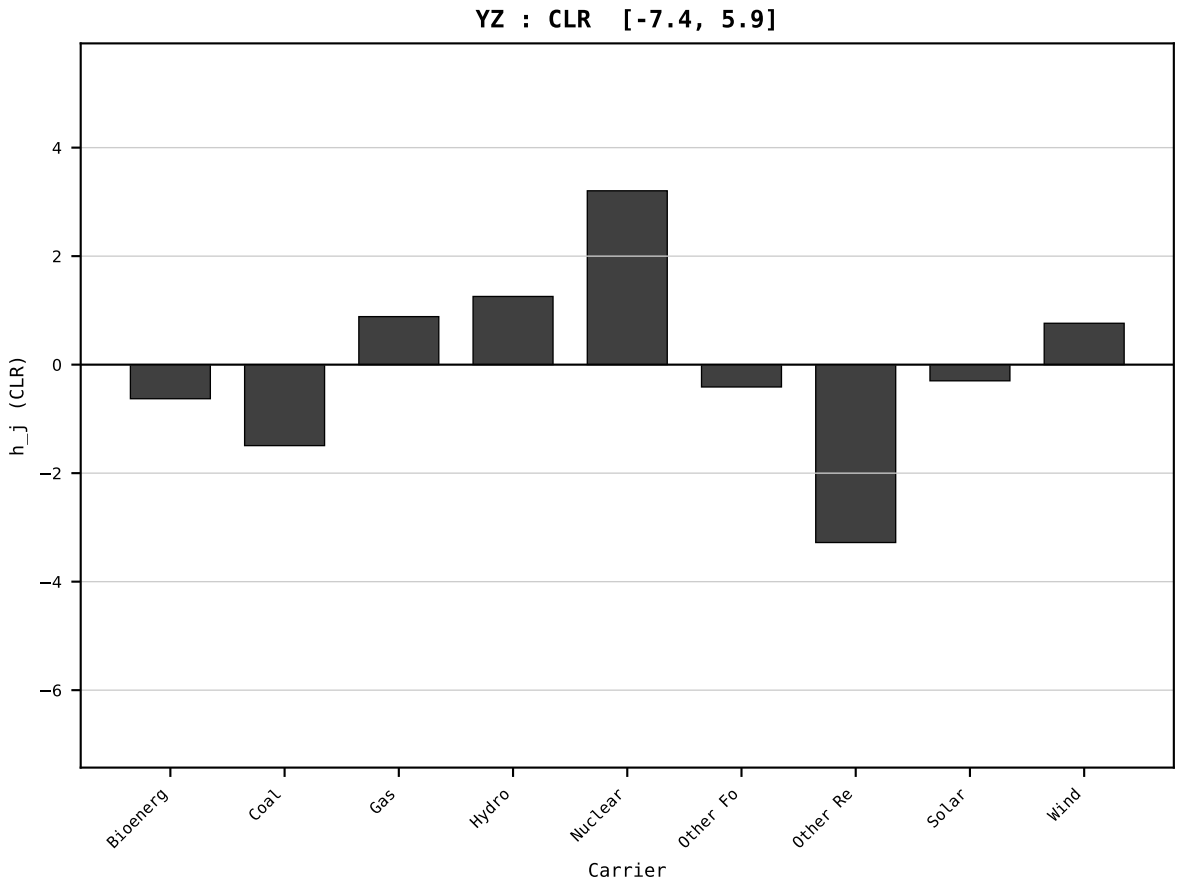
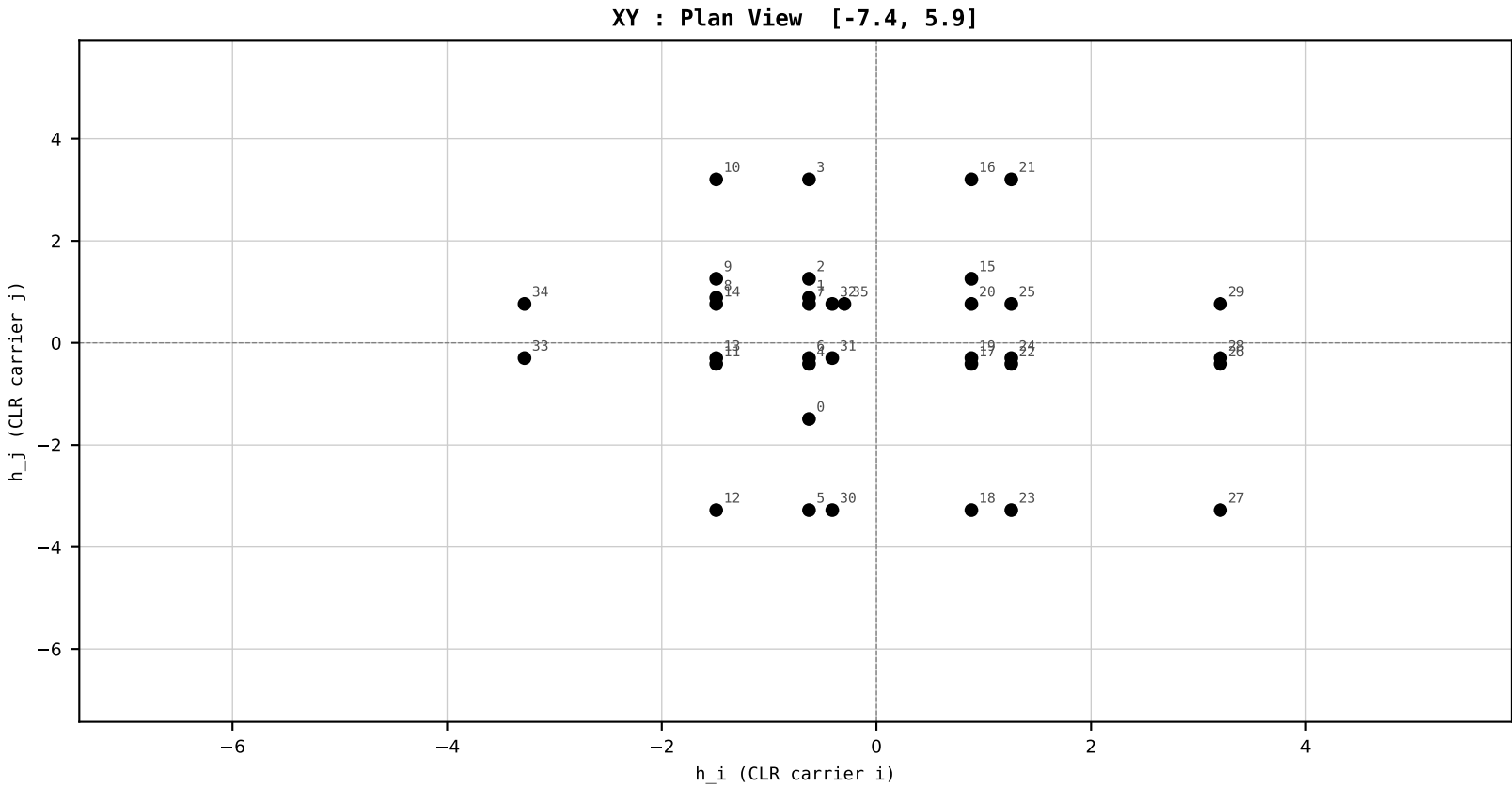
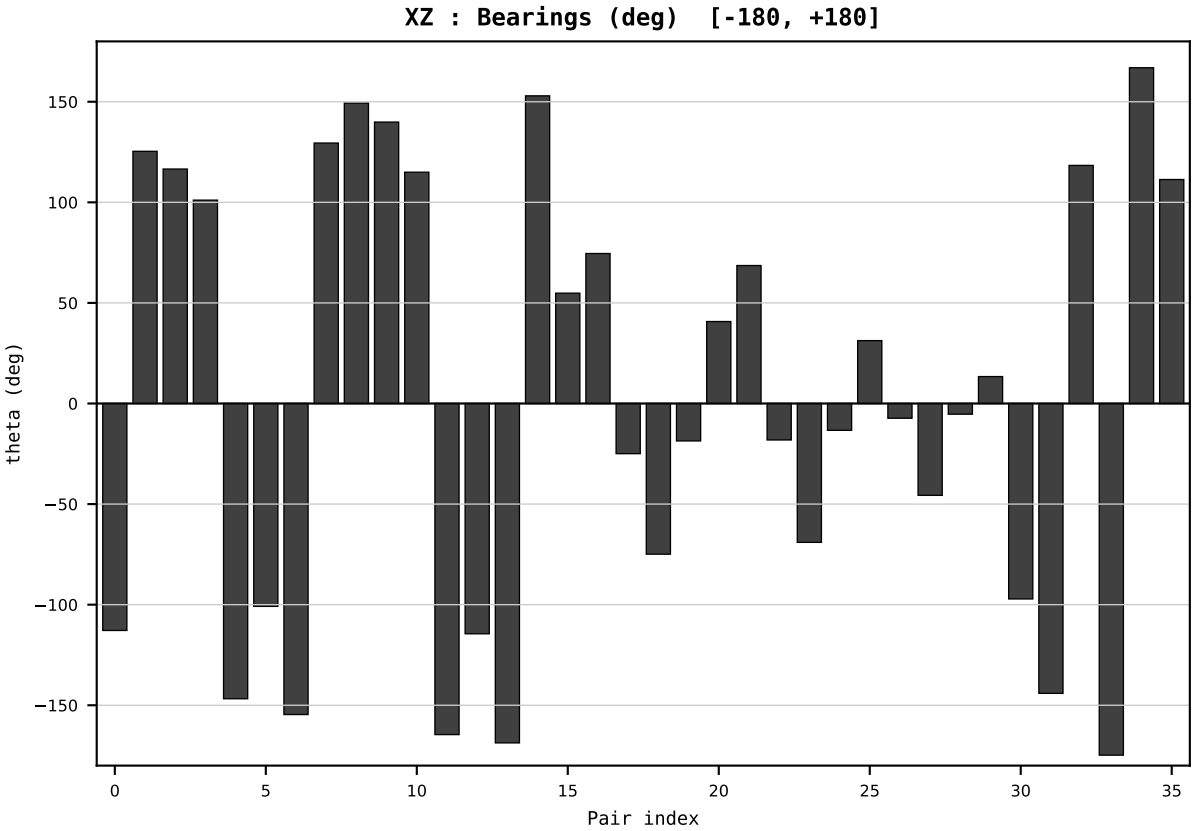
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_FRA_France_generation_TWh.csv
Year : 2019
t : 19 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.501913
Ring = Hs-4
E_metric= 26.8448
kappa_HS= 654.11
omega = 9.7988 deg
d_A = 0.887160
Helm = Coal
Helm d = -0.777550
DR = 6.483 HLR
DR ratio= 654.1x

XZ [-180,180]
CLR[-7.4,5.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=20 | Year 2020 | D=9 N=26 pairs=36

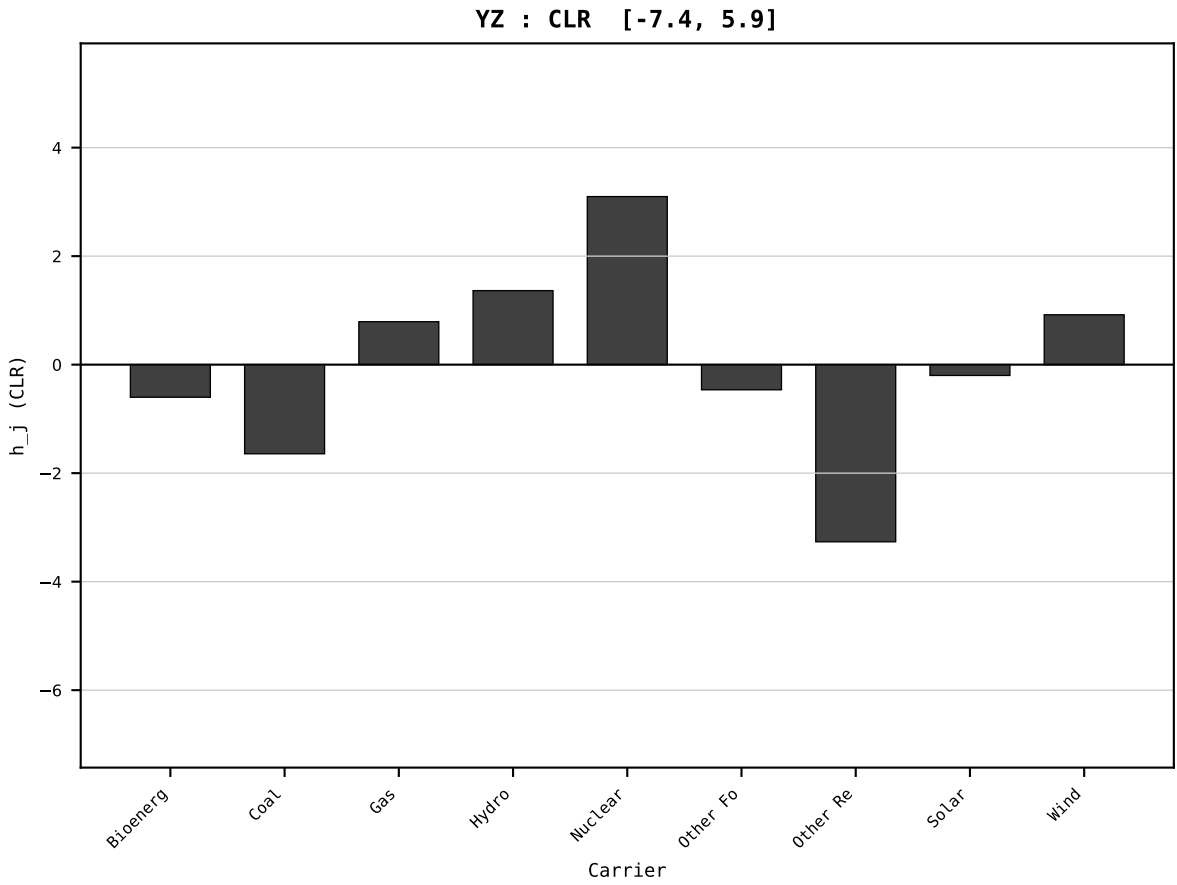
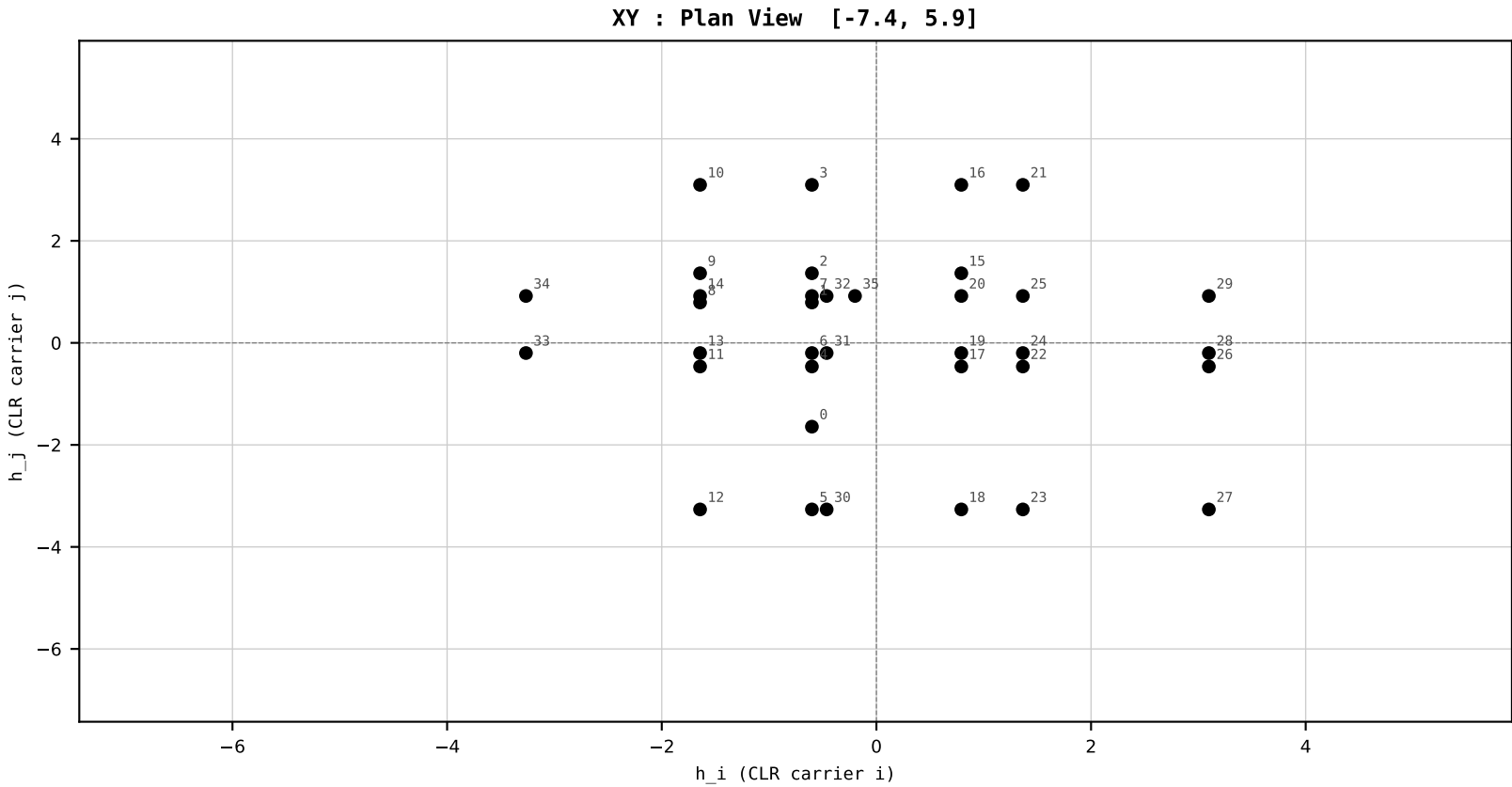
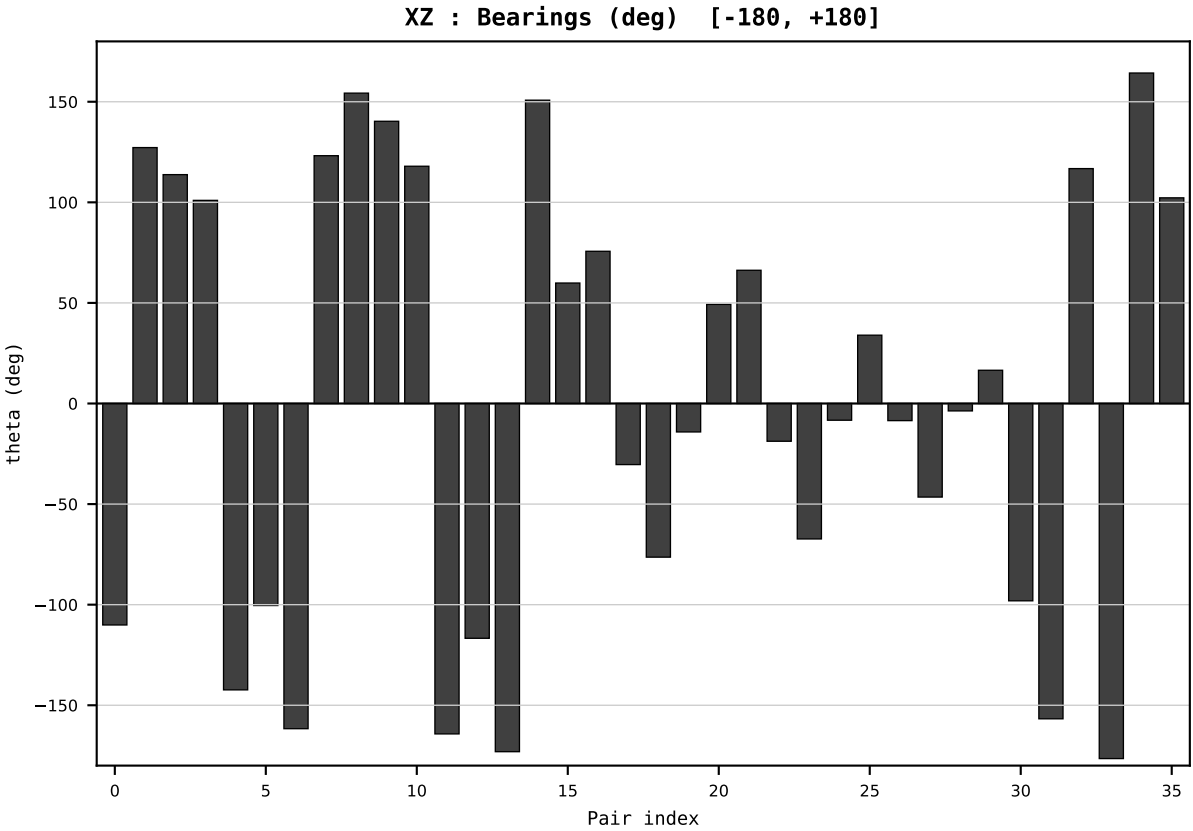
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_FRA_France_generation_TWh.csv
Year : 2020
t : 20 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.467323
Ring = Hs-3
E_metric= 26.9059
kappa_HS= 580.05
omega = 3.3546 deg
d_A = 0.303539
Helm = Wind
Helm d = +0.156233
DR = 6.363 HLR
DR ratio= 580.0x

XZ [-180,180]
CLR[-7.4,5.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=21 | Year 2021 | D=9 N=26 pairs=36

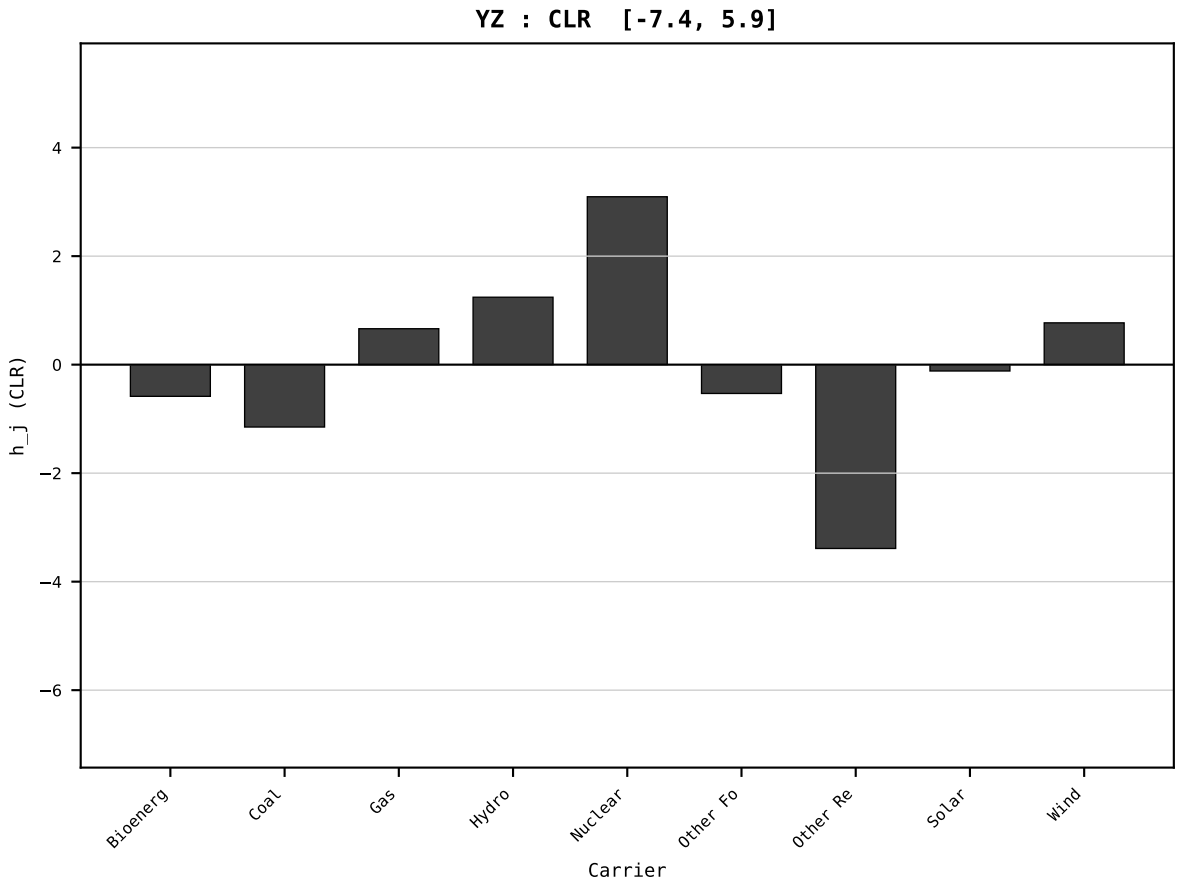
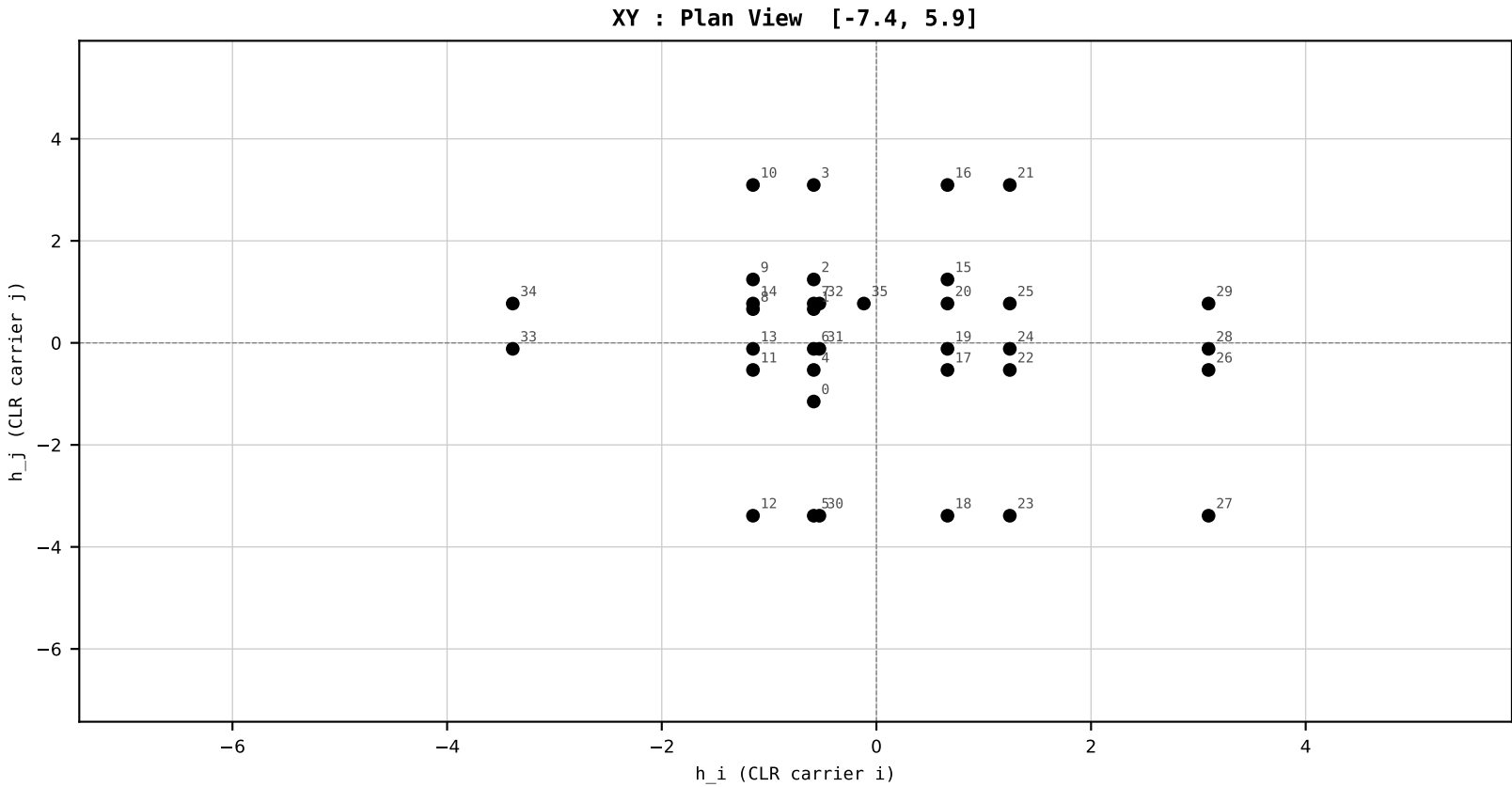
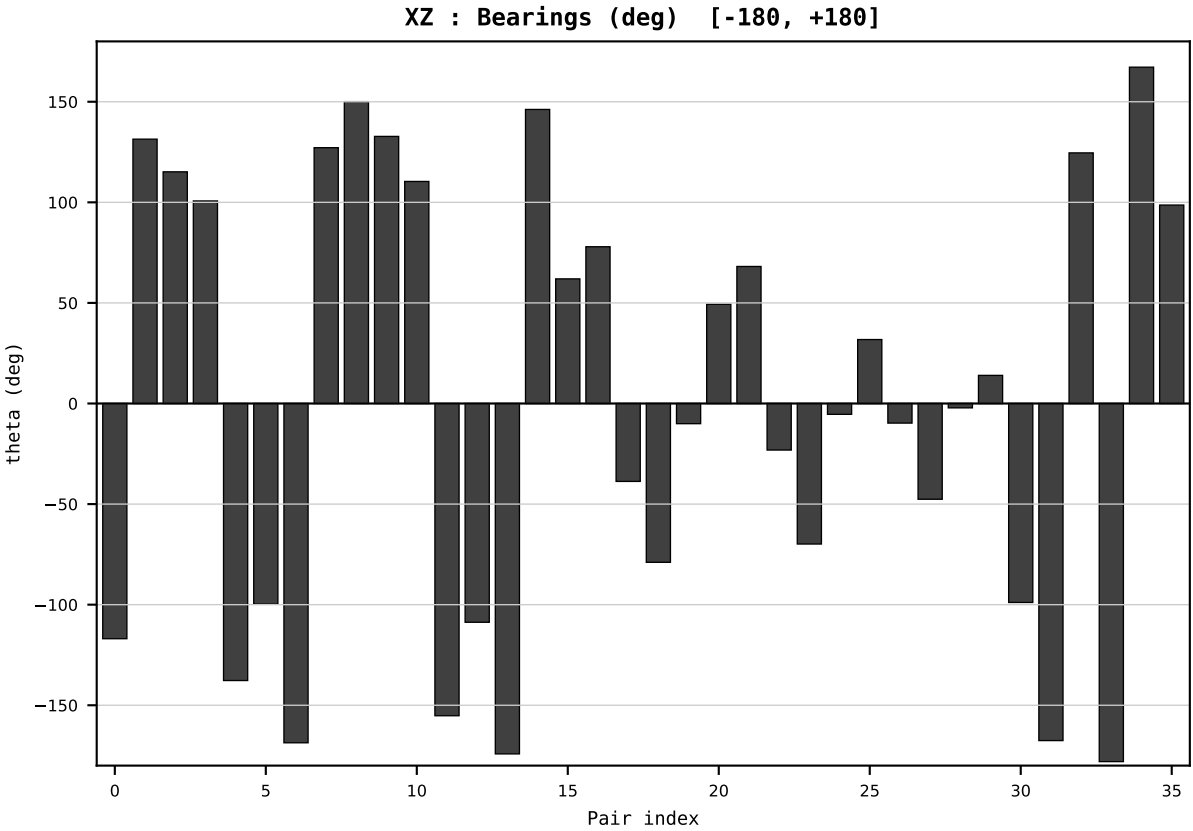
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_FRA_France_generation_TWh.csv
Year : 2021
t : 21 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.478912
Ring = Hs-3
E_metric= 25.5954
kappa_HS= 654.07
omega = 6.1997 deg
d_A = 0.568610
Helm = Coal
Helm d = +0.493100
DR = 6.483 HLR
DR ratio= 654.1x

XZ [-180,180]
CLR[-7.4,5.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



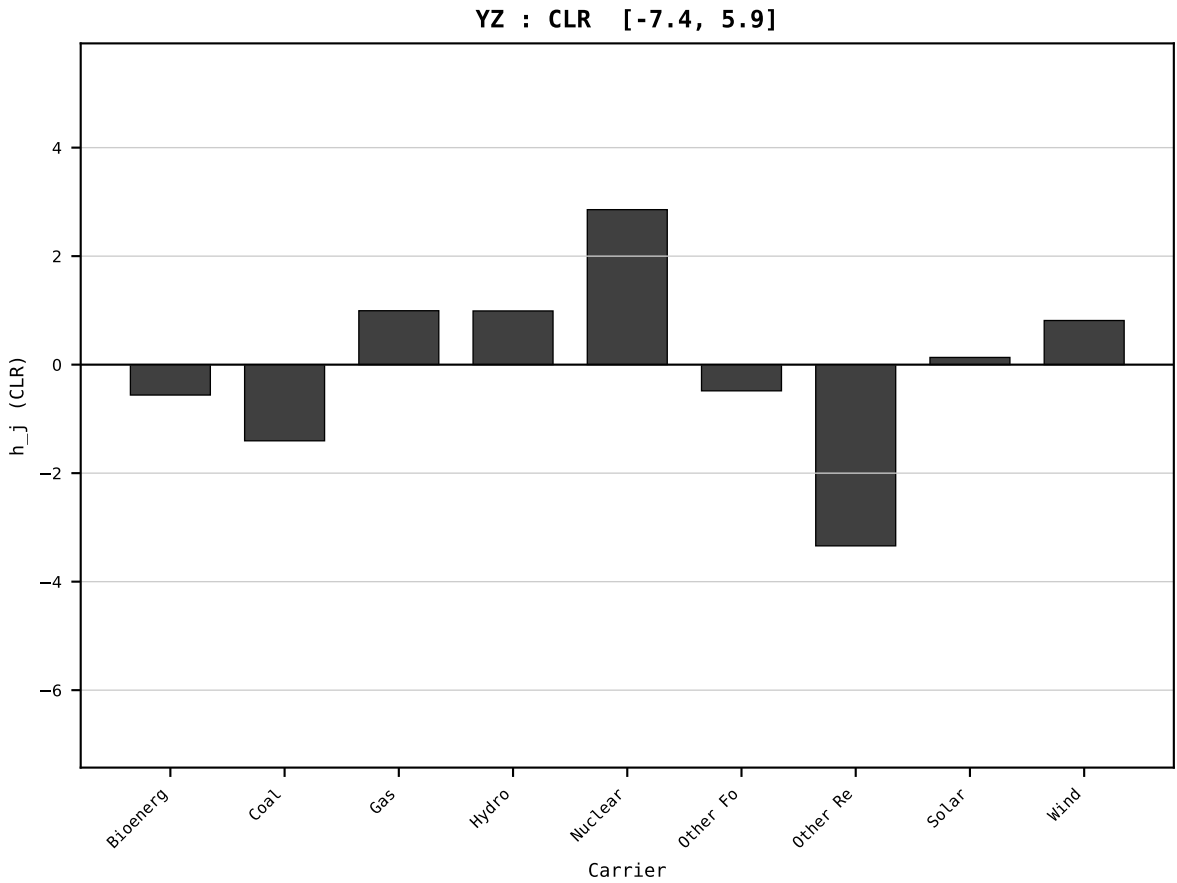
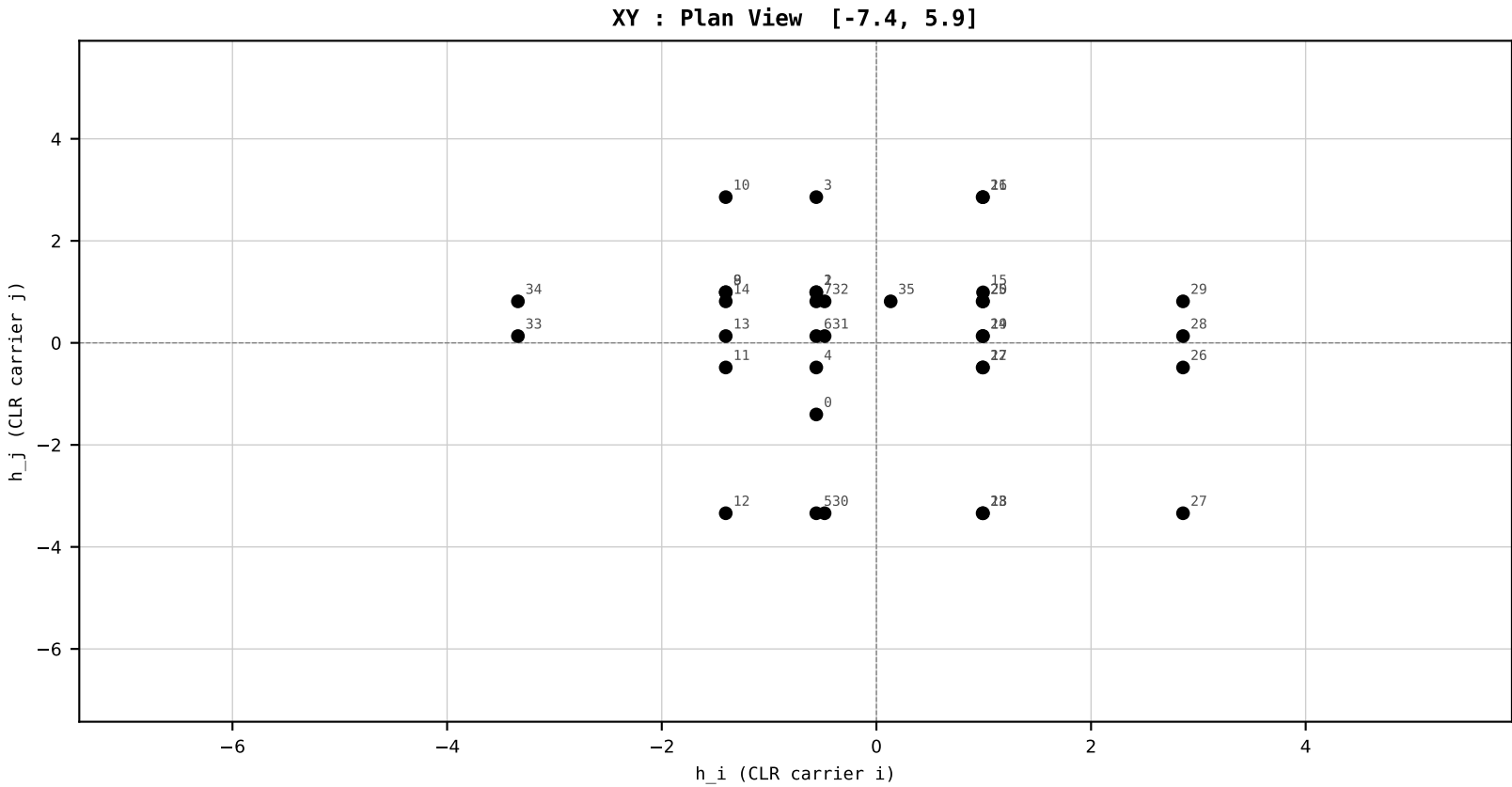
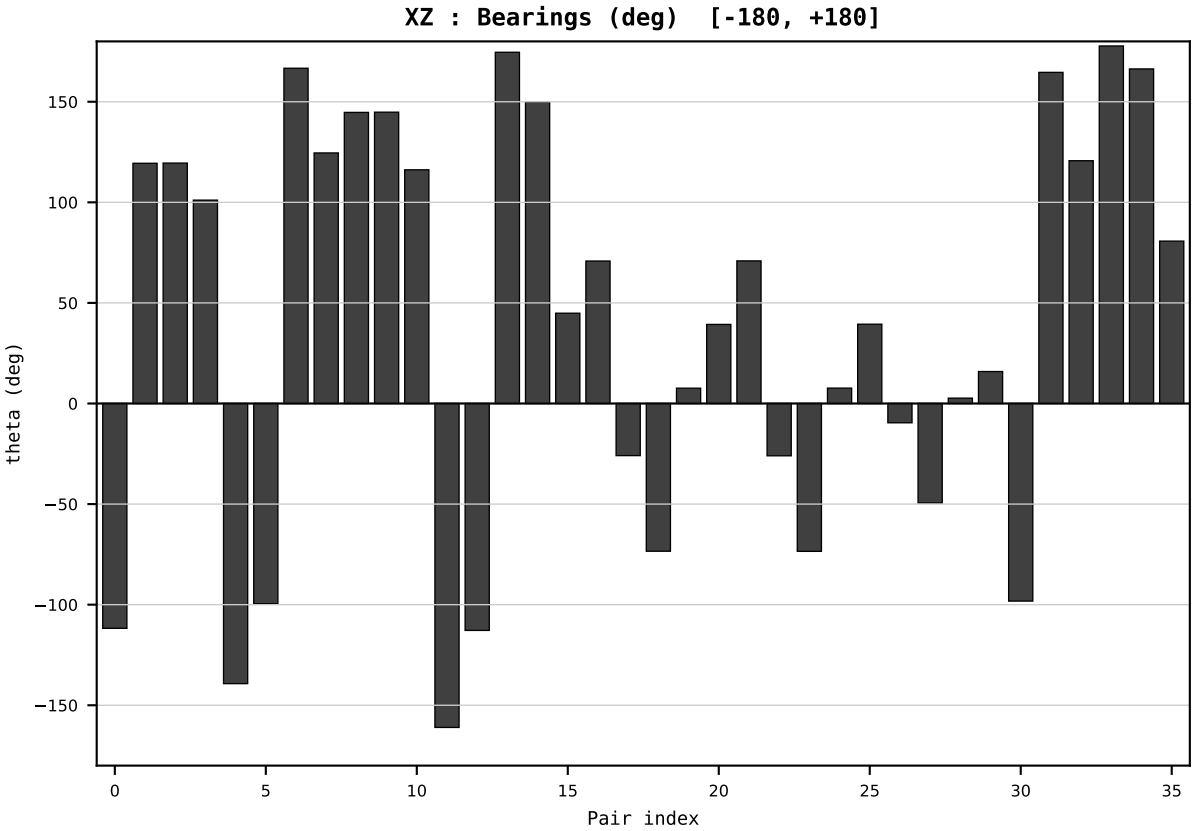
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_FRA_France_generation_TWh.csv
Year : 2022
t : 22 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.409828
Ring = Hs-3
E_metric= 24.4815
kappa_HS= 491.22
omega = 6.8017 deg
d_A = 0.603939
Helm = Gas
Helm d = +0.331511
DR = 6.197 HLR
DR ratio= 491.2x

XZ [-180,180]
CLR[-7.4,5.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



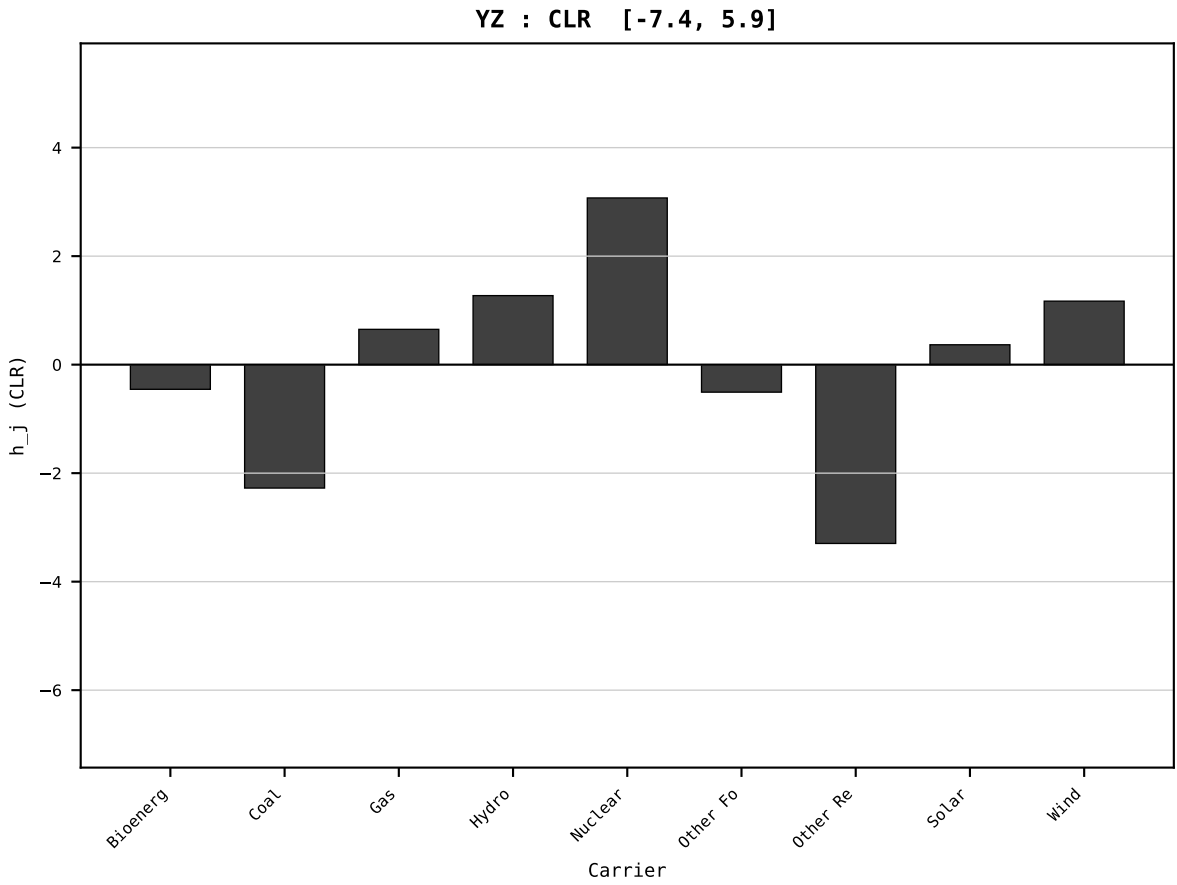
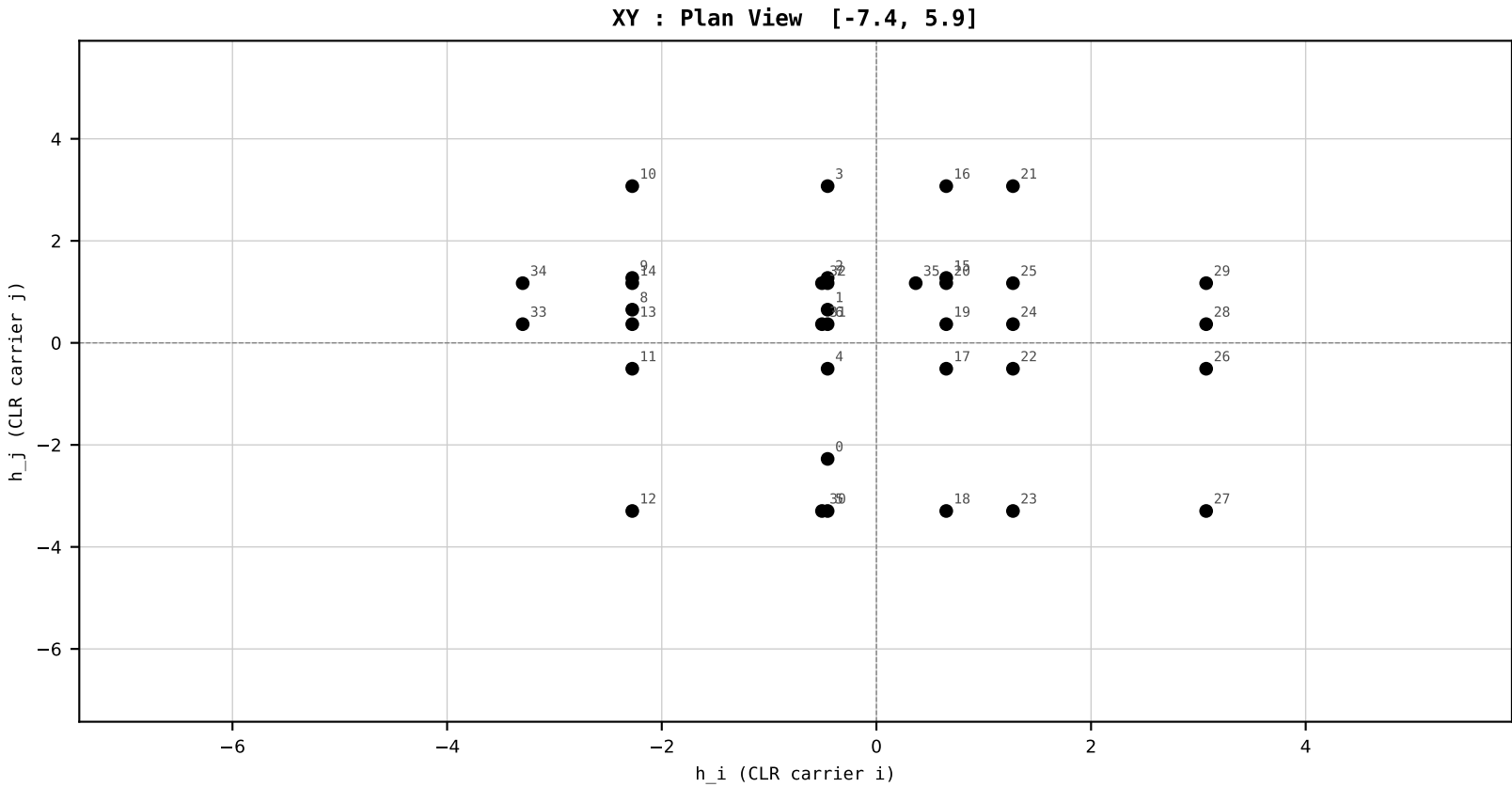
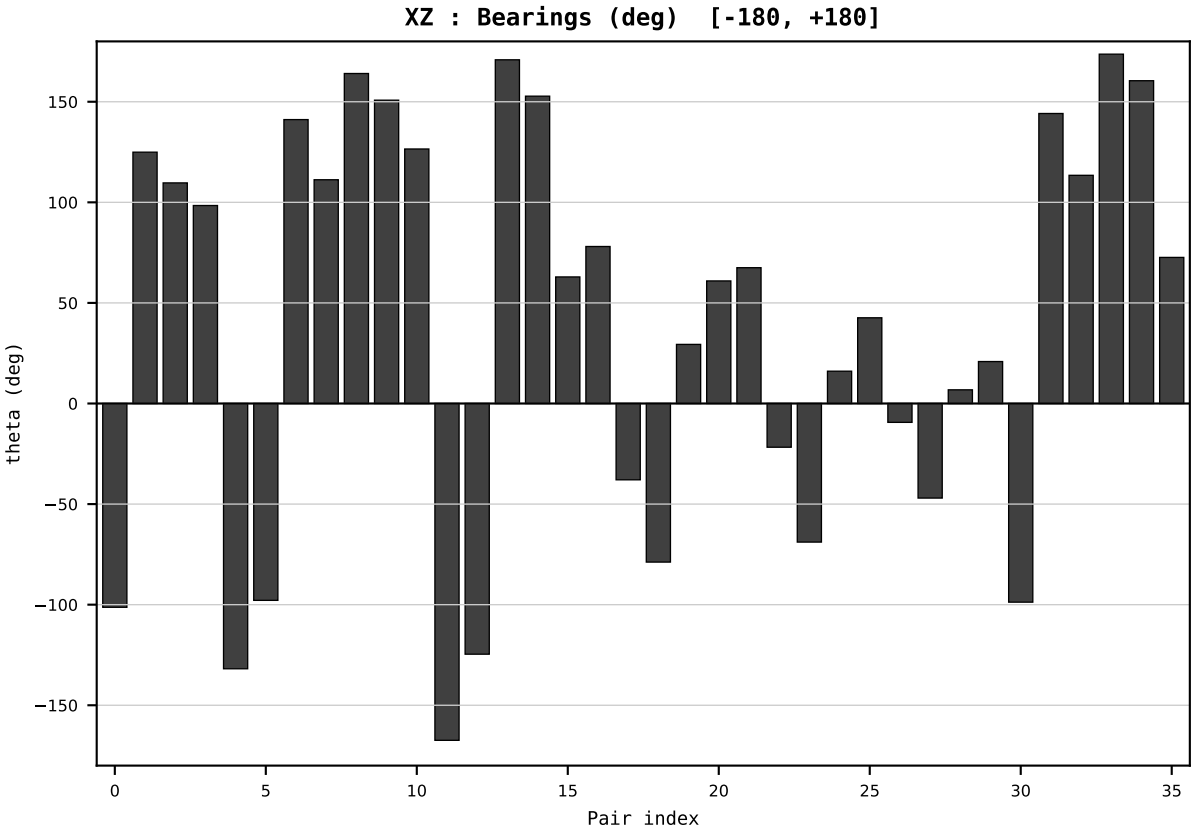
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_FRA_France_generation_TWh.csv
Year : 2023
t : 23 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.444201
Ring = Hs-3
E_metric= 29.4888
kappa_HS= 583.10
omega = 10.8790 deg
d_A = 1.094786
Helm = Coal
Helm d = -0.871139
DR = 6.368 HLR
DR ratio= 583.1x

XZ [-180,180]
CLR[-7.4,5.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



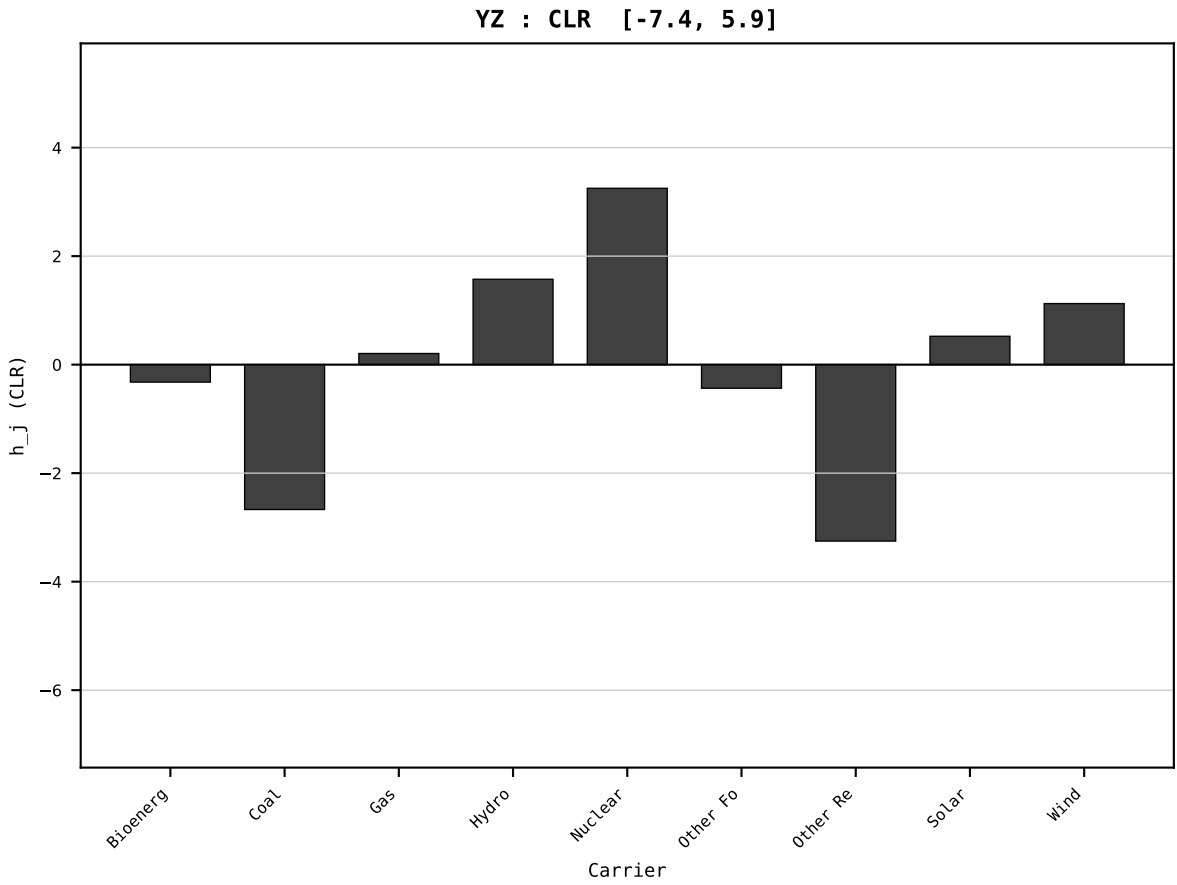
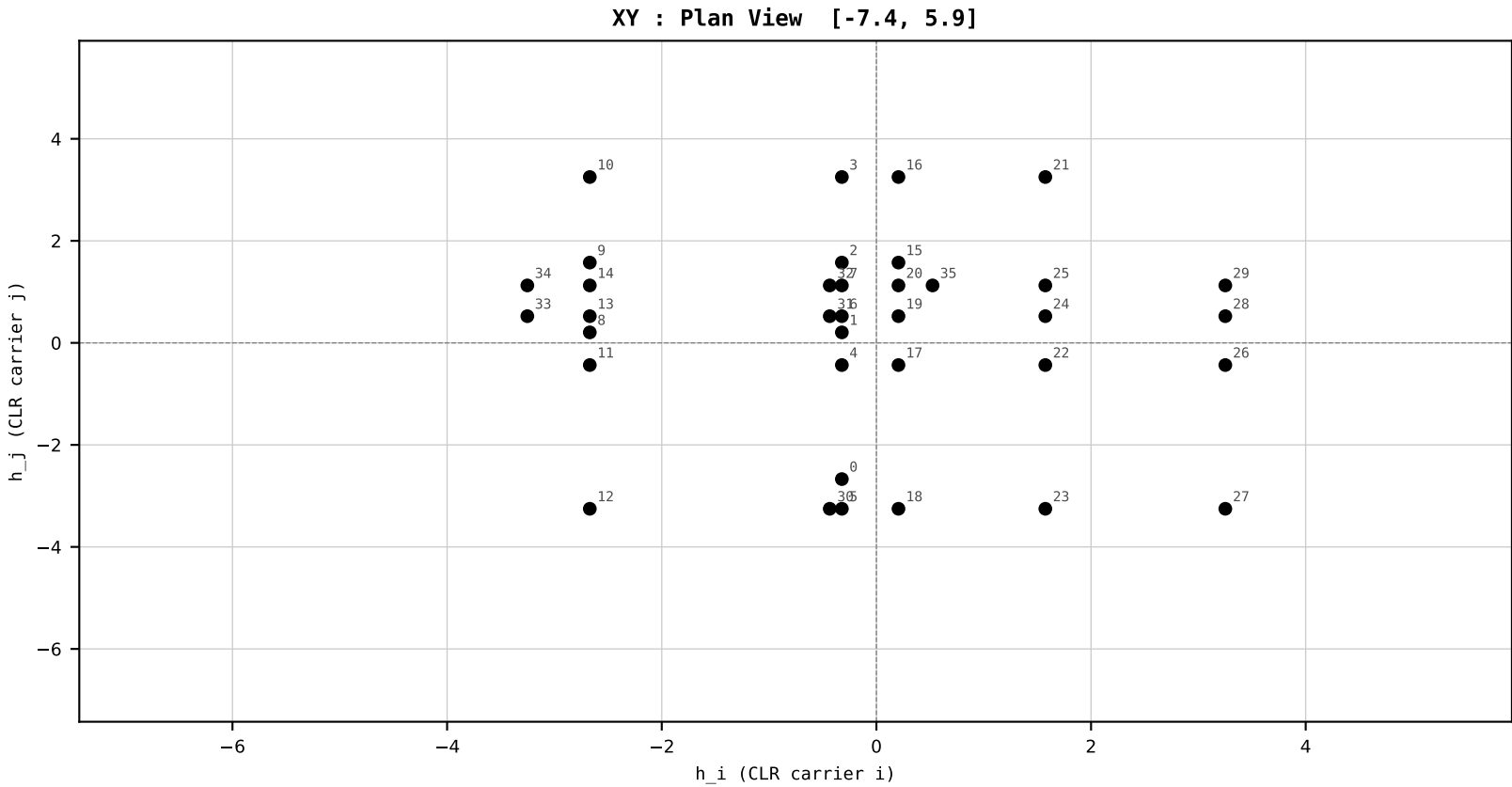
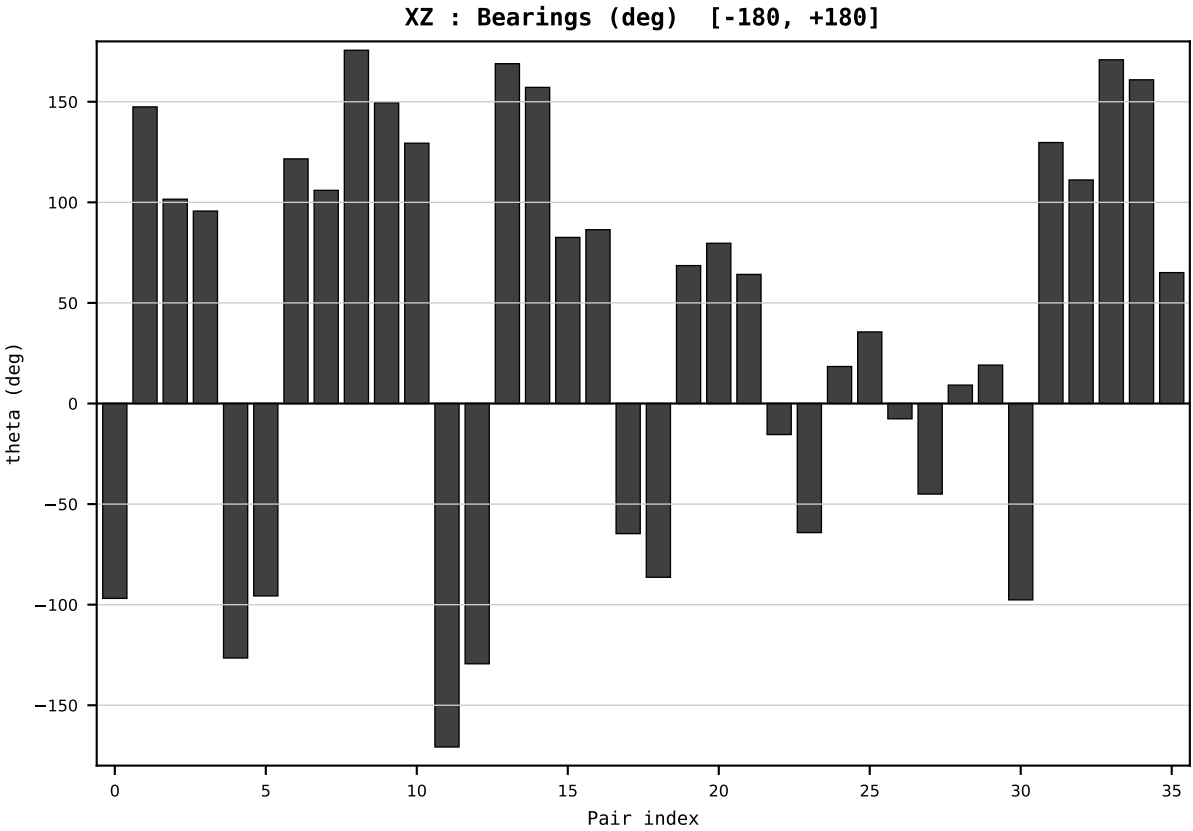
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_FRA_France_generation_TWh.csv
Year : 2024
t : 24 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.480863
Ring = Hs-3
E_metric= 32.6333
kappa_HS= 667.46
omega = 6.8989 deg
d_A = 0.727222
Helm = Gas
Helm d = -0.445163
DR = 6.503 HLR
DR ratio= 667.5x

XZ [-180,180]
CLR[-7.4,5.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=25 | Year 2025 | D=9 N=26 pairs=36

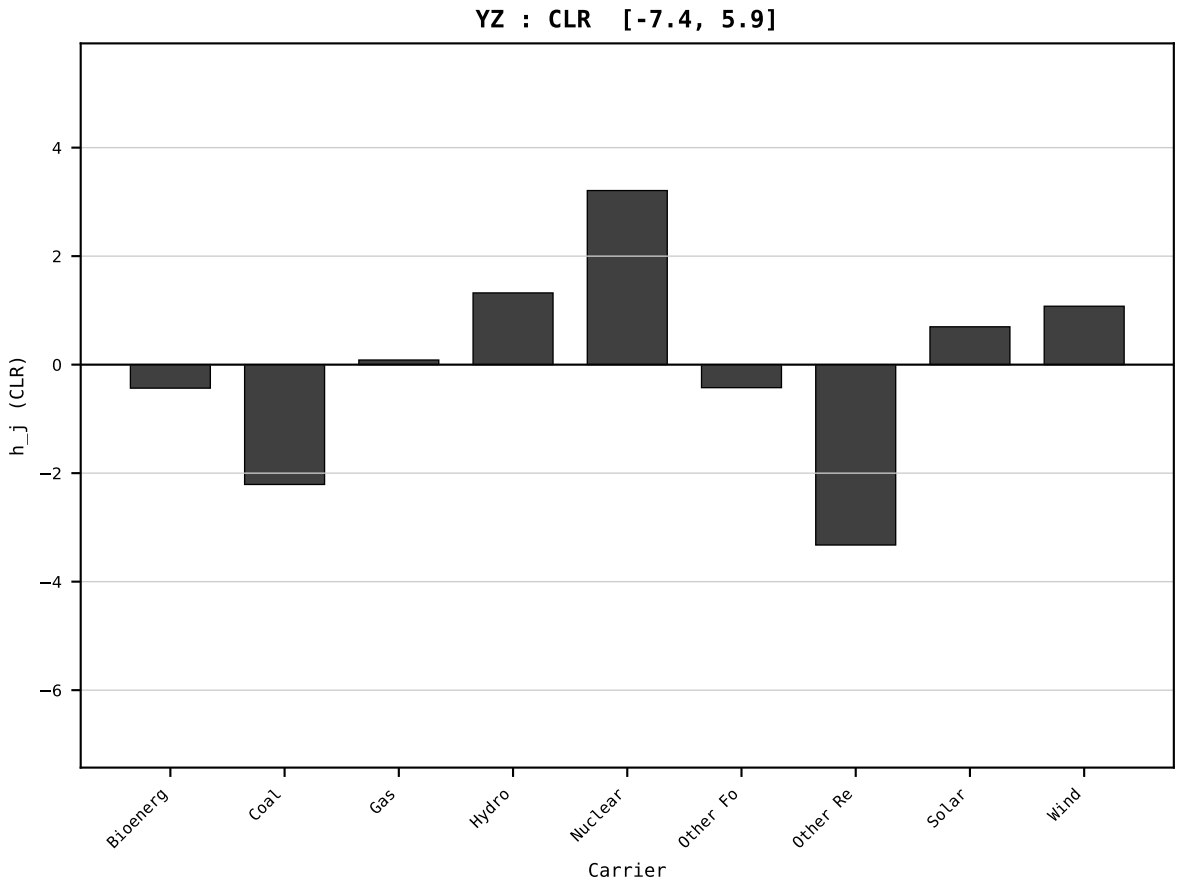
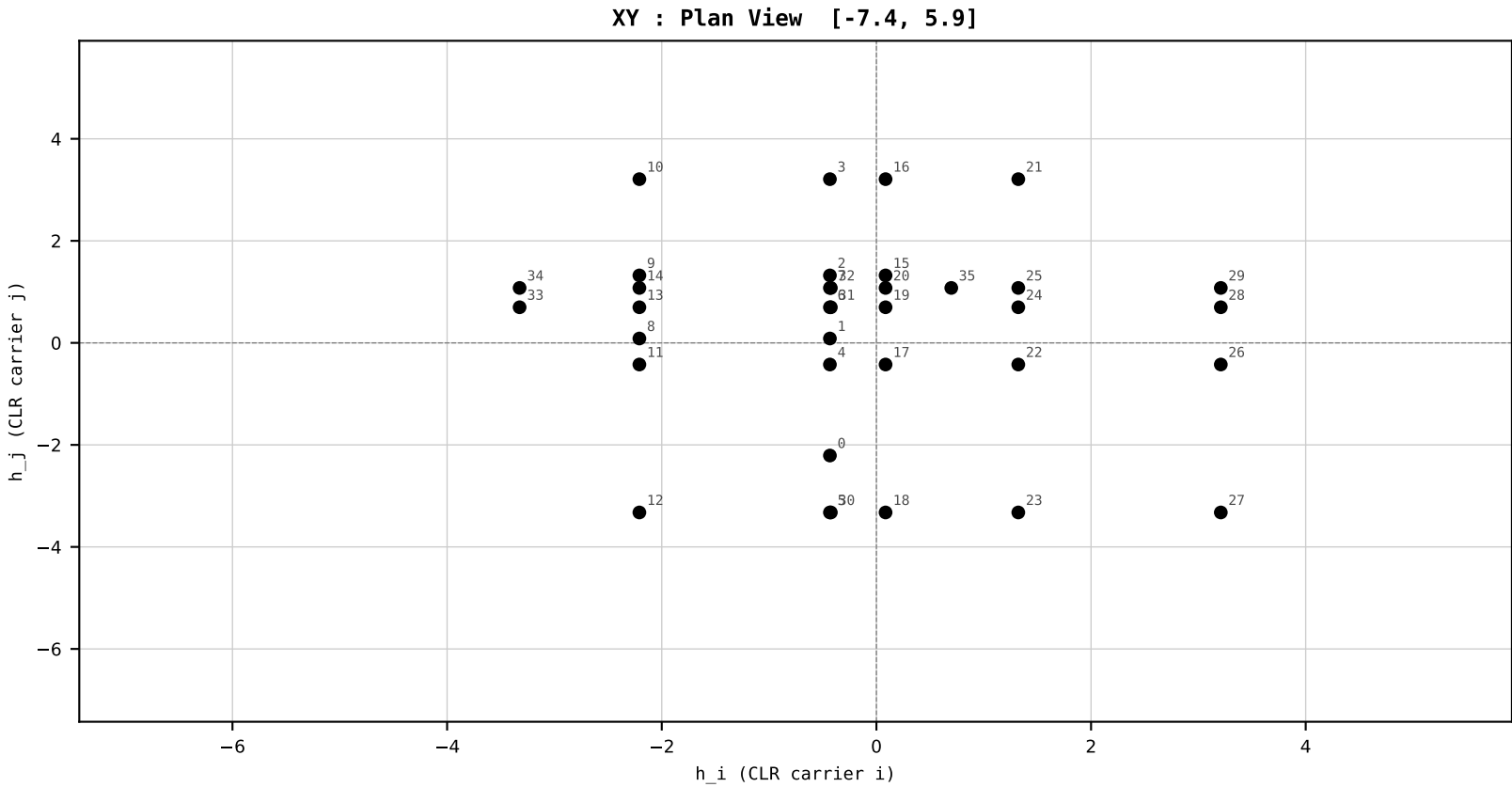
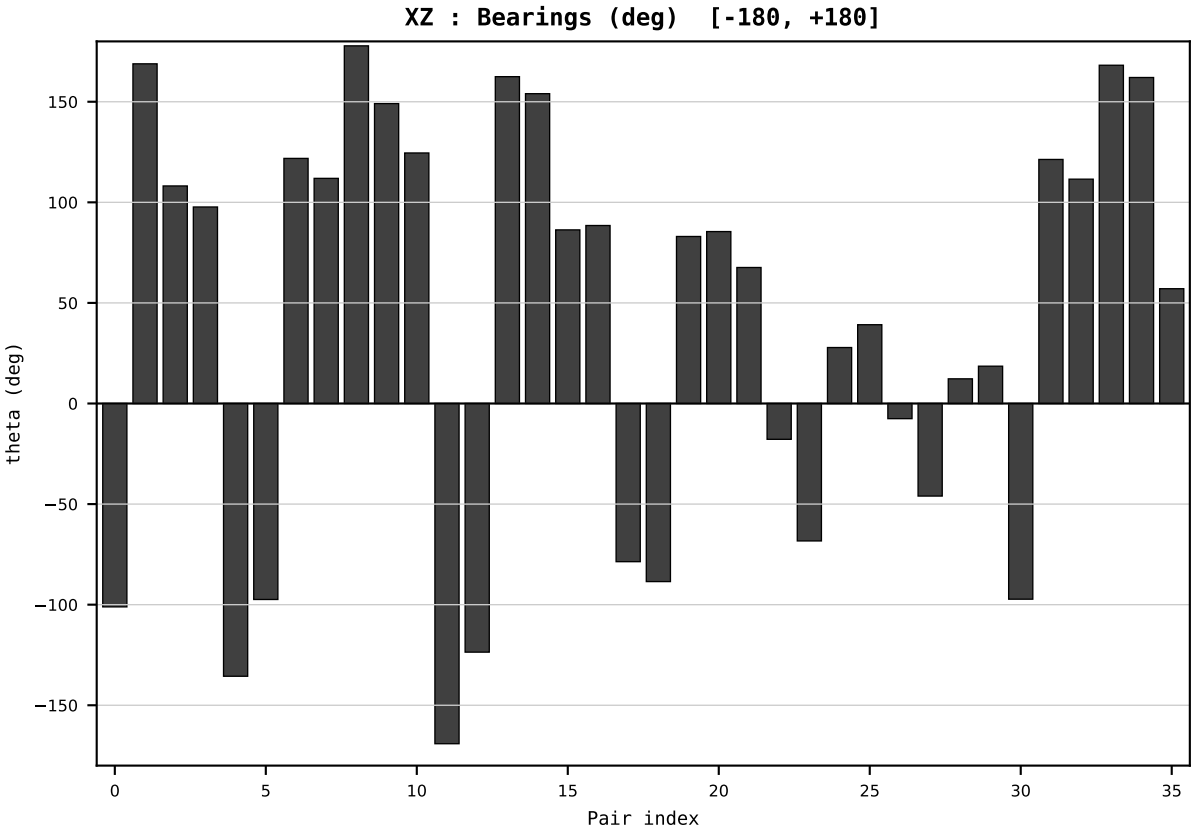
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

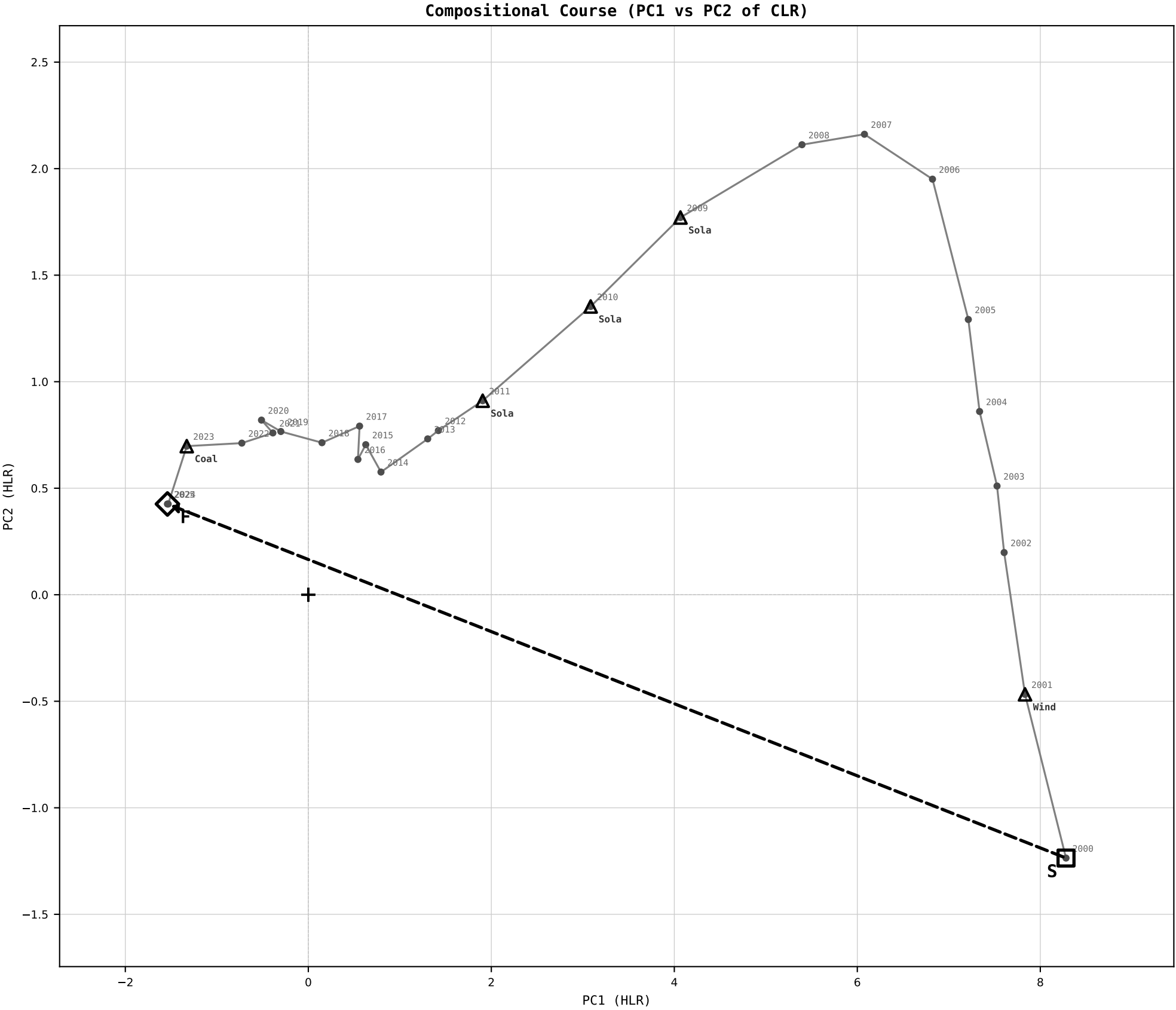
Dataset : ember_FRA_France_generation_TWh.csv
Year : 2025
t : 25 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.483790
Ring = Hs-3
E_metric= 29.9972
kappa_HS= 687.84
omega = 5.5004 deg
d_A = 0.586196
Helm = Coal
Helm d = +0.461974
DR = 6.534 HLR
DR ratio= 687.8x

XZ [-180,180]
CLR[-7.4,5.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind





COURSE METRICS

Scale Provenance:
Y_j → x_j=Y_j/T → h_j=CLR
→ PC projection
Unit: HLR (Higgins Log-Ratio)

Net distance = 10.0035 HLR
Path length = 17.9314 HLR
Directness = 0.5579

DR start = 10.634 HLR
ratio = 41516.0x
DR final = 6.534 HLR
ratio = 687.8x

E_{metric} start = 94.4242
E_{metric} final = 29.9972

PC1 loadings:
Solar -0.768
Coal +0.387
Wind -0.385

PC2 loadings:
Wind +0.841
Solar -0.448
Hydro -0.160

Top displacement events:
2009 Solar d=1.437
2011 Solar d=1.337
2010 Solar d=1.129
2023 Coal d=1.095
2001 Wind d=0.998

The instrument reads.
The expert decides.
The loop stays open.

View B

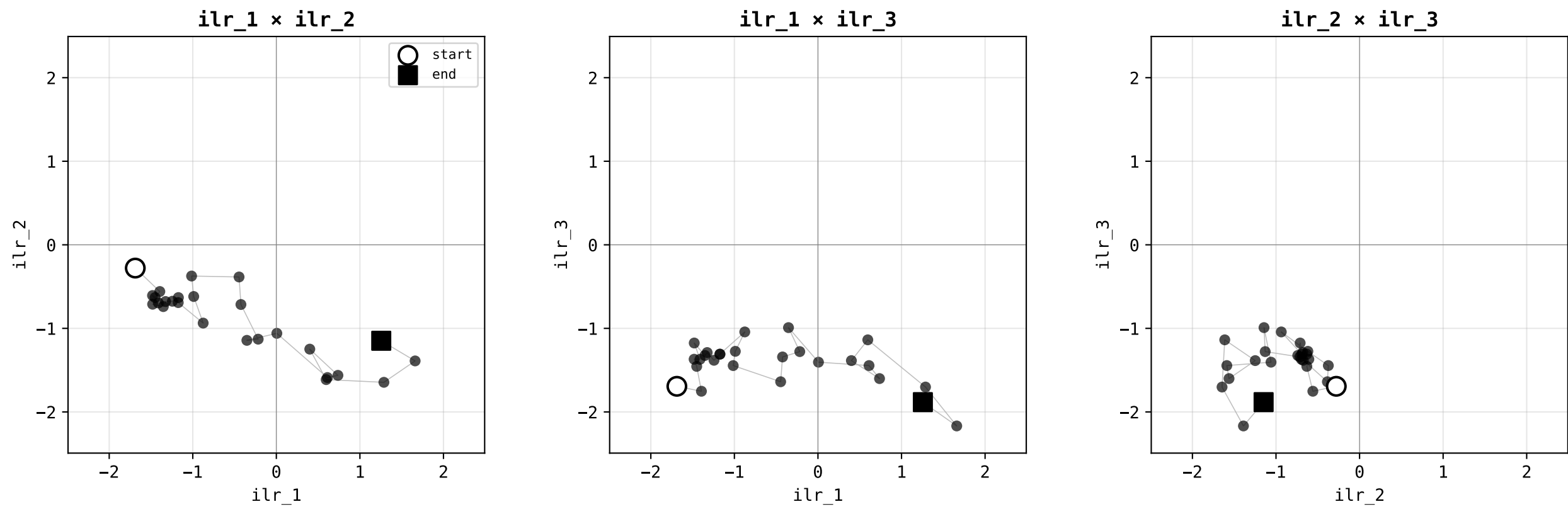
ILR-Helmert Orthogonal Triplet Plate

Orthonormal reading · three scatter faces of first three Helmert ILR axes

§ FRA France

ILR-Helmert Orthogonal Triplet – FRA France · Stage 1 Order-1 plate

Three orthogonal scatter projections of CLR onto the first three Helmert ILR axes · N=26 years · D=9 carriers



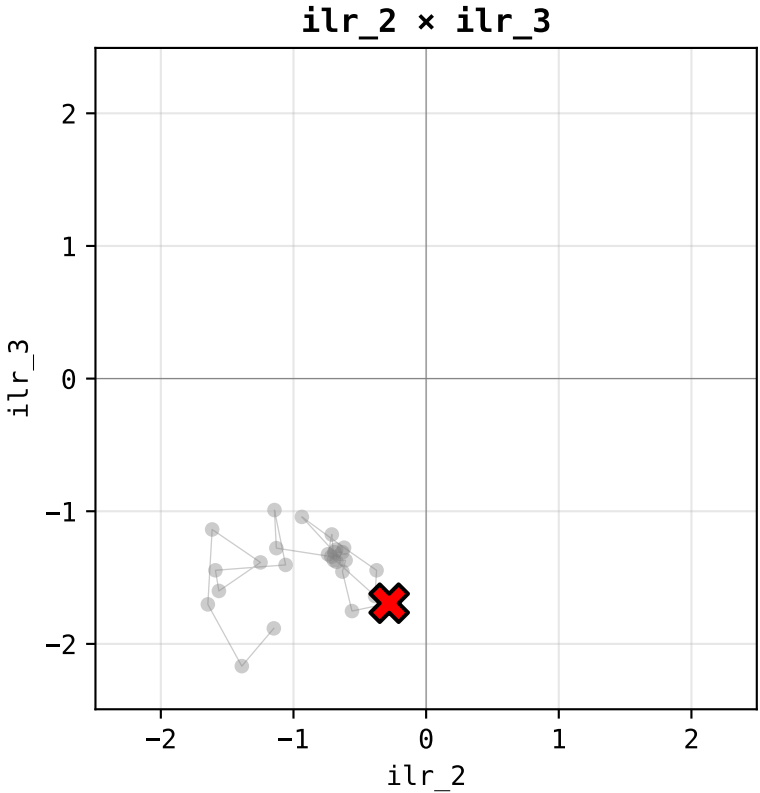
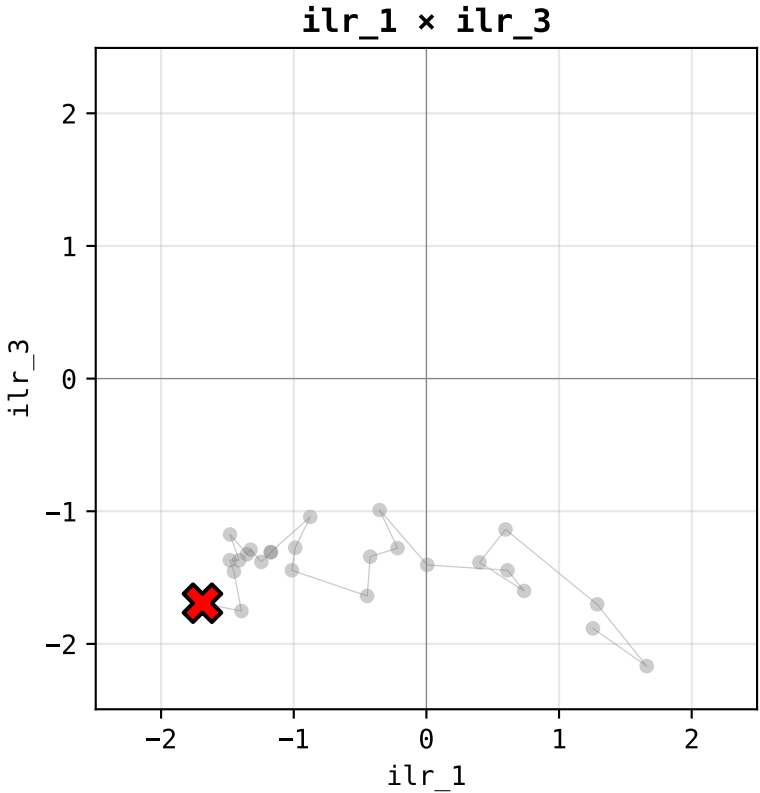
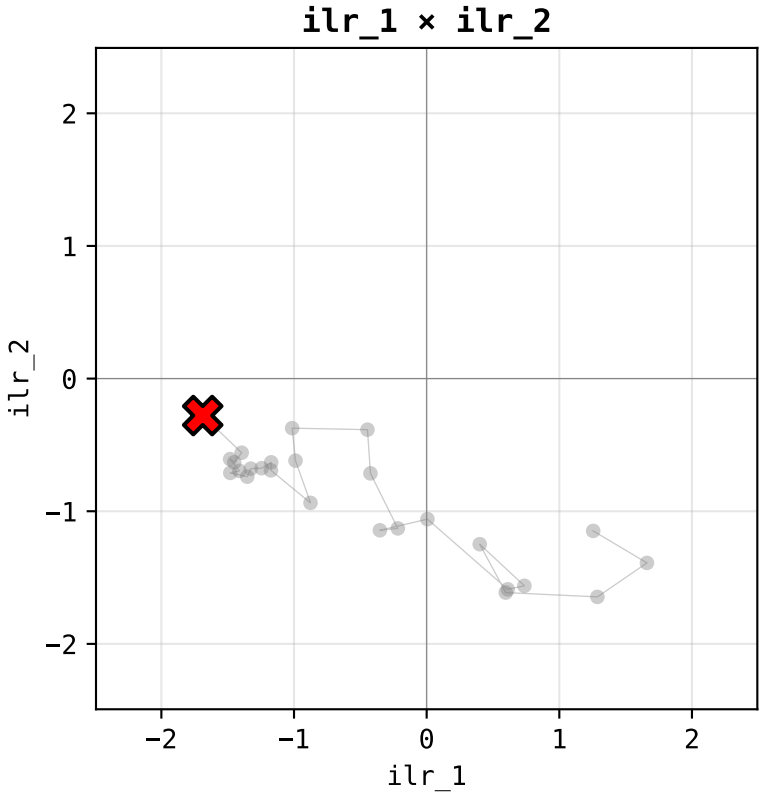
Helmert basis loadings (which carriers each ILR axis contrasts):

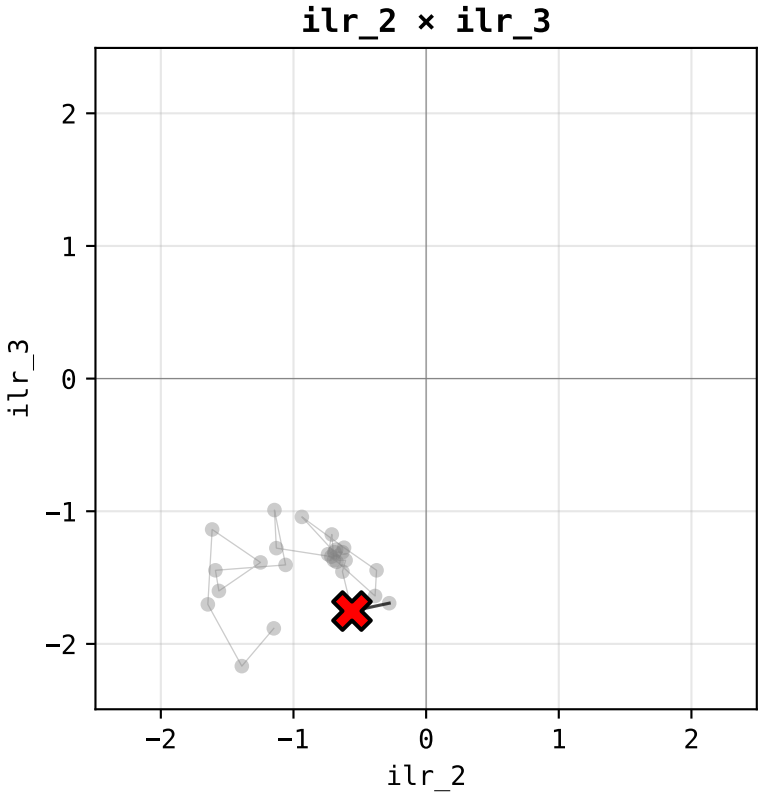
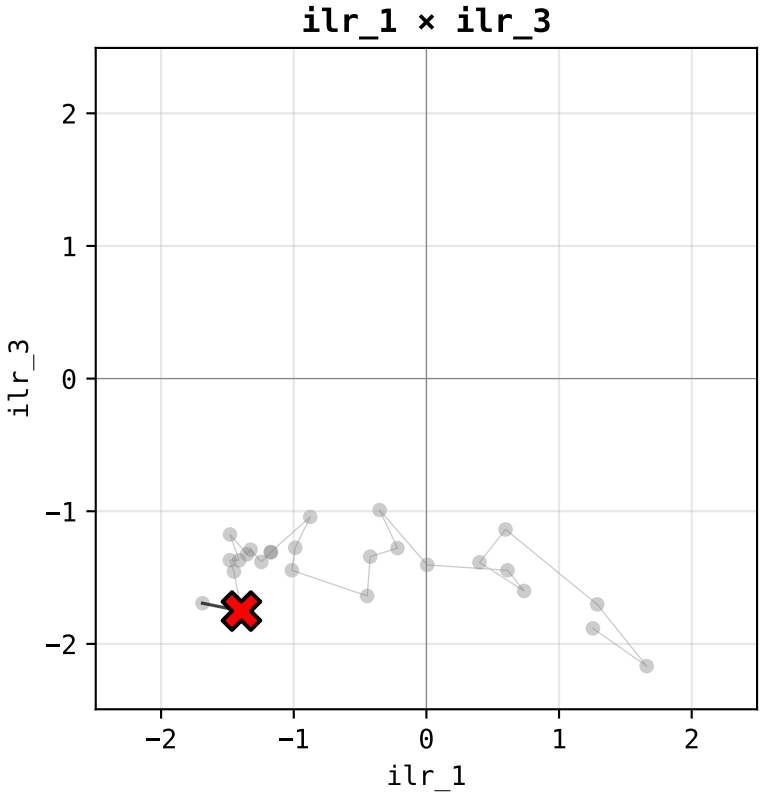
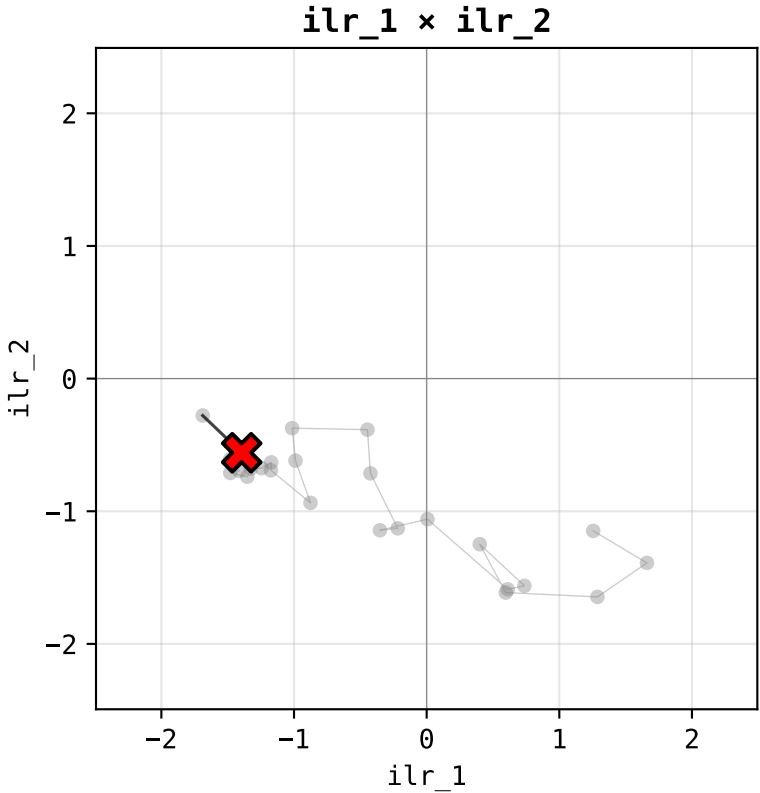
ilr_1: (Bioenergy) vs Coal
ilr_2: (Bioenergy + Coal) vs Gas
ilr_3: (Bioenergy + Coal + Gas) vs Hydro

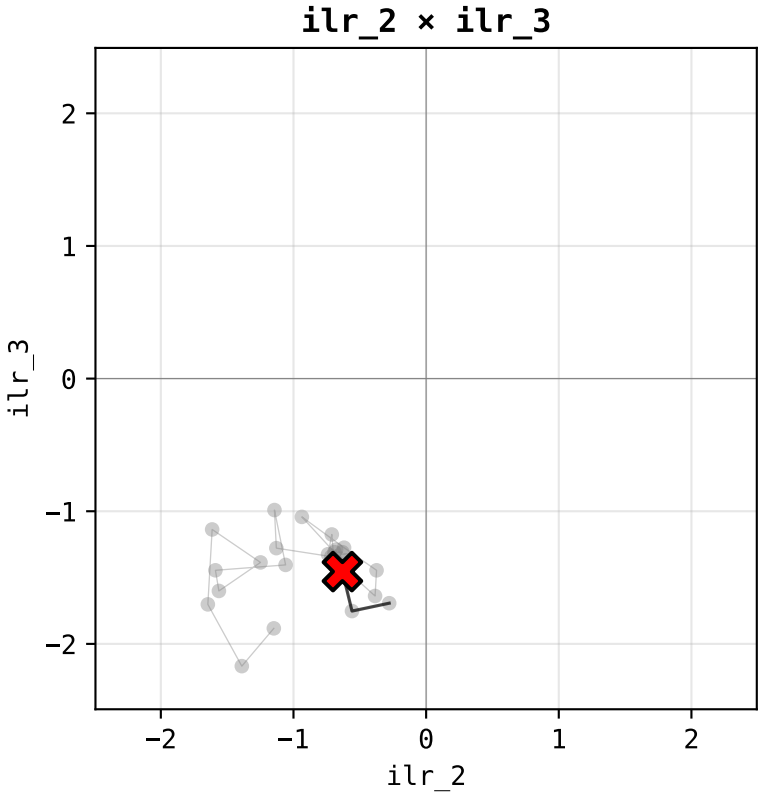
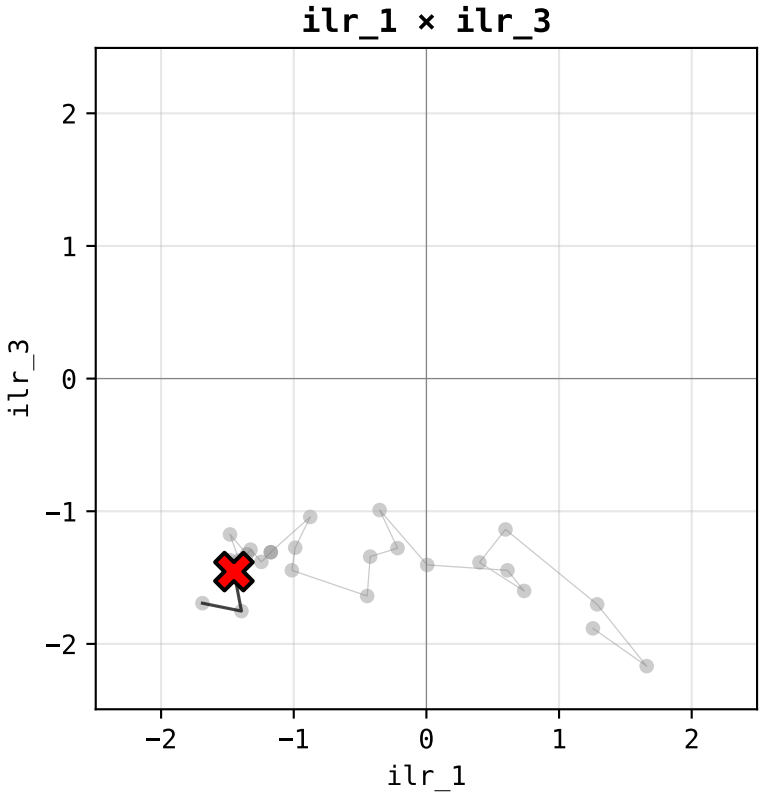
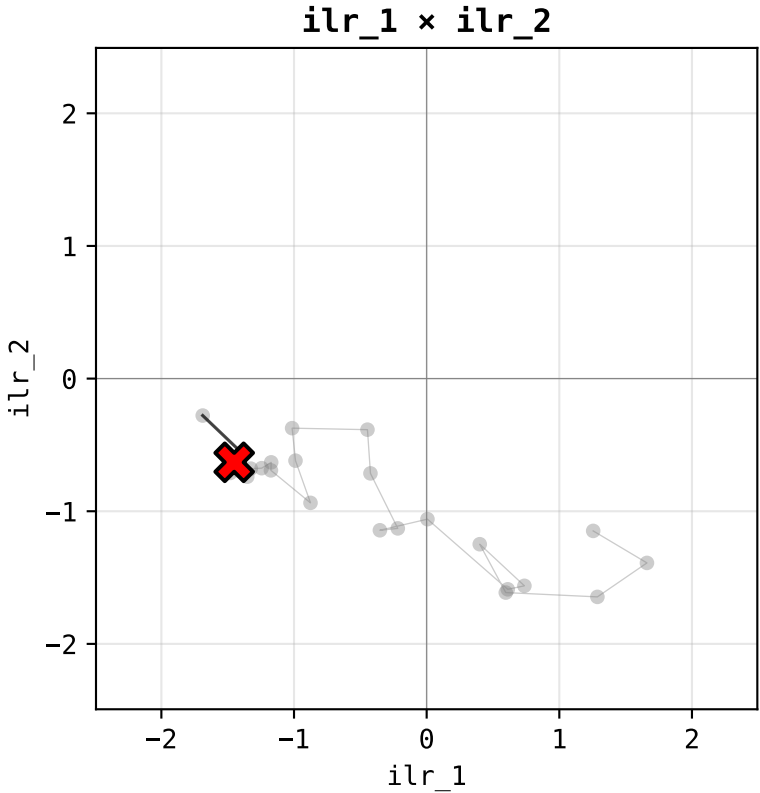
Trajectory:
start year: 2000
end year: 2025
N readings: 26

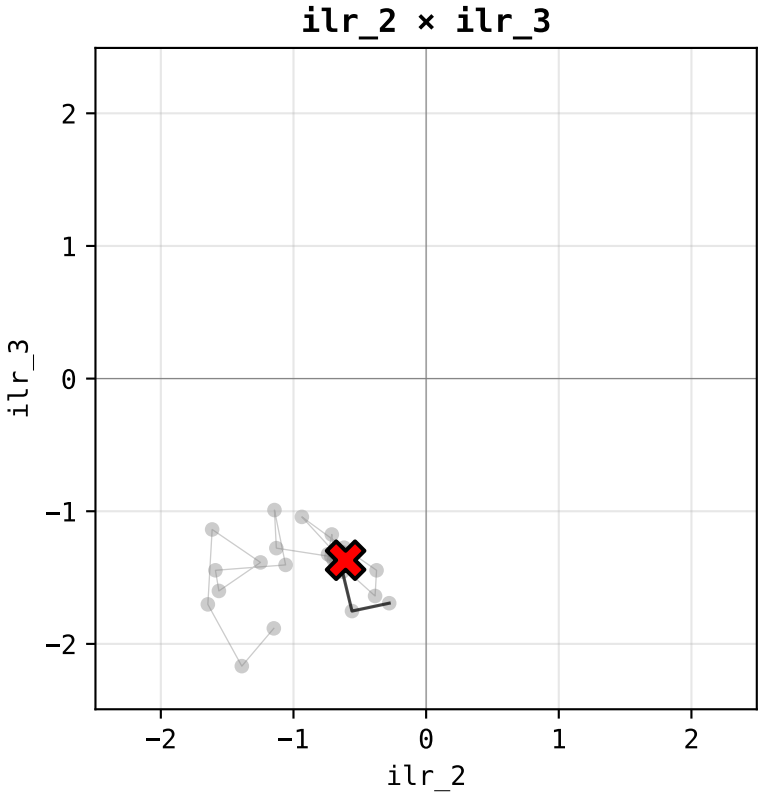
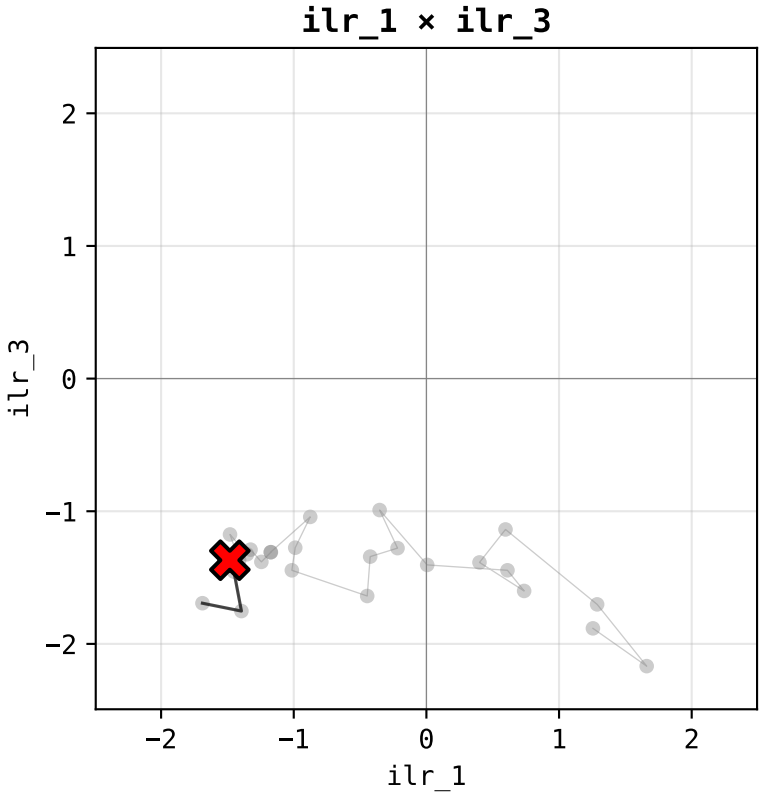
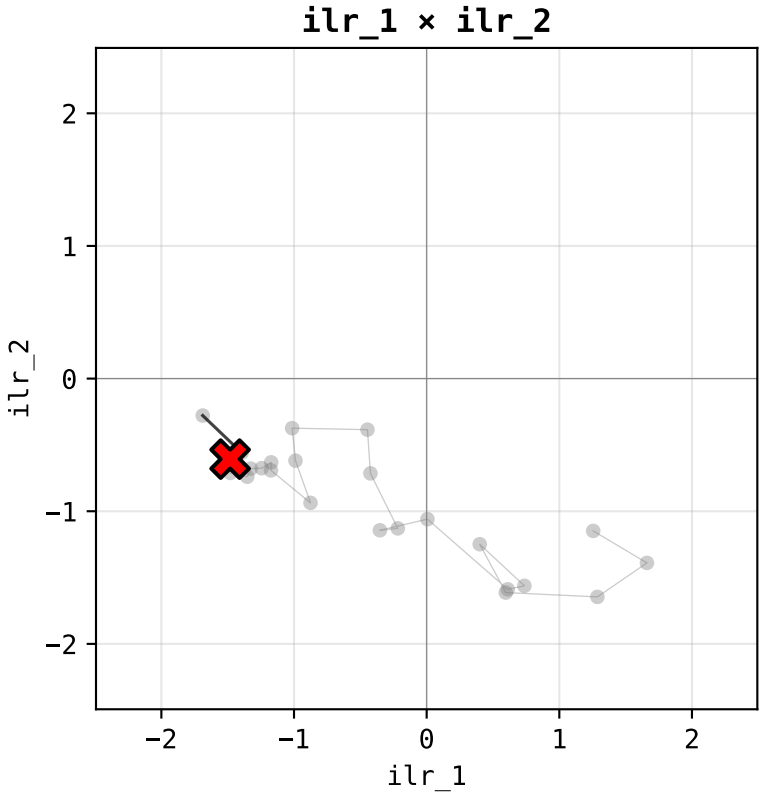
Display:
fixed scale ±2.49 ILR
Order 1 (first principles)
per Output Doctrine v1.0

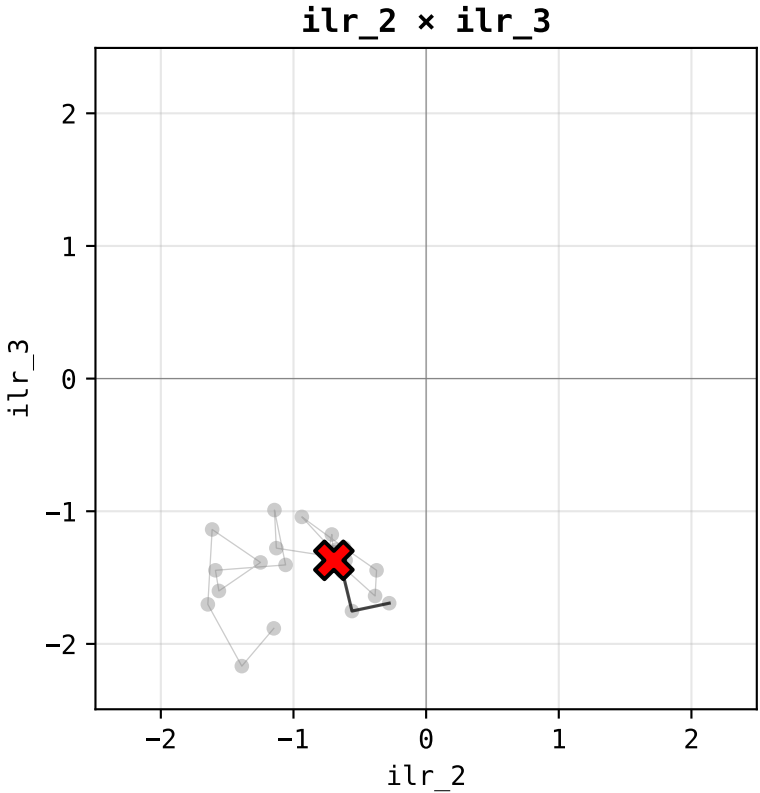
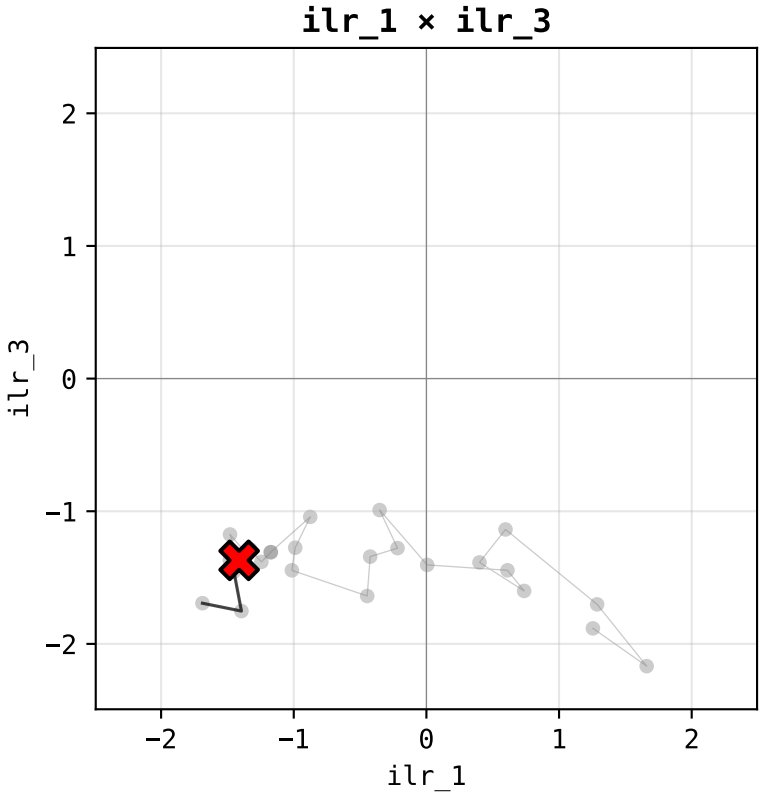
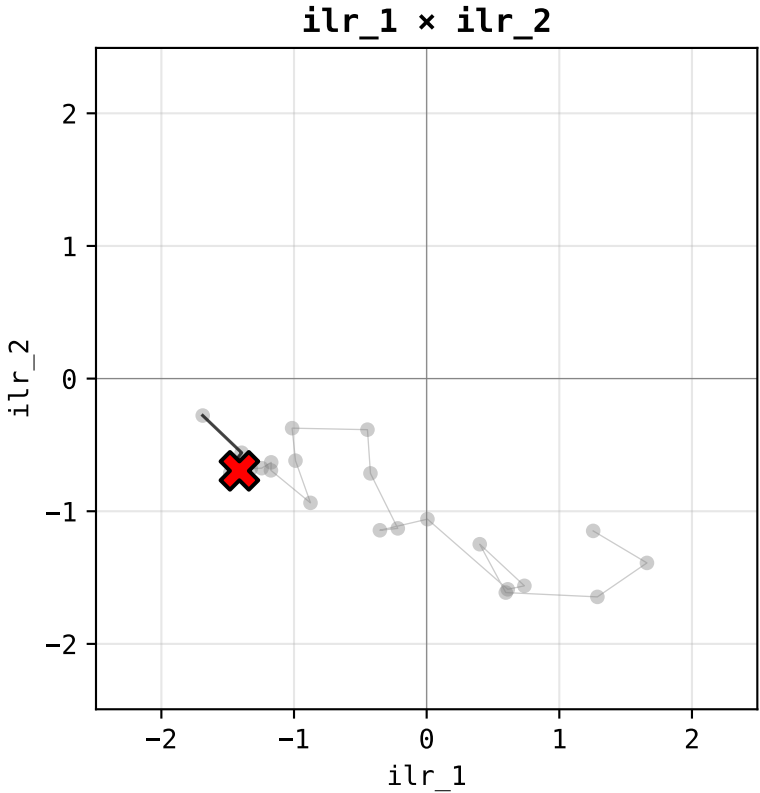
Symbology:
○ start · ■ end
· intermediate years

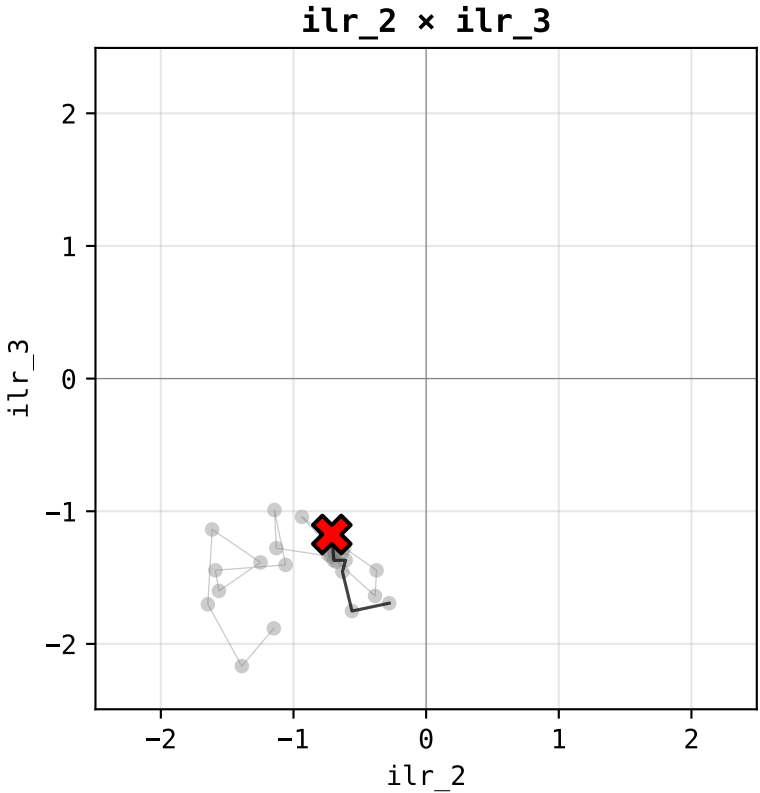
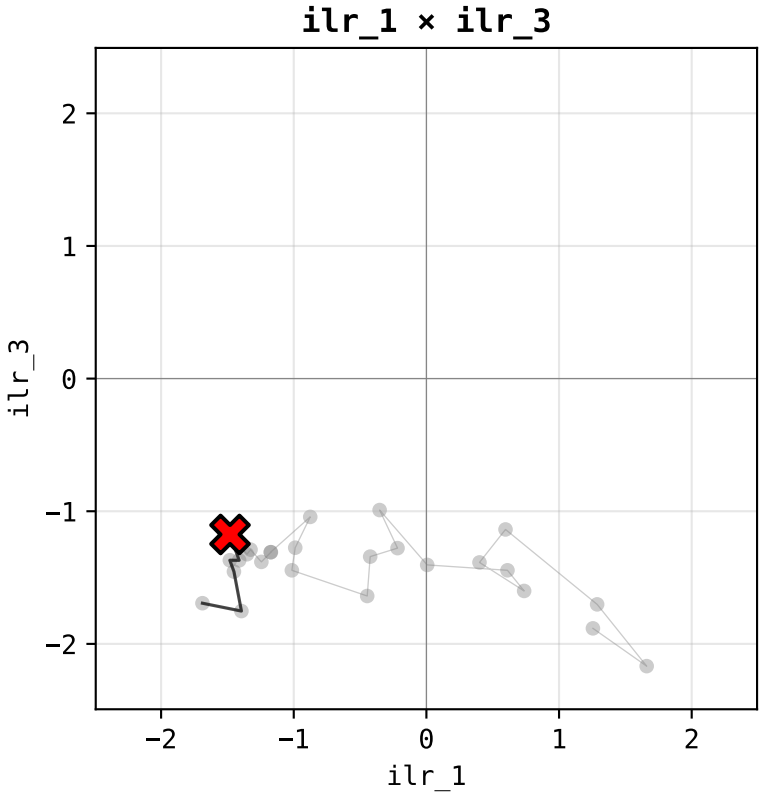
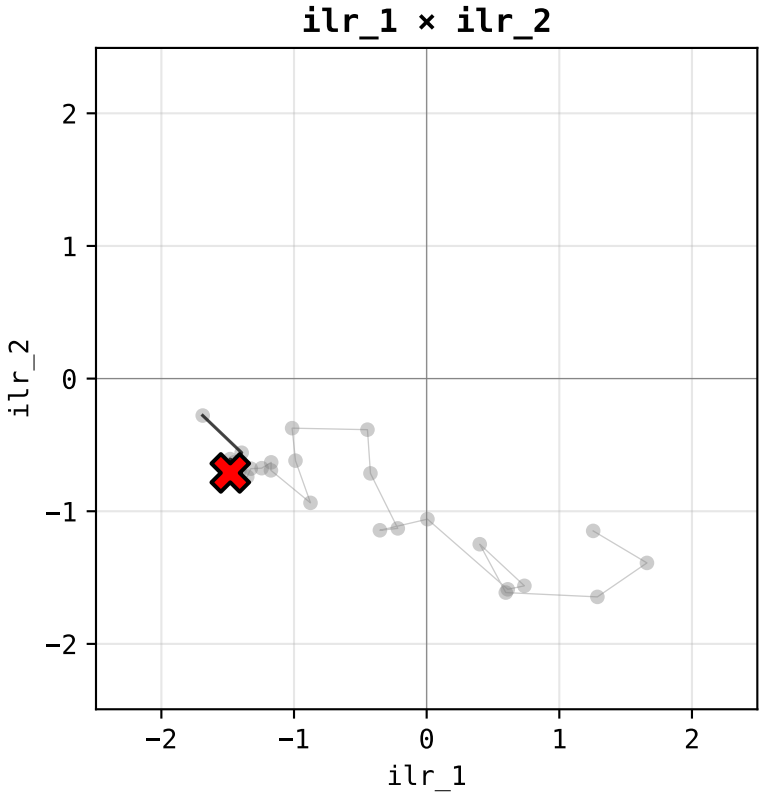


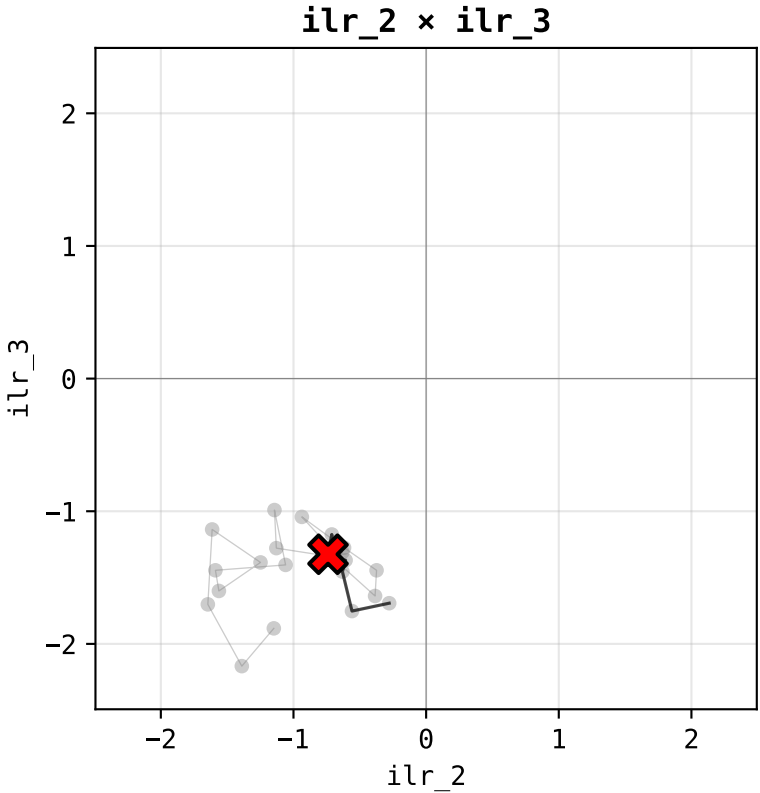
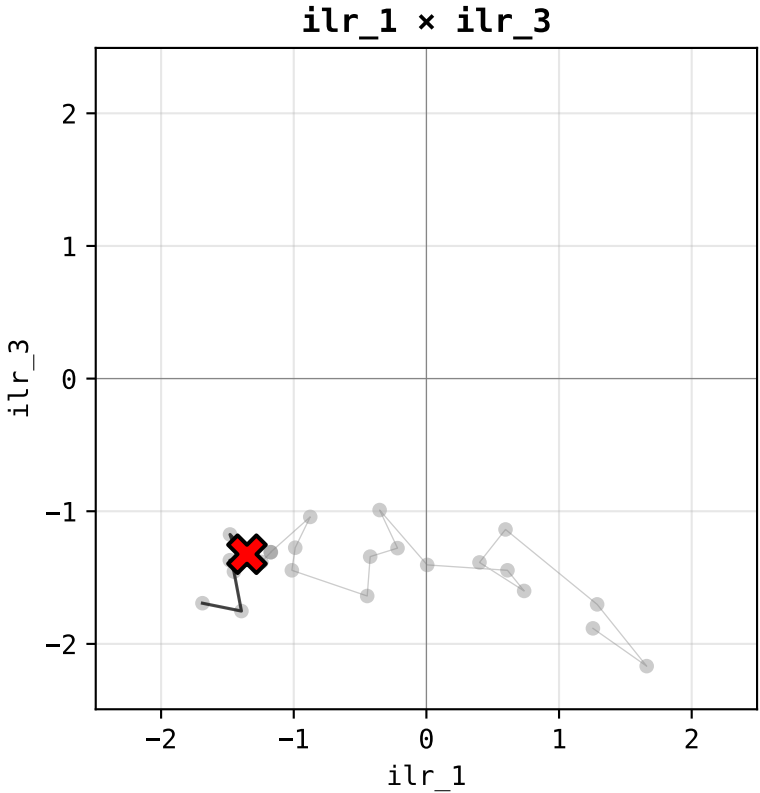
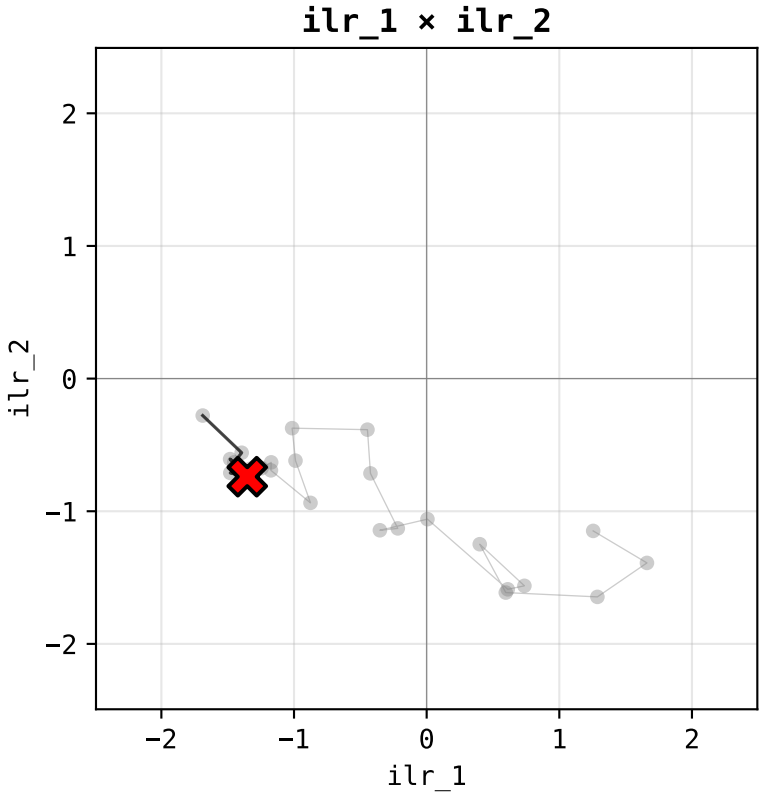


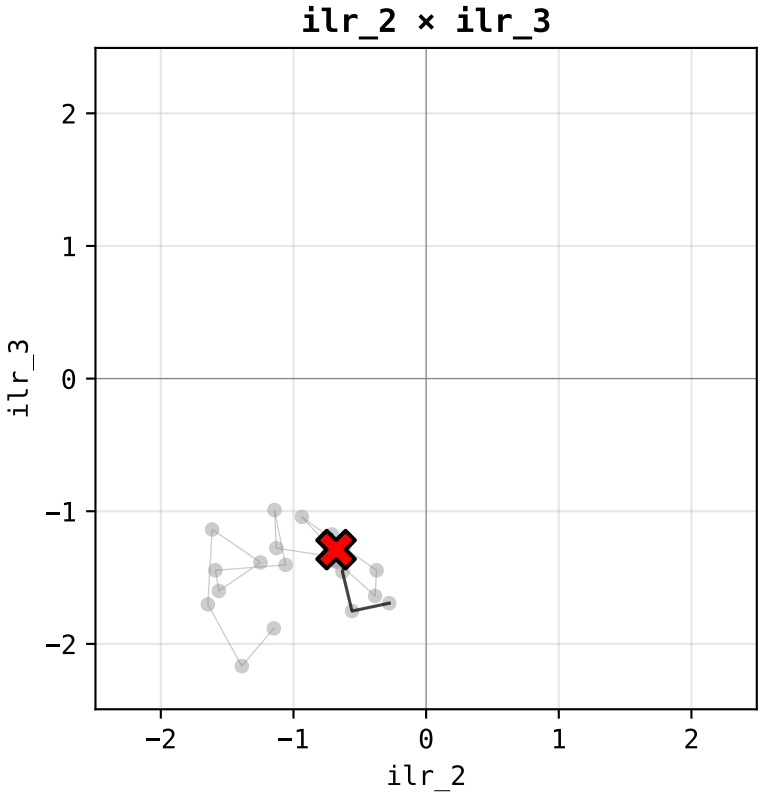
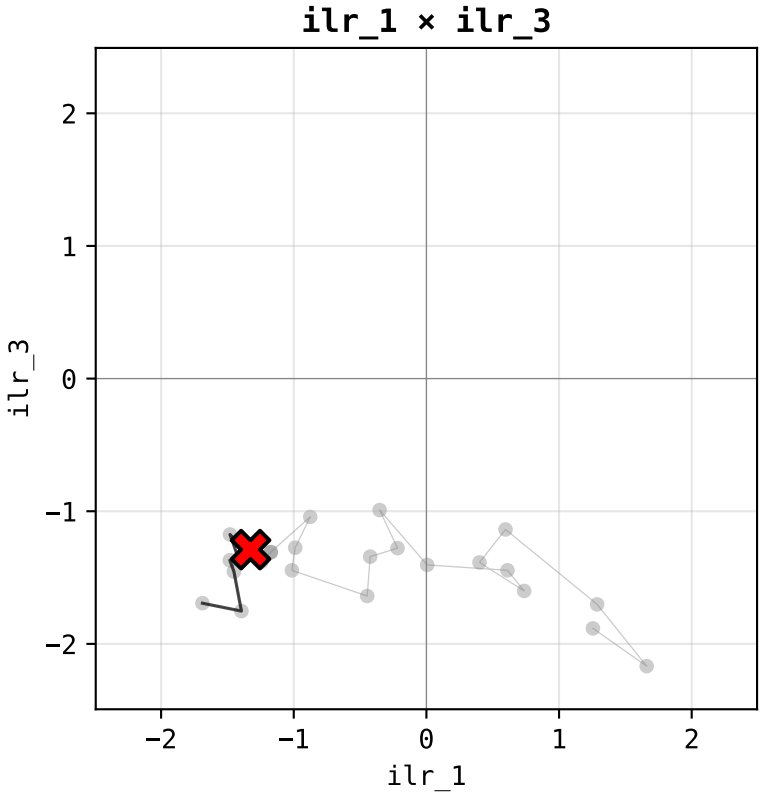
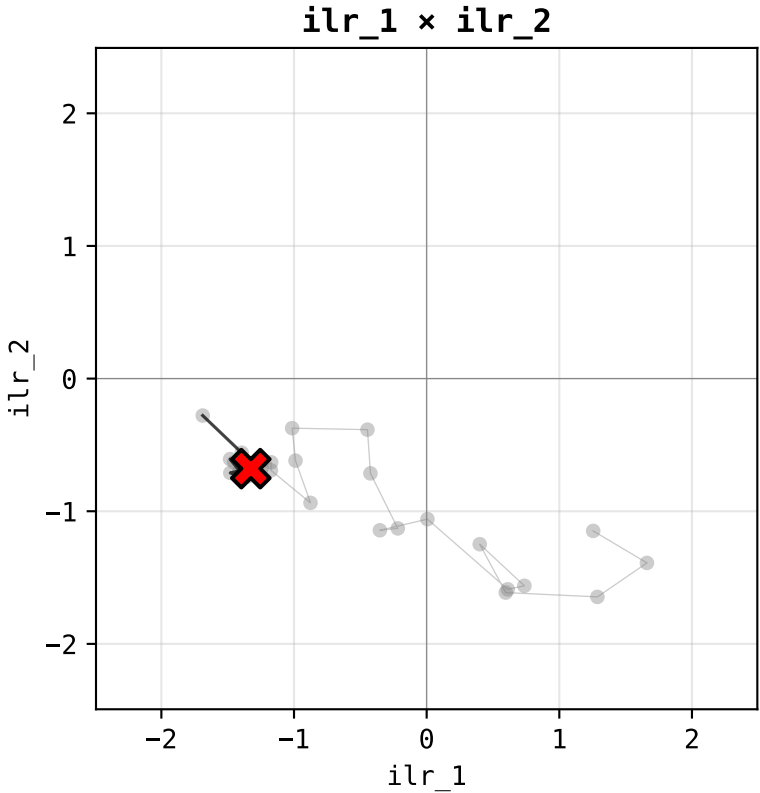


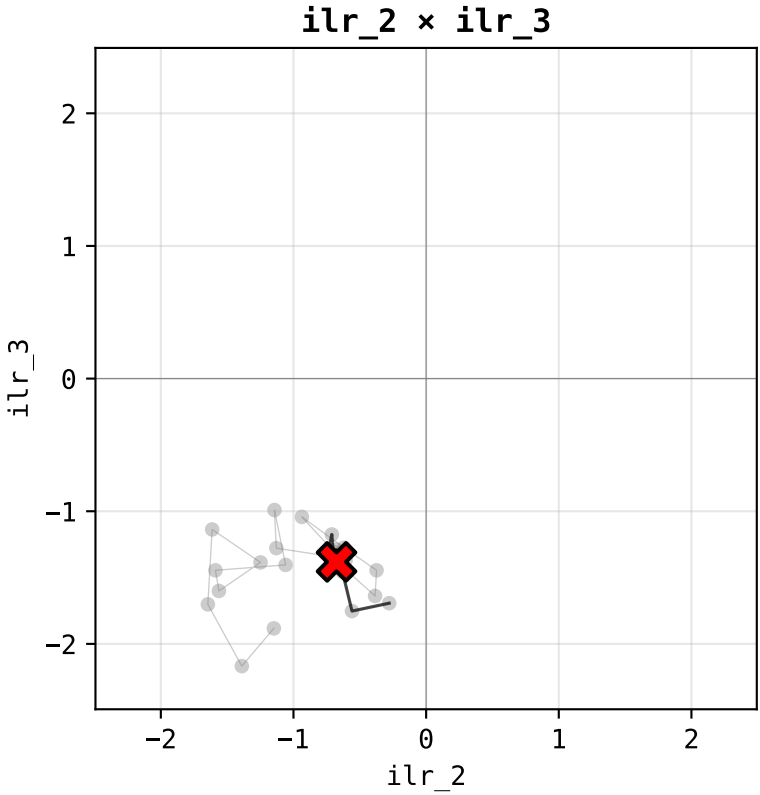
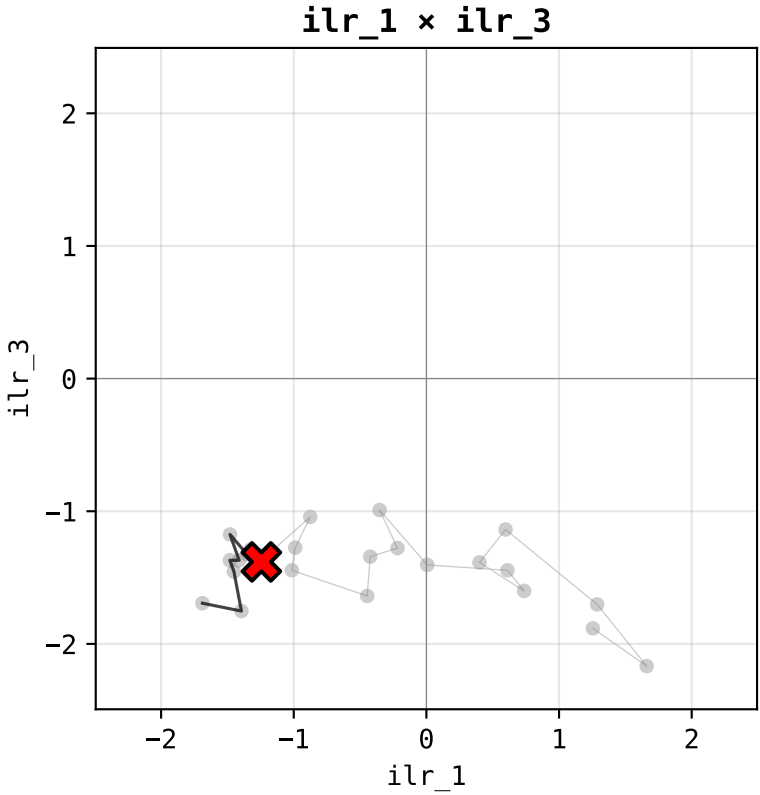
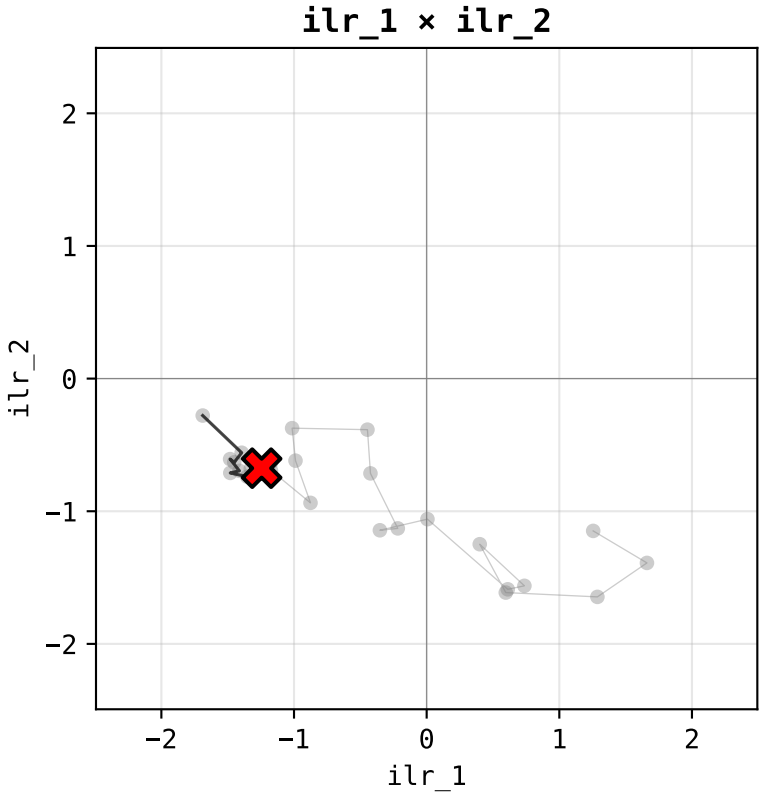




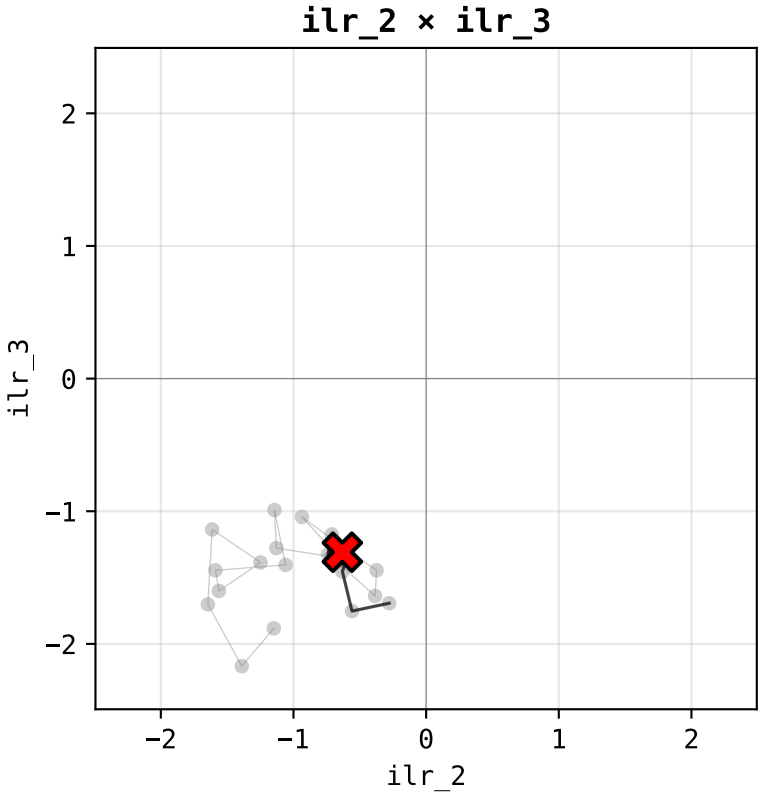
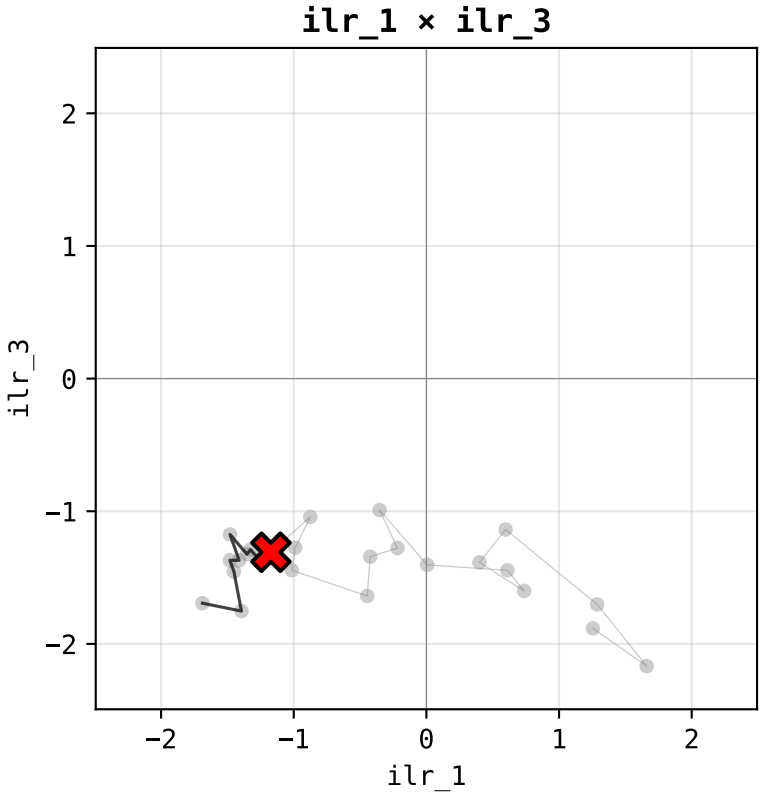
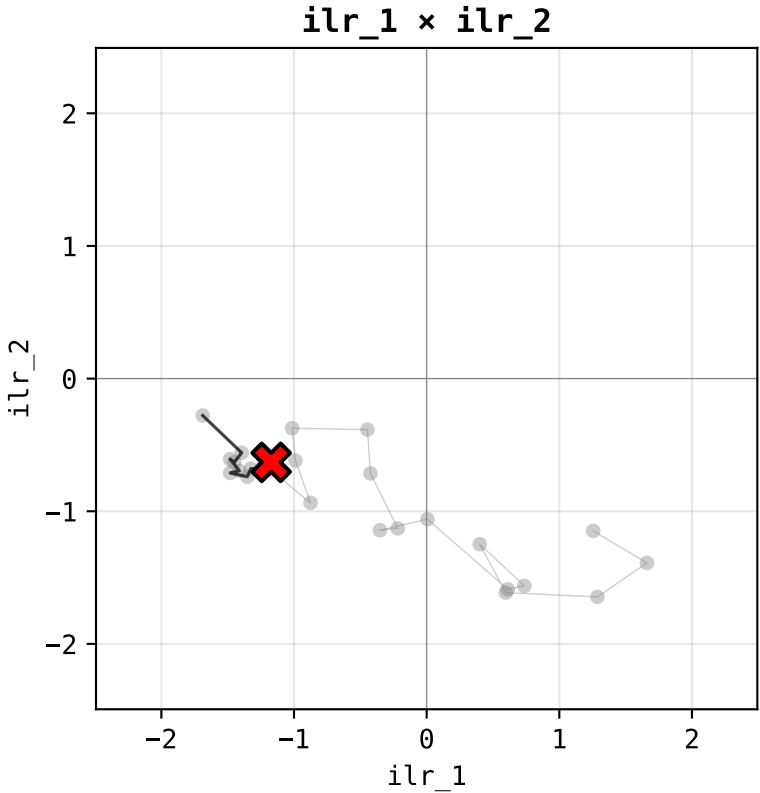




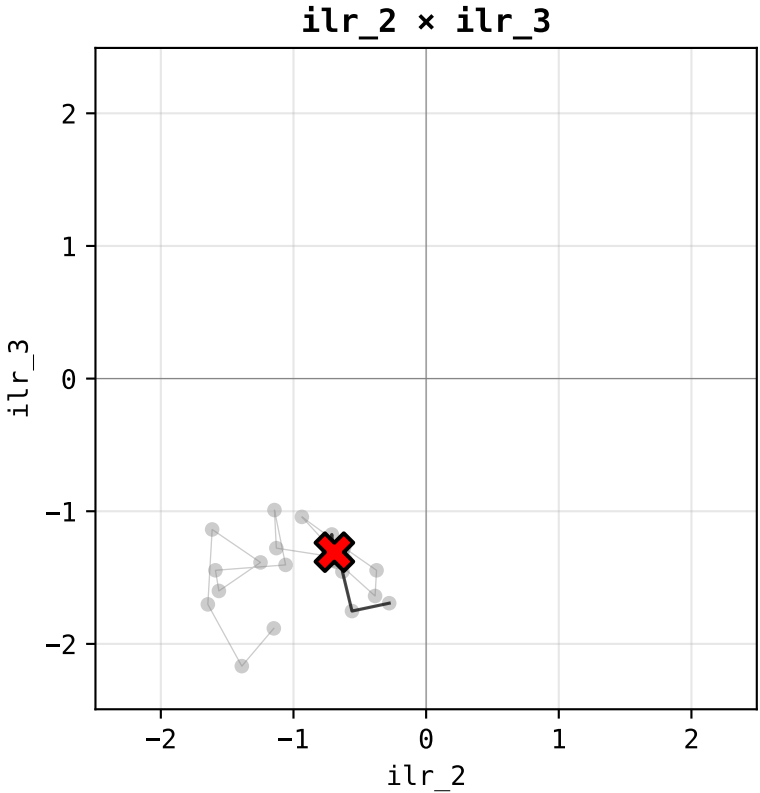
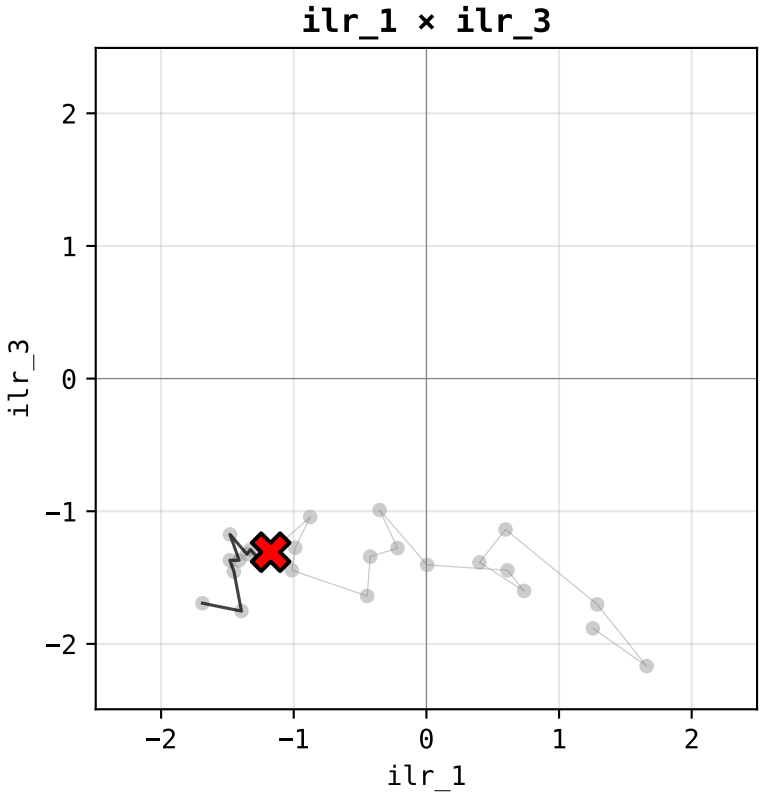
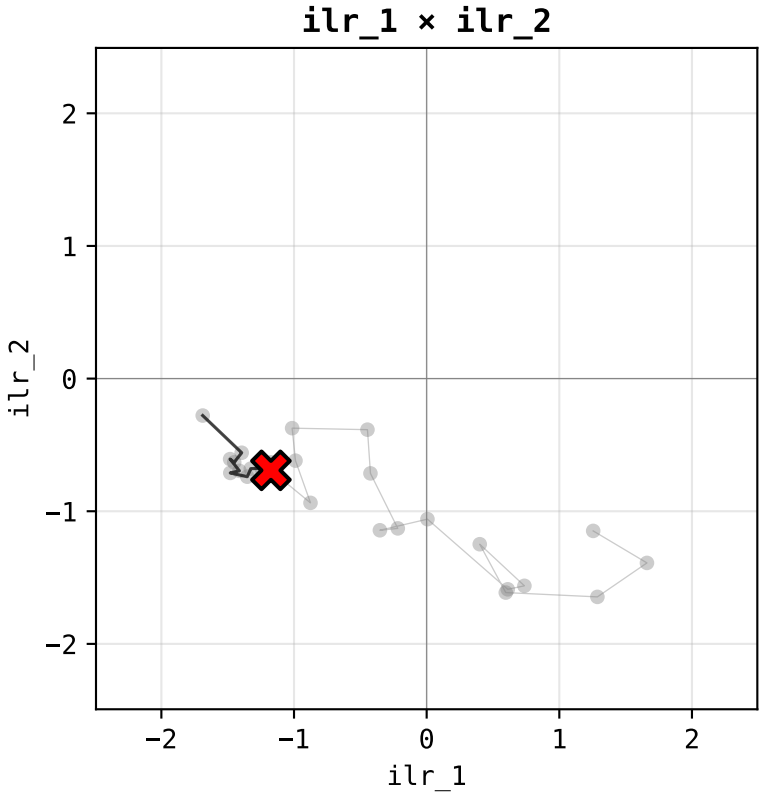


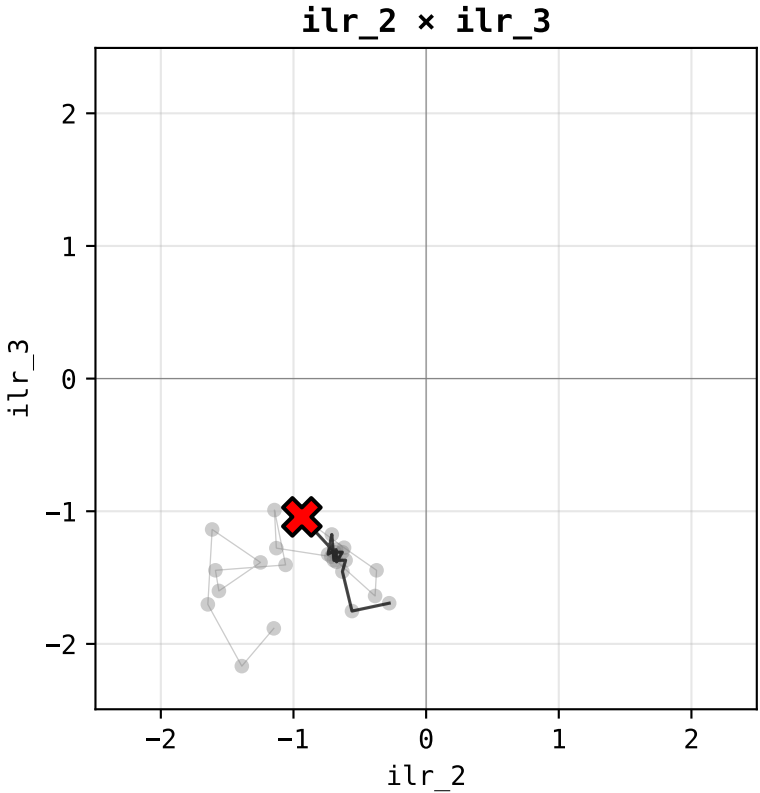
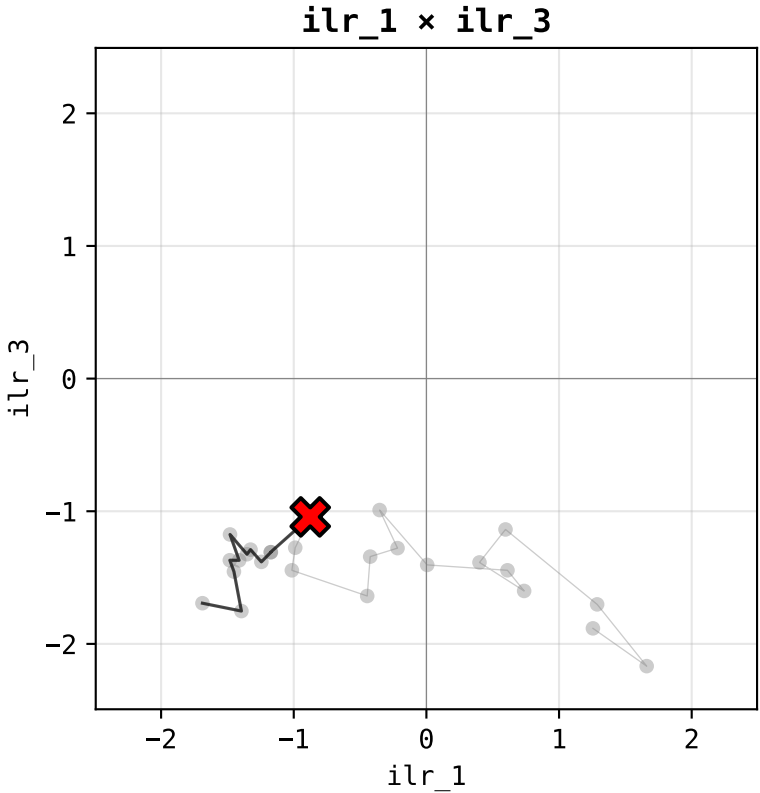
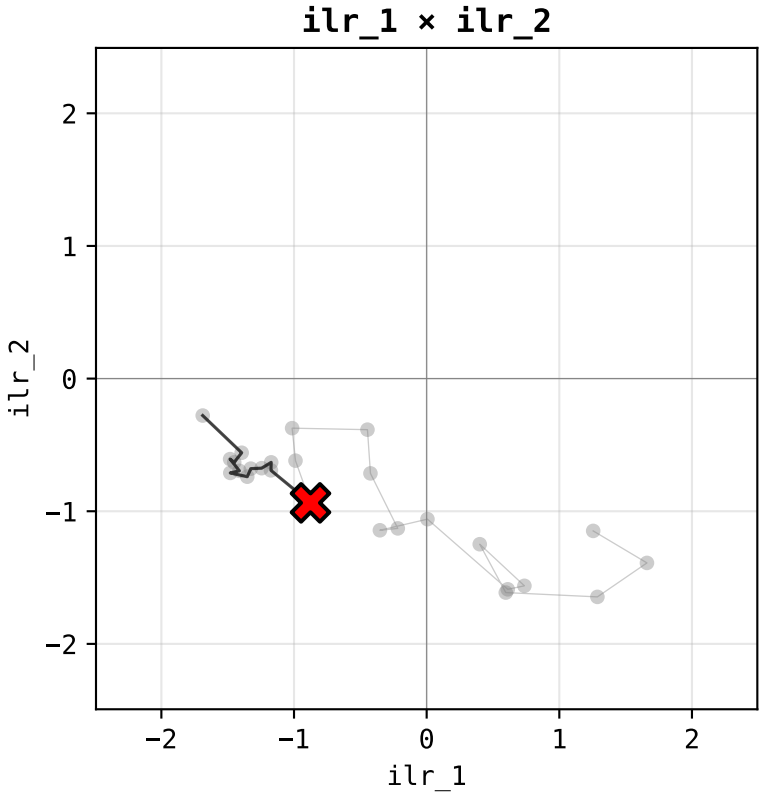


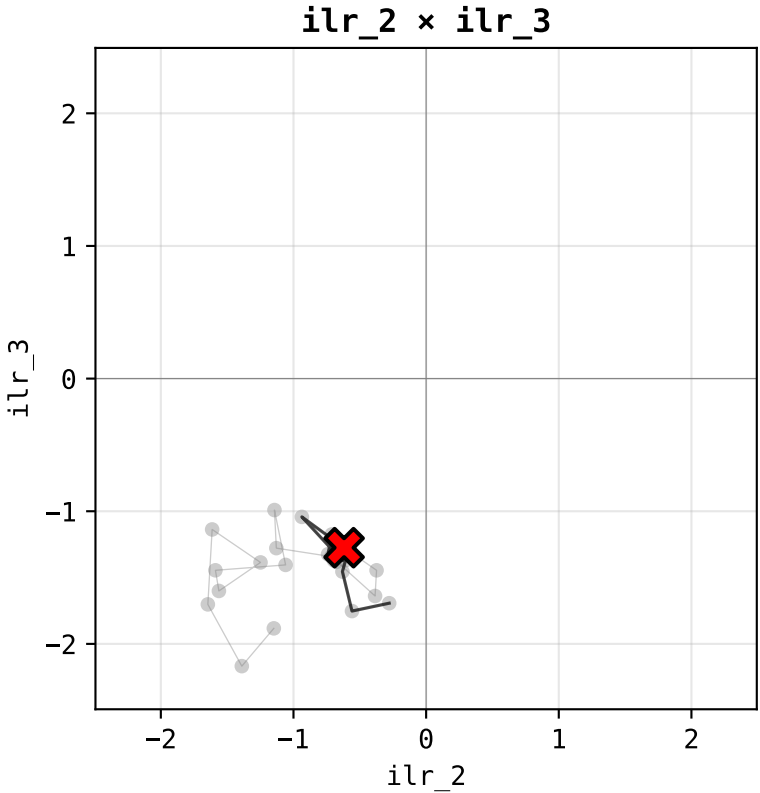
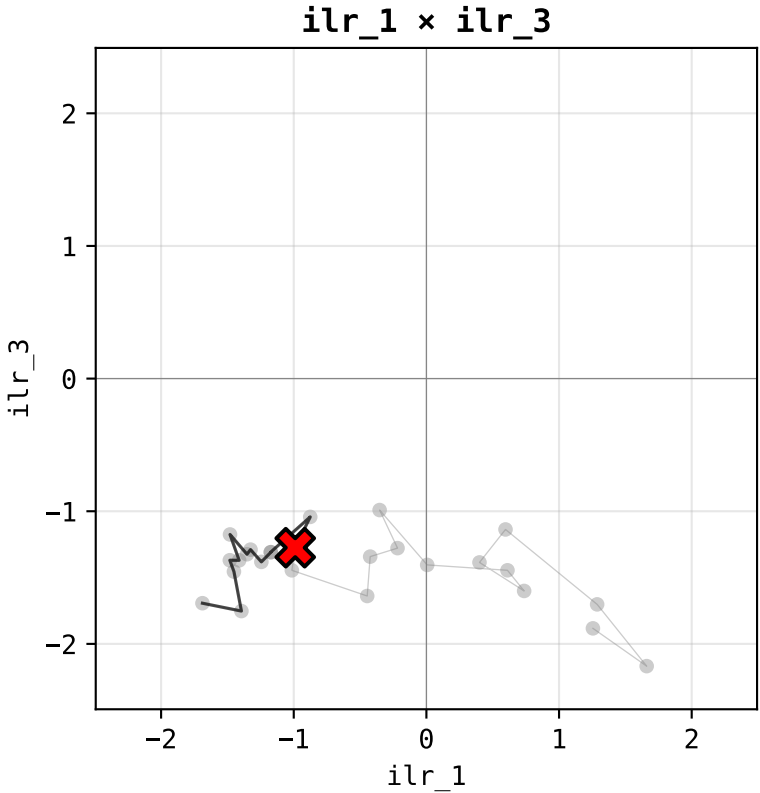
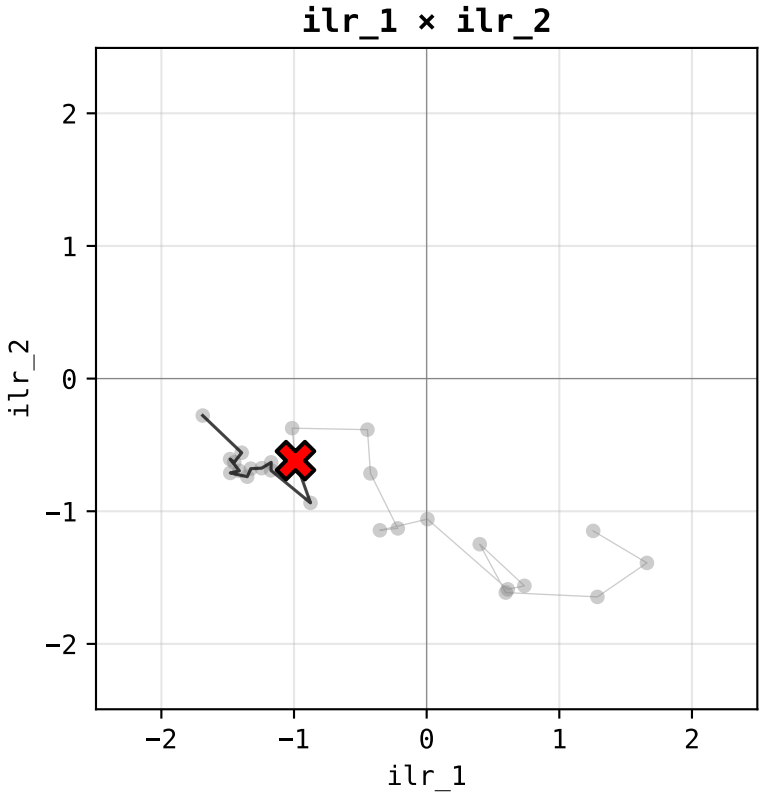
ILR Triplet – FRA France · year 2009 (10/26)

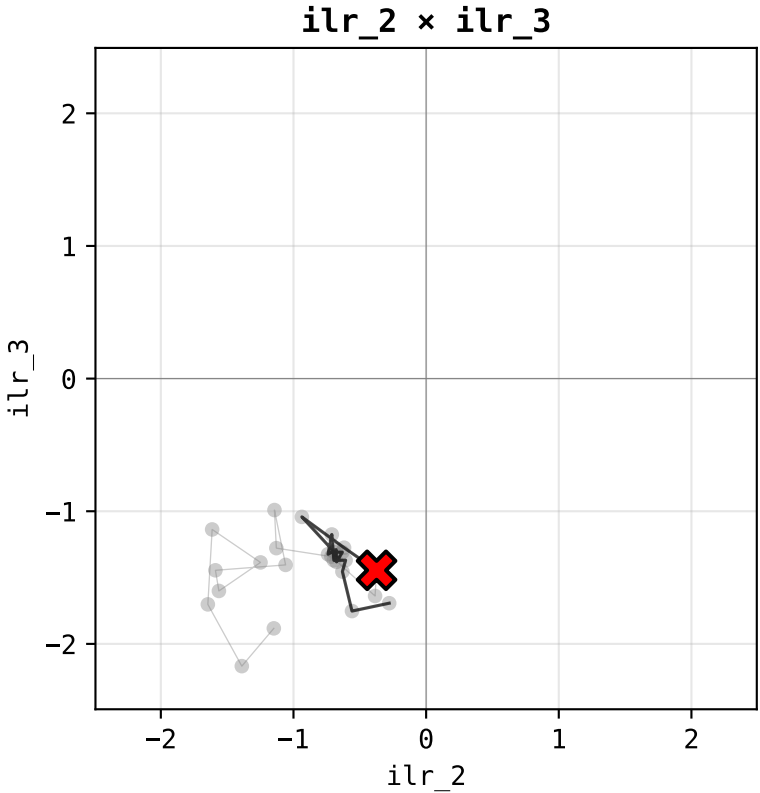
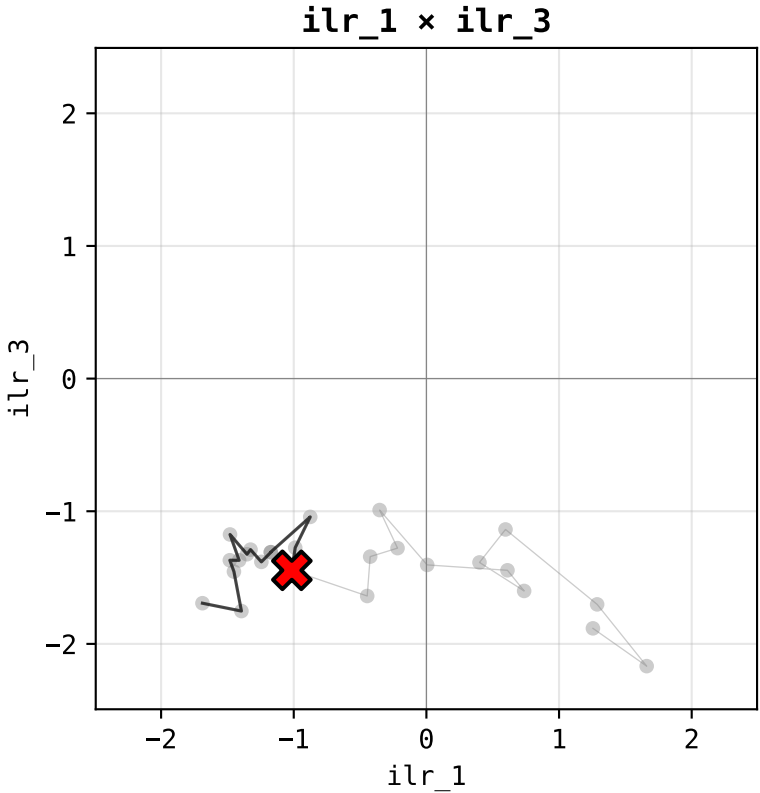
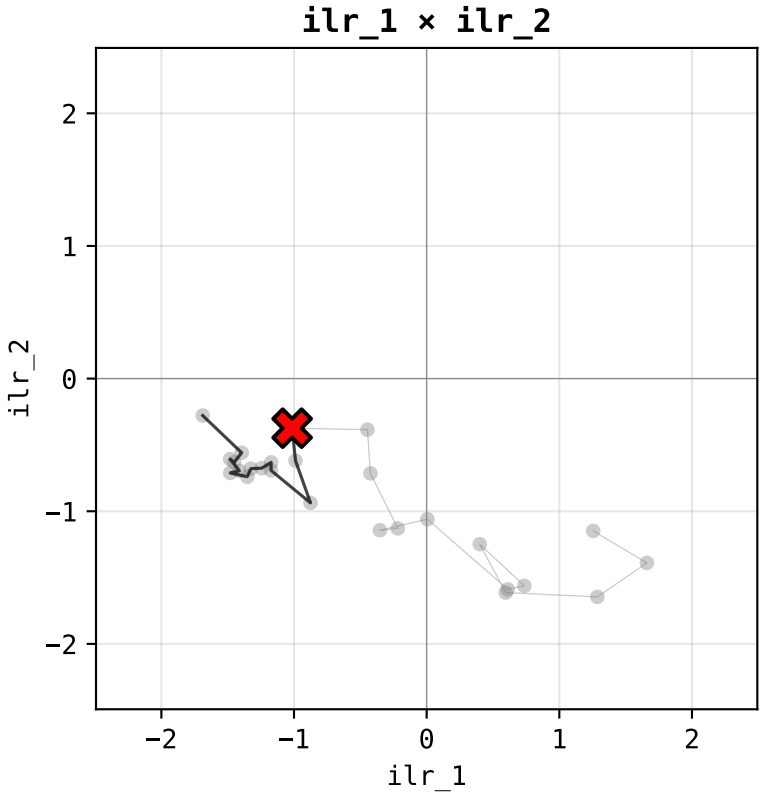


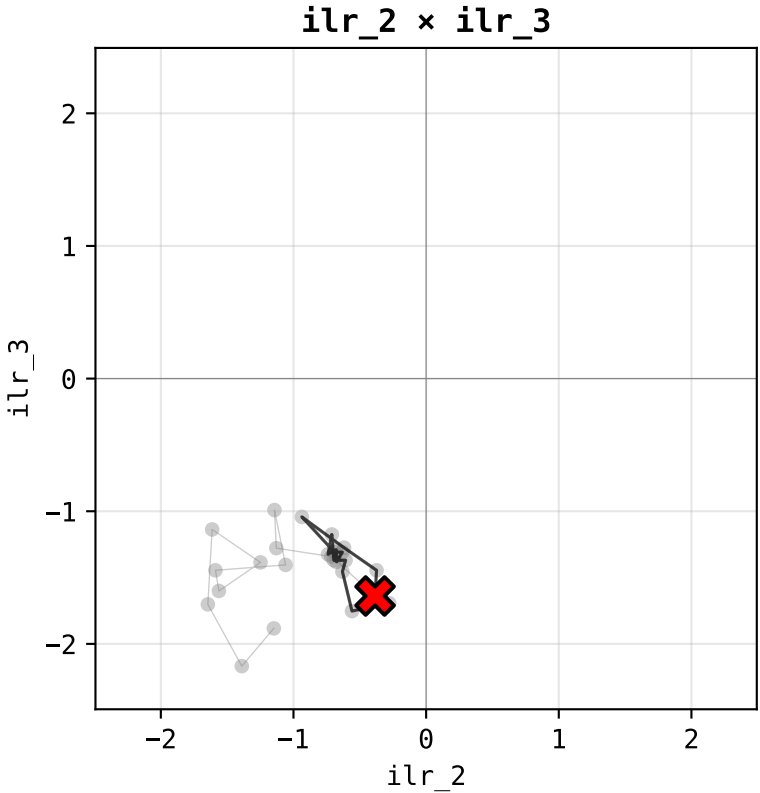
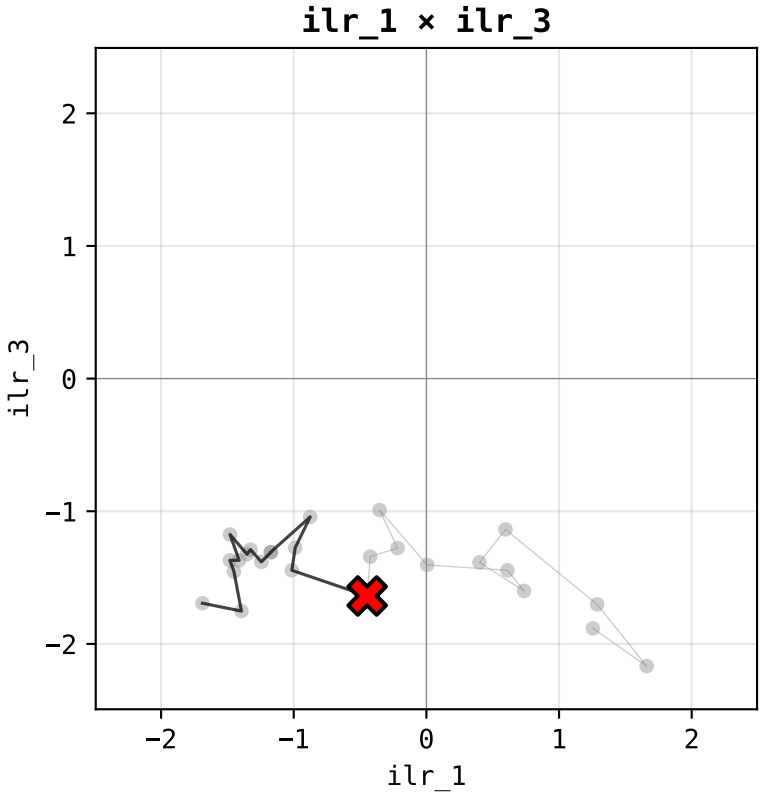
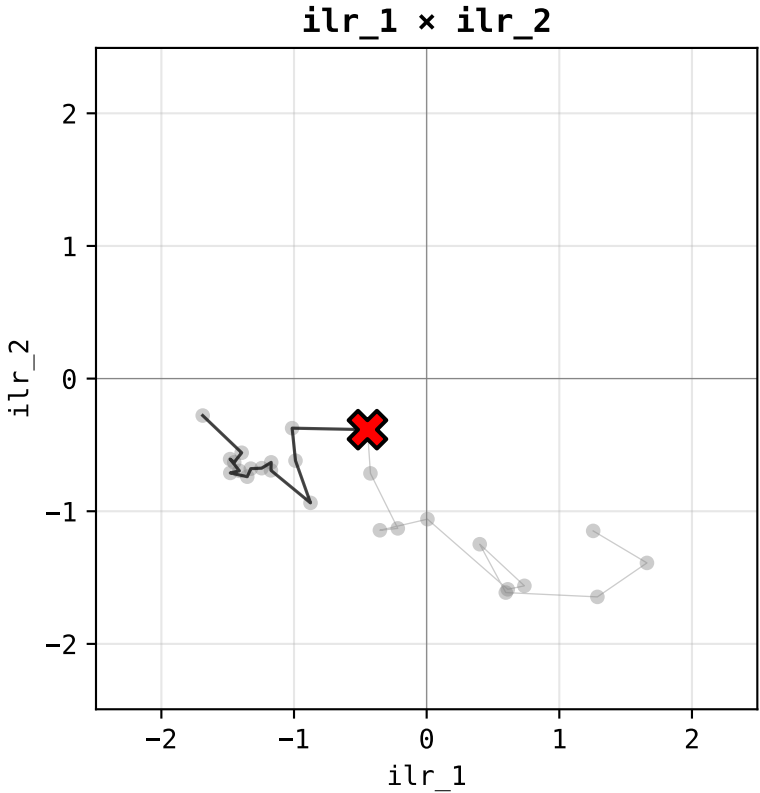
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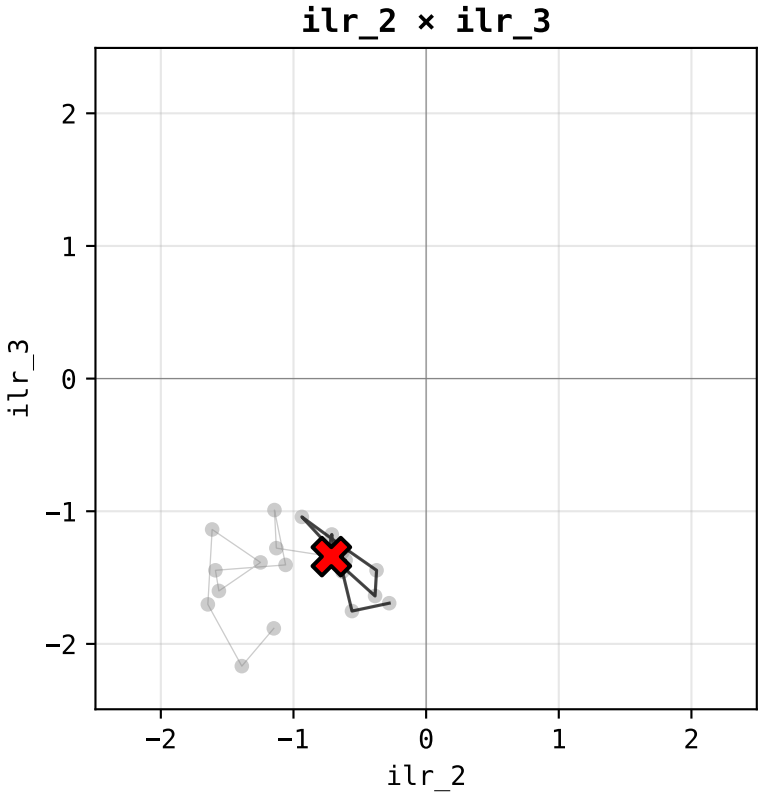
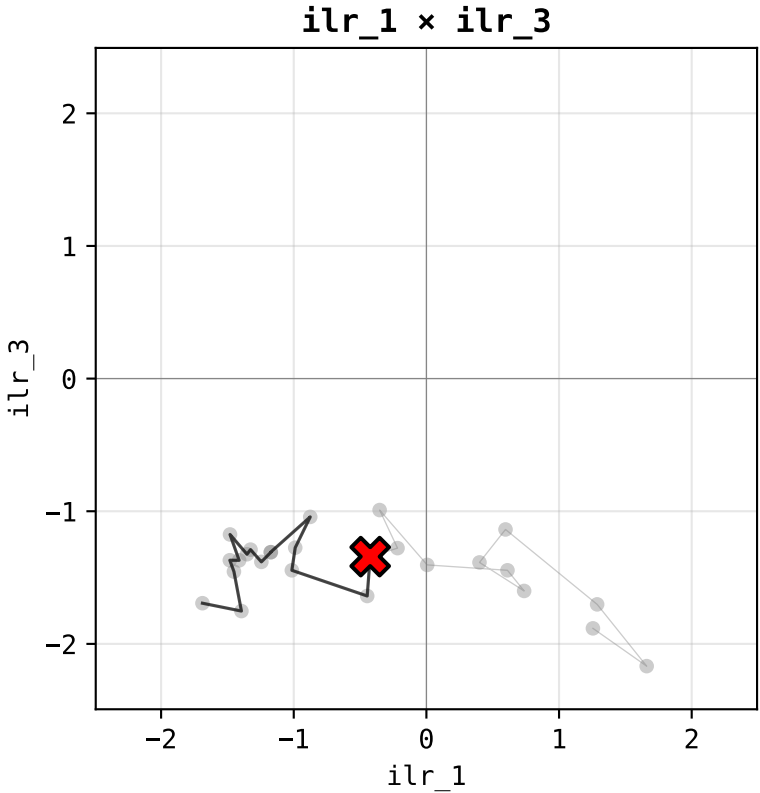
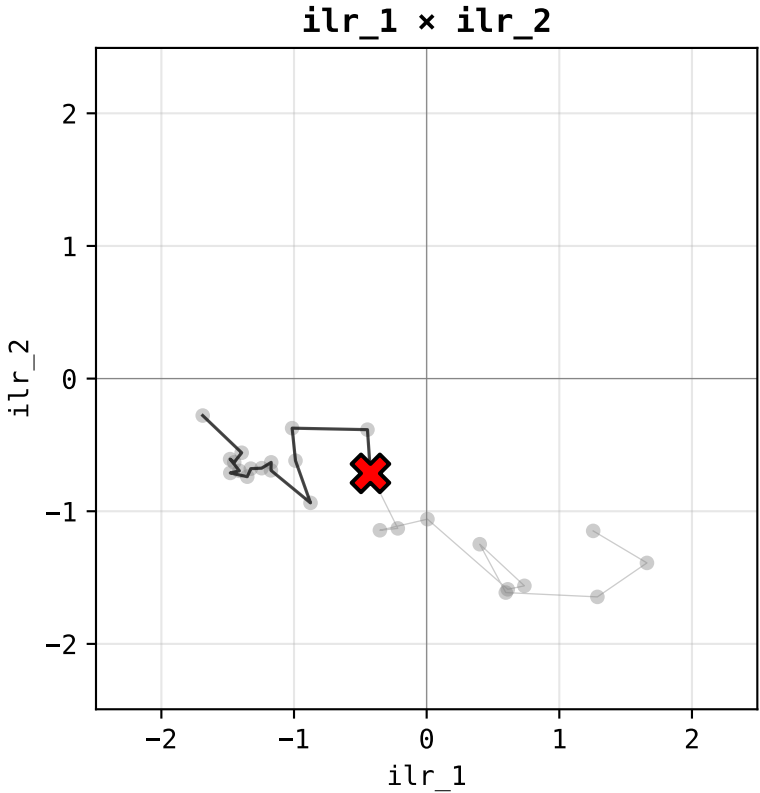


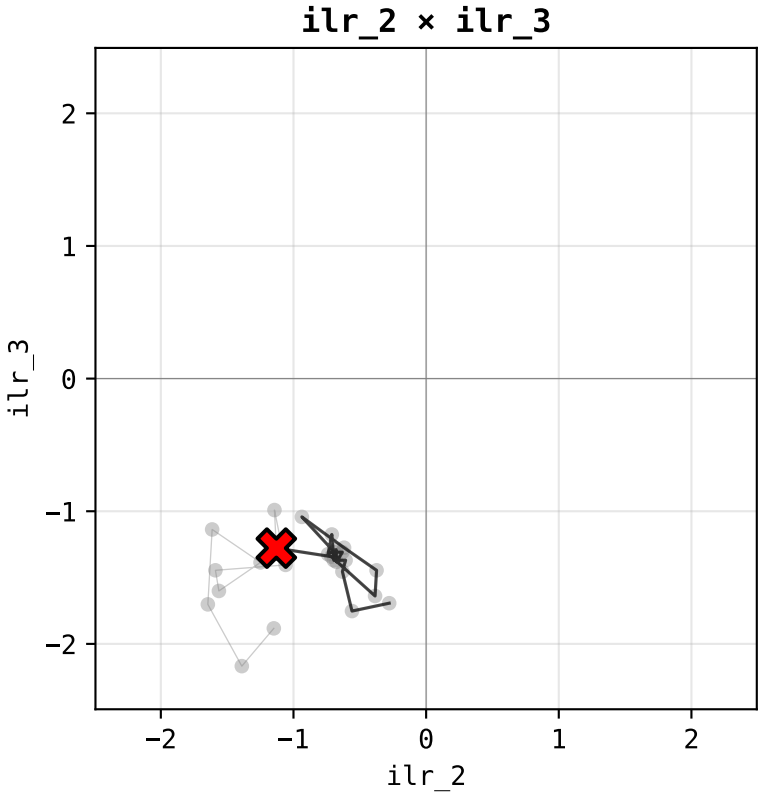
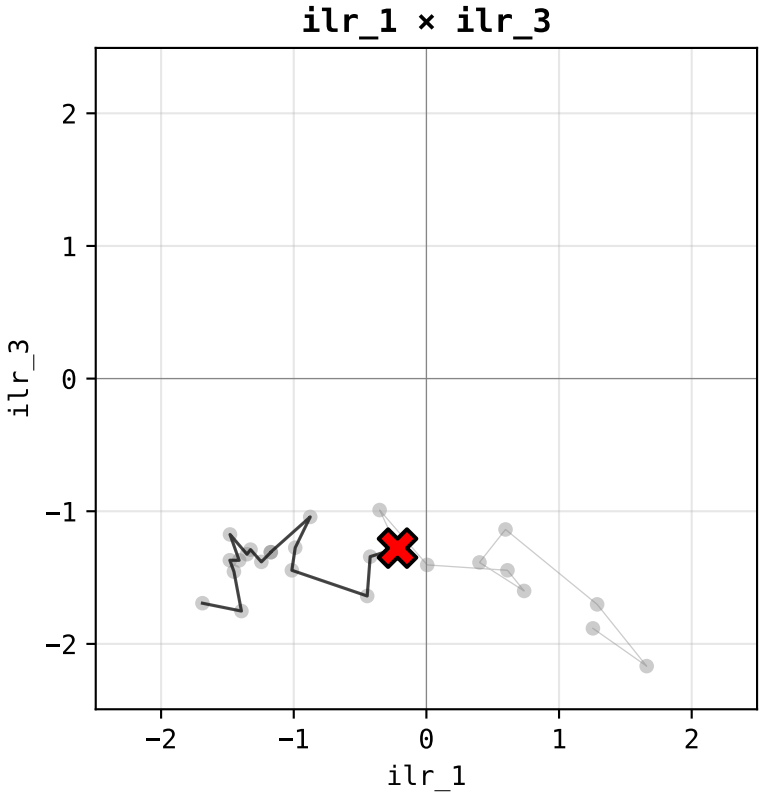
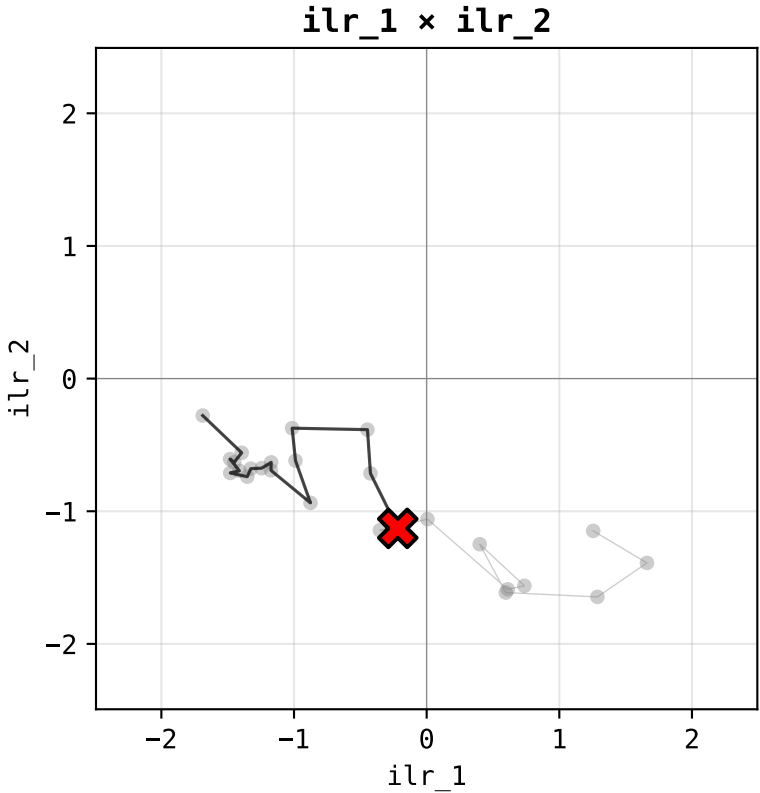


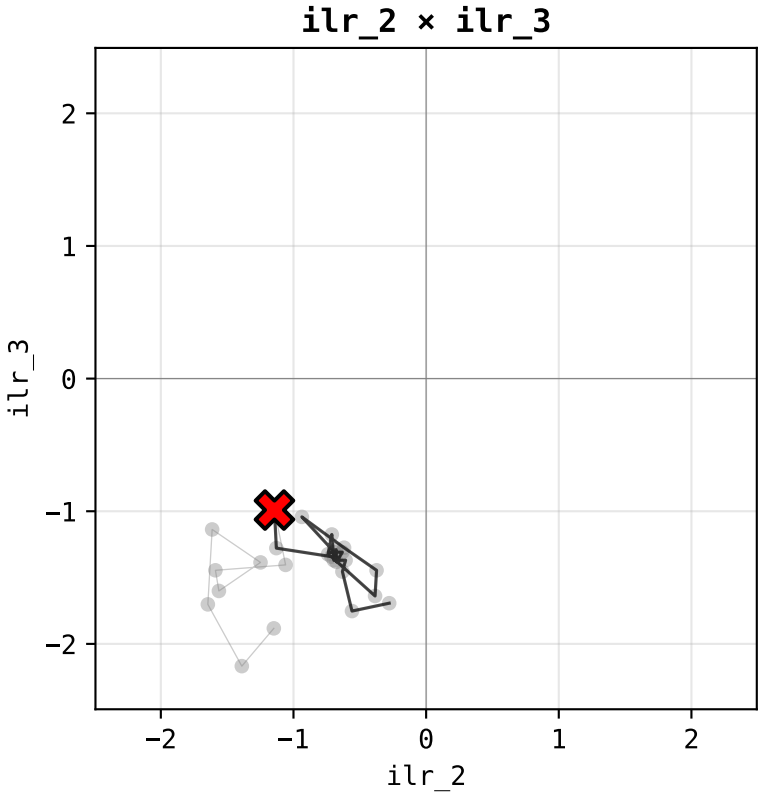
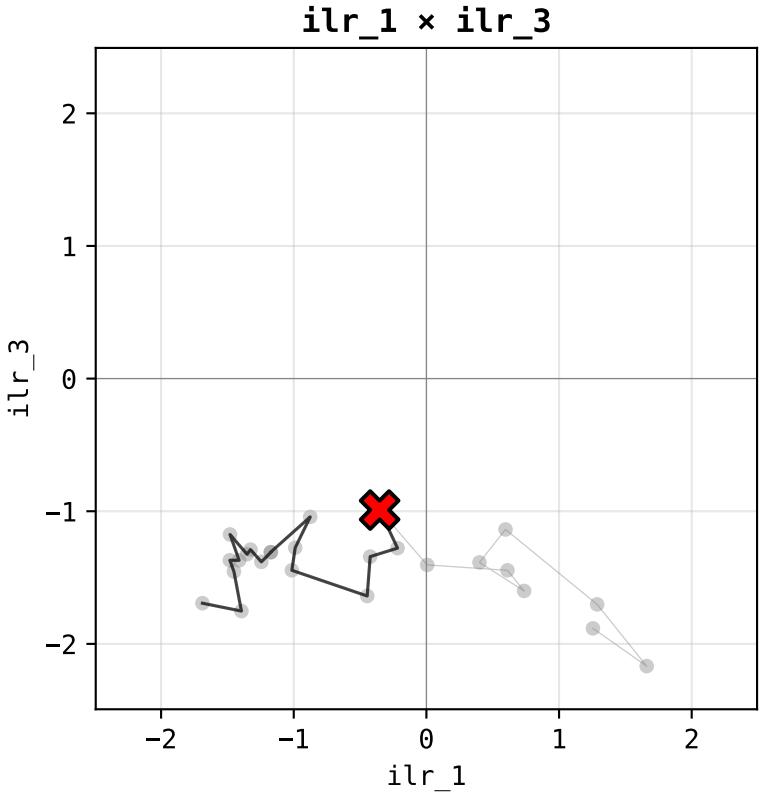
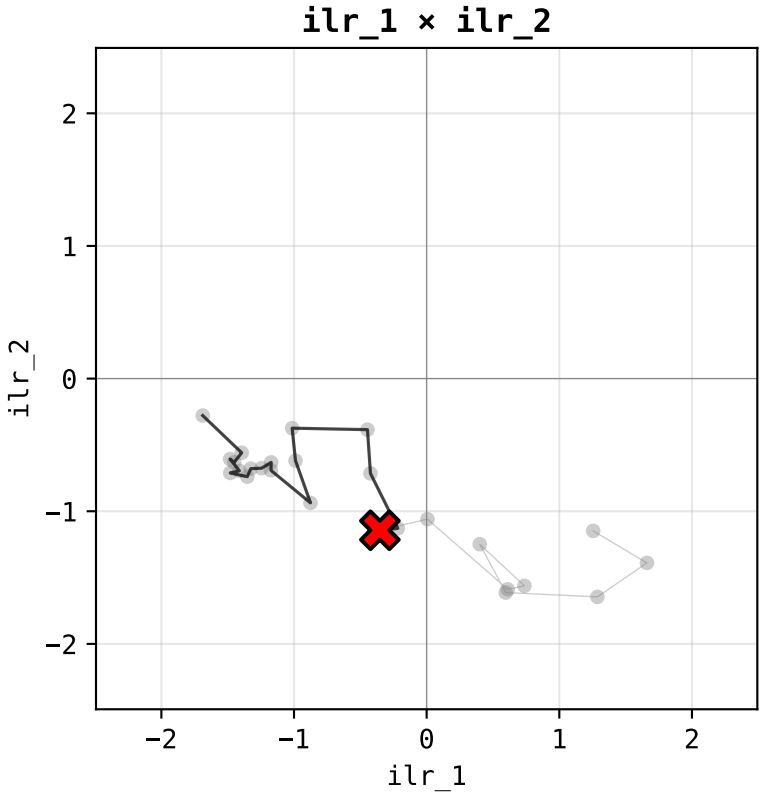












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